

Rational approximation for Zolotarev sign and ratio problems

C. Poussot-Vassal^{*1}, I-V. Gosea², and A.C. Antoulas³

¹ONERA, DTIS, Université de Toulouse, France
(charles.poussot-vassal@onera.fr)

²Max Planck Institute, CSC Group, Magdeburg, Germany
(gosea@mpi-magdeburg.mpg.de)

³Department of Electrical and Computer Engineering, Rice University, Houston
(aca@rice.edu)

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Abstract

We study algorithms to compute Zolotarev (3rd and 4th) problems rational approximation. First, we show that the Loewner framework (LF) is appropriate to rapidly and with no iterations, approximate the Zolotarev problems by compressing the (numerous) interpolation conditions. Second, we compare the approximation properties (e.g. coefficients and poles) of LF with the standard AAA, AAA-sign and its AAA-Lawson variants. We concentrate the study on a canonical example, namely the symmetric two-circles, for which the optimal solution is well documented. We show the numerical robustness of LF and its ability to recover the structure of the optimal solution. Additional geometries are also presented, emphasizing that LF is fast, reliable, and yields accurate approximants with no user intervention.

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^{*}Corresponding author: charles.poussot-vassal@onera.fr

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1 Introduction and contributions

About rational approximation. Approximation theory is a classic, well-established field of mathematics, split into a wide variety of sub-fields, out of which we are particularly interested in approximation by rational functions. This has a multitude of applications in areas of applied sciences. Some examples include computational fluid dynamics, solution and control of partial differential equations, data compression, electronic structure computation, systems and control theory (model order reduction of dynamical systems). For more details of the many aspects of approximation theory, we refer the reader to the classical books in [7, 21], and to the more recent books [23, 12] that also put emphasis on practical issues such as software implementation of the methods and applications, i.e., on data analysis for the latter.

Since rational functions are ratios of two polynomials, they have proven to be more versatile in approximating complex functions than simple polynomials. Furthermore, rational functions are extensively used in signal processing, systems, and control theory since the input-output behavior of linear dynamical systems is described in the frequency domain by the so-called transfer function (which is a rational function). Finally, the location of the poles of transfer functions determines important system properties such as asymptotic stability, transient behavior, damping, and harmonics.

Preliminaries on the Zolotarev 3rd and 4th problems. We are concerned with rational approximation, i.e., approximation of any type of functions, by means of rational ones. Here, we consider sign and ratio Zolotarev functions whose solution is intertwined. More precisely, the original function to be approximated falls into the category of a special sign function, defined on the complex domain, i.e., one that is defined on a reunion of two domains E and F , for which the value is constant on each, e.g., equal to ± 1 (hence the name Zolotarev sign problem, Z4, problem 2). To find the solution to the Zolotarev ratio function, one may appropriately transform the previously obtained solution. What is obtained is a function that is large on one domain (F), and small on the other (E), and thus, their ratio is minimized (hence the name Zolotarev ratio problem, Z3, problem 1).

Moreover, the methods we use to solve Zolotarev sign problems do not require access to the exact closed-form of the original function, but only to evaluations of it on a particular domain. Hence, these methods are *data-driven*.

In this work, we apply two established data-driven rational approximation methods: the Loewner Framework (LF) introduced in [16], and the Adaptive Antoulas Anderson (AAA) algorithm introduced in [18], and further developed in [19, 24]. The methods mentioned above have some things in common: they both compute approximants in the form of rational functions, they

exclusively use measurements of the original function (and not the function itself), and they involve, in one way or another, Loewner matrices.

1.1 Contributions

We summarize below the main contributions of this paper, which were confirmed through a series of comprehensive numerical experiments.

- (i) we show that the LF solves Zolotarev problems by compressing the number of interpolation points. This approach does not need neither any iterations nor user intervention, and yields solutions quite close to the optimal ones in a very low computational time. Consequently, the LF constitutes a valid alternative for addressing these problems;
- (ii) we show that the rational functions computed with LF come very close to optimal approximants and, in some specific cases (notably the symmetric ones), recover the structure of the optimal solution as well as its symmetry property (e.g., the distribution of eigenvalues and zeros, the alternance of polynomials coefficients, realness, etc.), whereas the other methods tend to add spurious and oddly distributed poles and zeros non-trivial artifacts that can't be easily explained. Such a feature is crucial for discovering the properties of the topology when only data values are available¹;
- (iii) we conduct an extensive numerical study² and comparison of the performance of different methods³ w.r.t. to computing time, accuracy of fit and interpretability, for approximating several sign functions defined on various domains showing, among others, the computational advantage of LF;

1.2 Paper structure

This paper is organized as follows: after the introduction in section 1, in section 2, the Zolotarev sign and ratio problems are introduced. Then, section 3 describes the main rational approximation tools to be used later on, e.g., the Loewner Framework in section 3.1 and the AAA algorithm in section 3.2. We keep the exposition brief, since both are established methods, and we refer to the original manuscripts for details. Then, in section 4 together with exact solutions for a special topology treated also in the work of [24], i.e., for which the E and F domains are two symmetric circles. Furthermore, a numerical treatment is also performed to expose the format of the numerators and denominators for low approximation orders. These numerical experiments are performed for various test cases (including the ones in [24]), and a detailed analysis is carried out in ???. Conclusions are presented in section 6.

2 The Zolotarev problems

In [24], a detailed historical account is presented, which is summarized next. Additionally, we refer the reader to the review paper [17] for further considerations.

2.1 Reminder and general procedure

For integers $m, n \geq 0$, let $R_{m,n}$ (or simply R_n if $m = n$) be the set of rational functions with numerator degree m and denominator of degree n , which means that any rational function $r \in R_{m,n}$ can be written as $p(z)/q(z)$ for some polynomials $p(z)$ and $q(z)$ of degrees at most m and n , respectively. The problem studied here is to find $r_{m,n}^* \in R_{m,n}$ such that

$$r_{m,n}^* := \arg \min_{r \in R_{m,n}} \frac{\max_{z \in E} |r(z)|}{\min_{z \in F} |r(z)|}. \quad (1)$$

¹Up to the author's knowledge, this property has not been highlighted in previous works.

²Results can be re-computed using the code available at <https://github.com/cpoussot/zolotarev>.

³Other methods are computed by means of the numerical tools proposed in [24].

Through normalization, i.e., assuming that $\min_{z \in F} |r(z)| = 1$, we end up with the problem of minimizing $|r(z)|$ solely over set E . This is known in the literature as the third Zolotarev problem, or sometimes, also the *Zolotarev ratio problem*.

Problem 1 (Zolotarev ratio problem - Z3) Find $r_{m,n}^* \in R_{m,n}$, satisfying condition $\min_{z \in F} |r_{m,n}^*(z)| = 1$, and that attains the minimum (where $\|\cdot\|_E$ denotes the supremum norm over E)

$$\sigma_n = \min_{r \in R_{m,n}} \|r\|_E. \quad (2)$$

As mentioned in [24], it is known that a solution $r_{m,n}^*$ to (2) exists, though it need not be unique, and that σ_n satisfies $0 < \sigma_n \leq 1$.

Then, the *sign function* relative to E and F is of relevance in what follows, and is defined as follows:

$$\text{sign}_{E/F}(z) = \begin{cases} -1 & z \in E, \\ +1 & z \in F. \end{cases} \quad (3)$$

It is to be noted that for $z \in \mathbb{C} \setminus \{E \cup F\}$, $\text{sign}_{E/F}(z)$ remains undefined.

Next, we introduce the problem of rational minimax approximation of $\text{sign}_{E/F}$ over E and F , or problem Z4:

Problem 2 (Zolotarev sign problem - Z4) Find $\hat{r}_{m,n} \in R_{m,n}$ that attains the minimum (where $\|\cdot\|_{E \cup F}$ denotes the supremum norm over $E \cup F$)

$$\tau_n = \min_{r \in R_{m,n}} \|r - \text{sign}_{E/F}\|_{E \cup F}. \quad (4)$$

As mentioned in [24], the proof showing that the problems Z3 and Z4 are equivalent in the case for which E and F are real intervals is due to the work of [1]. Then, in [13], it was shown that the equivalence generalizes to the complex case. We reproduce (with appropriate notation changes) in what follows their theorem.

Theorem 1 Every solution $r_{m,n}^*$ of Problem Z3 is related to a solution $\hat{r}_{m,n}$ of Problem Z4 by

$$\hat{r}_{m,n}(z) = \frac{1 - \sigma_n}{1 + \sigma_n} \frac{r_{m,n}^*(z) - \sqrt{\sigma_n}}{r_{m,n}^*(z) + \sqrt{\sigma_n}}, \quad r_{m,n}^*(z) = \sqrt{\sigma_n} \frac{(1 - \sigma_n)/(1 + \sigma_n) + \hat{r}_{m,n}(z)}{(1 - \sigma_n)/(1 + \sigma_n) - \hat{r}_{m,n}(z)}.$$

The minimal values of the two problems satisfy

$$\tau_n = \frac{2\sqrt{\sigma_n}}{1 + \sigma_n}, \quad \sigma_n = \left(\frac{\tau_n}{1 + \sqrt{1 - \tau_n^2}} \right)^2,$$

and the set of extremal points $M = \{z \in E \cup F, |\hat{r}_{m,n}(z) - \text{sign}_{E/F}(z)| = \tau_n\}$ is the union of $M_1 = \{z \in E, |r_{m,n}^*(z)| = \sigma_n\}$ and $M_2 = \{z \in F, |r_{m,n}^*(z)| = 1\}$.

The proposed algorithms for solving problem 1 (Z3) actually consist of solving problem 2 (Z4) first, and then transforming from $\hat{r}_{m,n}$ and τ_n to $r_{m,n}^*$ and σ_n . Following theorem 1, the authors of [24] approach Problem Z3 in the following two-step procedure: (i) solve Problem Z4 using a rational approximation method; then (ii) convert to a solution of Problem Z3 using theorem 1 and theorem 1.

The main challenge of the algorithm proposed in [24] is given by step (i), i.e., the computation of the degrees (m, n) rational best approximation $\hat{r}_{m,n}$ to the sign function $\text{sign}_{E/F}$ defined by the sets E and F . In [24], the proposed procedure is to find $\hat{r}_{m,n}$ by means of the AAA approximation technique. However, this has proven quite challenging. The first difficulty reported in [24] is that AAA encounters particular challenges when applied to sign functions. The second is the importance of having not just good rational approximations but nearly optimal ones, requiring the use of the AAA-sign and AAA-Lawson algorithm [19], which had a known problem of non-convergence in certain cases. These difficulties and their proposed solutions are discussed in [24,

§3.1 and §3.2]. Basically, the Lawson iteration "drives" the standard AAA approximant close to the optimal approximation value by performing several least squares steps involving the weights. Although seemingly effective, this approach has proven to be quite costly, mostly in terms of running time, but also in terms of the structure of the resulting polynomials, as will become evident in the next sections. This is particularly relevant when choosing a high degree (m, n) for the approximant.

2.2 Applications of the Z3 and Z4 problems

Zolotarev ratiion problems (Z3).

Zolotarev sign problems (Z4).

3 The main approximation methods

3.1 The Loewner framework

The Loewner Framework (LF) is a data-driven model identification, approximation, and reduction tool introduced in [16]. It provides a simple solution through rational approximation by means of interpolation. For more details, we refer the reader to the extensive tutorial/handbook papers in [14, 10, 6], the review paper [5], to the papers [11, 4, 15] that address extensions of the LF to some special classes of parametric, nonlinear or hybrid dynamical systems, and to chapter 4 in the recent book of interpolation and model reduction [3].

Given $N > 0$ distinct sampling points $\mathcal{T} = \{\tau_1, \tau_2, \dots, \tau_N\}$ together with N function evaluations at these points denoted with $\{f_1, f_2, \dots, f_N\}$, let $\mathfrak{D} = \{(\tau_\ell; f_\ell) | \ell = 1, \dots, N\}$ to be the problem data. Assume that $N = k + q$, where k and q are positive integers. The given data $\mathfrak{D}$ is first partitioned into the following two disjoint subsets,

$$\mathfrak{D} = \mathfrak{L} \cup \mathfrak{R}, \text{ where } \begin{cases} \text{right data : } \mathfrak{R} = \{(\lambda_i; w_i) | i = 1, \dots, q\}, \\ \text{left data : } \mathfrak{L} = \{(\mu_j; v_j) | j = 1, \dots, k\}. \end{cases} \quad (5)$$

The elements of the subsets $\mathfrak{L}$ and $\mathfrak{R}$ are as follows (for $1 \leq i \leq q$ and $1 \leq j \leq k$):

1. $\lambda_i, \mu_j \in \mathbb{C}$ are the right, and, respectively, left *sample/interpolation points*.
2. $w_i, v_j \in \mathbb{C}$ are the right, and, respectively, left *sample values*.

The problem can be formulated as follows: find a rational function $\mathbf{R}_{\text{Loew}}(z)$ of order $r \ll k$, such that the following interpolation conditions are (approximately) fulfilled:

$$\mathbf{R}_{\text{Loew}}(\lambda_i) = w_i, \quad \mathbf{R}_{\text{Loew}}(\mu_j) = v_j. \quad (6)$$

The next step is to arrange data (5) into matrix format. The **Loewner matrix** $\mathbb{L} \in \mathbb{C}^{k \times q}$ and the **shifted Loewner matrix** $\mathbb{S} \in \mathbb{C}^{k \times q}$ are defined as

$$\mathbb{L} = \begin{bmatrix} \frac{v_1 - w_1}{\mu_1 - \lambda_1} & \dots & \frac{v_1 - w_q}{\mu_1 - \lambda_q} \\ \vdots & \ddots & \vdots \\ \frac{v_k - w_1}{\mu_k - \lambda_1} & \dots & \frac{v_k - w_q}{\mu_k - \lambda_q} \end{bmatrix}, \quad \mathbb{S} = \begin{bmatrix} \frac{\mu_1 v_1 - w_1 \lambda_1}{\mu_1 - \lambda_1} & \dots & \frac{\mu_1 v_1 - w_q \lambda_q}{\mu_1 - \lambda_q} \\ \vdots & \ddots & \vdots \\ \frac{\mu_k v_k - w_1 \lambda_1}{\mu_k - \lambda_1} & \dots & \frac{\mu_k v_k - w_q \lambda_q}{\mu_k - \lambda_q} \end{bmatrix}, \quad (7)$$

while the data vectors $\mathbf{V}, \mathbf{W}^\top \in \mathbb{R}^k$ are introduced as

$$\mathbf{V} = [v_1 \ v_2 \ \dots \ v_k]^\top \in \mathbb{C}^k, \quad \mathbf{W} = [w_1 \ w_2 \ \dots \ w_q] \in \mathbb{C}^q. \quad (8)$$

Lemma 1 *If $k = q$ the matrix pencil $(\mathbb{S}, \mathbb{L})$ is regular, then one can directly compute the interpolation function as below:*

$$\mathbf{R}_{\text{Loew}}(z) = \mathbf{W}(\mathbb{S} - z\mathbb{L})^{-1}\mathbf{V}. \quad (9)$$

In applications (see, e.g., [10]), the pencil $(\mathbb{S}, \mathbb{L})$ is often singular. In these cases, one needs to perform an SVD of the Loewner pencil, constructed from the row- and column-wise concatenation of matrices $\mathbb{L}$ and $\mathbb{S}$.

By choosing a truncation value r , we can write the truncated SVDs as follows (with $\mathbf{X}_r = \mathbf{X}_r^{(1)} \in \mathbb{C}^{k \times r}$, $\mathbf{Y}_r = \mathbf{Y}_r^{(2)} \in \mathbb{C}^{q \times r}$)

$$\begin{bmatrix} \mathbb{L} & \mathbb{S} \end{bmatrix} \approx \mathbf{X}_r^{(1)} \mathbf{S}_r^{(1)} \mathbf{Y}_r^{(1)*}, \quad \begin{bmatrix} \mathbb{L} \\ \mathbb{S} \end{bmatrix} \approx \mathbf{X}_r^{(2)} \mathbf{S}_r^{(2)} \mathbf{Y}_r^{(2)*}. \quad (10)$$

Lemma 2 *The rational function corresponding to the reduced-order Loewner model is derived as:*

$$\mathbf{R}_{\text{Loew}}(z) = \hat{\mathbf{C}}_{\text{Loew}}(z\hat{\mathbf{E}}_{\text{Loew}} - \hat{\mathbf{A}}_{\text{Loew}})^{-1}\hat{\mathbf{B}}_{\text{Loew}} = \mathbf{W}\mathbf{Y}_r \left(\mathbf{X}_r^*(\mathbb{S} - z\mathbb{L})\mathbf{Y}_r \right)^{-1} \mathbf{X}_r^* \mathbf{V},$$

where the system matrices corresponding to a reduced Loewner model are:

$$\hat{\mathbf{E}}_{\text{Loew}} = -\mathbf{X}_r^* \mathbb{L} \mathbf{Y}_r, \quad \hat{\mathbf{A}}_{\text{Loew}} = -\mathbf{X}_r^* \mathbb{S} \mathbf{Y}_r, \quad \hat{\mathbf{B}}_{\text{Loew}} = \mathbf{X}_r^* \mathbf{V}, \quad \hat{\mathbf{C}}_{\text{Loew}} = \mathbf{W} \mathbf{Y}_r. \quad (11)$$

3.2 The AAA algorithm

The AAA (Adaptive Antoulas-Anderson) algorithm proposed in [18] represents an adaptive extension of the method originally introduced by A.C. Antoulas and B.D.O. Anderson in [2]. The AAA algorithm is a rational approximation data-based method that makes use of the barycentric form of rational functions and uses an iteration to add more interpolation points (in a greedy fashion) until a certain level of approximation is attained. As exposed in section 3.1, it only requires as input evaluations of the function on a set of points denoted with $\mathcal{T}$. We refer to [20] for a recent review paper on applications of AAA.

Consider $N > 0$ distinct sampling points in the set $\mathcal{T} = \{\tau_1, \tau_2, \dots, \tau_N\}$ together with N function evaluations at these points, denoted with $\{f_1, f_2, \dots, f_N\}$. The AAA algorithm is based on an iteration. Let the order of the rational approximant $\mathbf{R}_{\text{AAA}}(z)$ that is constructed through the algorithm in [18] at step $\ell \geq 1$ be (ℓ, ℓ) (degree of numerator and denominator), which is written in the following barycentric format:

$$\mathbf{R}_{\text{AAA}}(z) = \frac{\sum_{k=0}^{\ell} \frac{\alpha_k w_k}{z - \lambda_k}}{\sum_{k=0}^{\ell} \frac{\alpha_k}{z - \lambda_k}}, \quad (12)$$

where $\{\lambda_0, \dots, \lambda_{\ell}\}$ are the interpolation points (selected from the set $\mathcal{T}$), $\{w_0, \dots, w_{\ell}\}$ are the corresponding function evaluations, and $\{\alpha_0, \dots, \alpha_{\ell}\}$ are the weights. The method enforces interpolation at the ℓ points $\{\lambda_0, \dots, \lambda_{\ell}\}$, i.e. $\mathbf{R}_{\text{AAA}}(\lambda_k) = w_k = F(\lambda_k)$, $\forall 0 \leq k \leq \ell$. Then, at each step, one needs to determine suitable weights $\alpha_0, \alpha_2, \dots, \alpha_{\ell}$ by formulating an approximation problem. Instead of solving the more general problem above (which is nonlinear in variables $\alpha_1, \alpha_2, \dots, \alpha_{\ell}$), one solves a relaxed problem by means of linearization. This procedure leads to a least squares minimization problem that involves a (rectangular) Loewner matrix of dimension $(N - \ell) \times \ell$. The solution relies on computing the SVD of the latter, and selecting the weights as the entries of the (normalized) last right singular vector. At the end of the iteration, the rational approximant is of order (r, r) ; one can also explicitly write the AAA function as $\mathbf{R}_{\text{AAA}}(z) = \hat{\mathbf{C}}_{\text{AAA}}(z\hat{\mathbf{E}}_{\text{AAA}} - \hat{\mathbf{A}}_{\text{AAA}})^{-1}\hat{\mathbf{B}}_{\text{AAA}}$, where the matrices are of dimension $r = \ell + 1$, given explicitly in terms of the selected support points $\{\lambda_0, \dots, \lambda_r\}$, the function evaluations $\{w_0, \dots, w_r\}$ and the weights $\{\alpha_0, \dots, \alpha_r\}$. Note that for the numerical examples presented in what follows, we will be using the AAA implementation available in the **Chebfun** tool [8]⁴.

⁴Also available at <https://github.com/chebfun/chebfun>.

3.2.1 Minimax algorithms and Lawson step extensions

Computing rational minimax approximations is a challenging task, especially when there are singularities on or near the interval of approximation. In the recent contribution [9], the authors propose different combinations of robust algorithms that are based on rational barycentric representations. One key feature is that the support points are chosen adaptively as the approximant is computed. The **Chebfun** minimax implementation is available in [8]. The AAA-Lawson algorithm in [19] uses an iteratively re-weighted least-squares method followed by a classical Remez algorithm to enforce quadratic convergence. Minimax approximants can be recognized by their equioscillatory error curves in real cases and nearly circular error curves in complex problems.

In the recent note [24], the authors present an algorithm based on AAA-Lawson approximation [19], enhanced by various modifications that make it suitable for dealing with functions of the flavor of $f(z) = \text{sign}(z)$ and for ensuring convergence of the Lawson phase. This variant/improvement of the standard AAA algorithm in [18] is meant as a reliable tool for numerical computations of solutions for the third and fourth Zolotarev problems, i.e., classical problems in approximation theory.

An attentive reader may realize that in [24], these developments are deemed a "breakthrough" since "a reliable method for computing these functions has not been available before". However, the LF is known since 1986 in its one-sided version [2] and since 2007 in its two-sided version [16] (which was used in this work).

4 The two symmetric circles case

We start the numerical exposition with a rather simplified configuration, e.g., for which the true optimal rational approximation solution for the Zolotarev problem Z4 is explicitly known (this configuration is denoted '1a' in [24]). Let us consider two symmetric circles (with respect to the imaginary axis), centered along the horizontal axis, with radius ρ and centered at $\pm\alpha$. In this case, the optimal rational r -th order approximant $r_r^*(z)$ derived in [22] is given by:

$$r_r^*(z) = \left(\frac{z - \sqrt{\alpha^2 - \rho^2}}{z + \sqrt{\alpha^2 - \rho^2}} \right)^r \quad \text{and} \quad \sigma_r = \left(\frac{1 - \sqrt{\alpha^2 - \rho^2}}{1 + \sqrt{\alpha^2 - \rho^2}} \right)^r. \quad (13)$$

4.1 Experimental setup and results

The case considered in what follows corresponds to the one in [24, Figure 1(a)], for which $\rho = 1/2$ and $\alpha = 1$. We now compare the rational approximation obtained for problem Z4 when using (i) the LF, (ii) the AAA-Lawson (with 200 iterations) and (iii) the AAA-Lawson (with 1000 iterations)⁵. In ?? we report the coefficients of both numerator and denominator of the rational approximation obtained with each different method, and compare to the optimal Z4 approximant of the same order, obtained with eq. (13) ($r = 2$). The maximum approximation errors σ_2 are displayed for all approximants.

4.2 Comments

The first comment for this $r = 2$ case and ?? is that the optimal rational approximation eq. (13) is attained with a real-valued function first columns) and a rational complexity of degree (1,2). One notices that:

- LF provides a solution with real-valued coefficients (up to machine precision), while both AAA-Lawson end up with non-negligible imaginary coefficients;

⁵The AAA-Lawson is obtained with optional arguments 'sign',1,'damping',.95,'lawson',200 and 'sign',1,'damping',.95,'lawson',1000, using the AAA code in Chebfun, available at <https://github.com/chebfun/chebfun.git>.

- by inspecting the numerator (first row), both AAA-Lawson suggest a non-null coefficient in s^2 , resulting in a rational complexity of $R_{2,2}$ while the LF indicates a null term, leading to a complexity $R_{1,2}$ (as the optimal one);
- by inspecting the numerator of degree 0 and denominator of degree 1, it appears that the optimal are real with null coefficients. Again, only LF recovers this property.
- by inspecting the σ_2 values for each method, AAA-Lawson provides a better approximant (i.e., with a lower σ_2) than LF (which is better than standard AAA); moreover, AAA-Lawson actually provides the near-optimal σ_2 .

By choosing higher order values as $r = 3, 4, \dots$, one may observe that the previous remarks still hold⁶. In addition, inspecting the coefficients of the denominator of the AAA and AAA-Lawson, the magnitude of the imaginary terms is even larger. In each case, this is quite unexpected and difficult to justify.

As an illustration, we pursue the investigation with $r = 6$ and $r = 8$, and obtain the coefficients displayed in table 53 and table 83. These two cases are of interest since, while for table 53 the above comment does not stand (indeed, both AAA-Lawson lead to almost null imaginary parts), they hold true again for table 83. In addition, for the latter case ($r = 8$), computing the poles and zeros (only for the optimal LF and AAA-Lawson approximants), we obtain ??, emphasizing that both the poles and zeros obtained with AAA-Lawson are quite far from the optimal ones, while the LF ones are very close. In addition, inspecting the right frame (poles), only seven values are visible. Indeed, the last one has the value: $3.53 \cdot 10^7 - i5.44 \cdot 10^9$. In this case, by inverting the numerator and denominator of the AAA-Lawson approximant, poles and zeros are more aligned (at least in the imaginary part) and coherent, which is again difficult to justify.

Basis	Opt., $\sigma_3 = 0.00037$		LF, $\sigma_3 = 0.0021$		AAA-sign-L200, $\sigma_3 = 0.00044$		AAA-sign-L1000, $\sigma_3 = 0.0005$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^3	0	0	0	0	-0.011	$6.5 \cdot 10^{-4}$	-70.0	12.0
z^2	2.6	0	3.7	$3.3 \cdot 10^{-16}$	2.6	$-2.2 \cdot 10^{-3}$	0.95	0.064
z	0	0	$-6.9 \cdot 10^{-15}$	$-4.1 \cdot 10^{-16}$	$9.9 \cdot 10^{-3}$	$1.0 \cdot 10^{-3}$	-150.0	27.0
1.0	0.65	0	1.5	$2.7 \cdot 10^{-16}$	0.65	$-6.1 \cdot 10^{-4}$	-0.36	0.082
z^3	1.0	0	1.0	0	1.0	0	1.0	0
z^2	0	0	$-4.3 \cdot 10^{-15}$	$-9.4 \cdot 10^{-16}$	$-6.6 \cdot 10^{-3}$	$2.1 \cdot 10^{-3}$	-180.0	31.0
z	2.2	0	4.2	$1.3 \cdot 10^{-15}$	2.3	$-1.9 \cdot 10^{-3}$	-0.38	0.084
1.0	0	0	$-2.8 \cdot 10^{-15}$	$-2.0 \cdot 10^{-16}$	$7.3 \cdot 10^{-3}$	$2.0 \cdot 10^{-4}$	-44.0	7.6

Table 1: Case 1a (cpv MACA64 MR2023b), $r = 3$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$		LF, $\sigma_6 = 3.2 \cdot 10^{-07}$		AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$		AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^6	0	0	0	0	$-1.2 \cdot 10^{-10}$	$-2.0 \cdot 10^{-10}$	$1.9 \cdot 10^{-12}$	$-8.0 \cdot 10^{-11}$
z^5	5.2	0	5.5	$-5.7 \cdot 10^{-15}$	5.2	$-6.3 \cdot 10^{-12}$	5.2	$-4.6 \cdot 10^{-12}$
z^4	0	0	$-2.3 \cdot 10^{-14}$	$-1.3 \cdot 10^{-14}$	$-1.4 \cdot 10^{-9}$	$-2.3 \cdot 10^{-9}$	$2.1 \cdot 10^{-11}$	$-9.2 \cdot 10^{-10}$
z^3	13.0	0	15.0	$-1.1 \cdot 10^{-14}$	13.0	$-4.7 \cdot 10^{-11}$	13.0	$-3.5 \cdot 10^{-11}$
z^2	0	0	$-4.2 \cdot 10^{-14}$	$-3.3 \cdot 10^{-14}$	$-1.1 \cdot 10^{-9}$	$-1.7 \cdot 10^{-9}$	$1.5 \cdot 10^{-11}$	$-7.0 \cdot 10^{-10}$
z	2.9	0	3.7	$6.4 \cdot 10^{-16}$	2.9	$-1.7 \cdot 10^{-11}$	2.9	$-1.3 \cdot 10^{-11}$
1.0	0	0	$-4.1 \cdot 10^{-15}$	$-2.9 \cdot 10^{-15}$	$-5.3 \cdot 10^{-11}$	$-8.7 \cdot 10^{-11}$	$7.8 \cdot 10^{-13}$	$-3.6 \cdot 10^{-11}$
z^6	1.0	0	1.0	0	1.0	0	1.0	0
z^5	0	0	$-7.0 \cdot 10^{-15}$	$-4.7 \cdot 10^{-15}$	$-6.5 \cdot 10^{-10}$	$-1.0 \cdot 10^{-9}$	$9.8 \cdot 10^{-12}$	$-4.2 \cdot 10^{-10}$
z^4	11.0	0	12.0	$-1.4 \cdot 10^{-14}$	11.0	$-2.7 \cdot 10^{-11}$	11.0	$-2.0 \cdot 10^{-11}$
z^3	0	0	$-4.5 \cdot 10^{-14}$	$-3.7 \cdot 10^{-14}$	$-1.6 \cdot 10^{-9}$	$-2.6 \cdot 10^{-9}$	$2.4 \cdot 10^{-11}$	$-1.1 \cdot 10^{-9}$
z^2	8.4	0	10.0	$-1.8 \cdot 10^{-16}$	8.4	$-4.0 \cdot 10^{-11}$	8.4	$-3.0 \cdot 10^{-11}$
z	0	0	$-2.0 \cdot 10^{-14}$	$-9.8 \cdot 10^{-15}$	$-3.6 \cdot 10^{-10}$	$-6.0 \cdot 10^{-10}$	$5.3 \cdot 10^{-12}$	$-2.5 \cdot 10^{-10}$
1.0	0.42	0	0.56	$-5.6 \cdot 10^{-16}$	0.42	$-3.0 \cdot 10^{-12}$	0.42	$-2.3 \cdot 10^{-12}$

Table 2: Case 1a (cpv MACA64 MR2023b), $r = 6$, Z4: numerator (first lines) and denominator (last lines) coefficients.

⁶We refer to the <https://github.com/cpoussot/zolotarev> page for an in-depth analysis.

Basis	Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$		LF, $\sigma_8 = 1.6 \cdot 10^{-09}$		AAA-sign-L200, $\sigma_8 = 7.1 \cdot 10^{-10}$		AAA-sign-L1000, $\sigma_8 = 7.1 \cdot 10^{-10}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^8	0	0	0	0	$-8.7 \cdot 10^5$	$1.7 \cdot 10^8$	$-5.1 \cdot 10^6$	$-7.9 \cdot 10^8$
z^7	6.9	0	7.2	$-1.8 \cdot 10^{-14}$	7.0	$-4.9 \cdot 10^{-4}$	7.0	$-3.4 \cdot 10^{-3}$
z^6	0	0	$-5.1 \cdot 10^{-14}$	$-1.6 \cdot 10^{-13}$	$-1.8 \cdot 10^7$	$3.7 \cdot 10^9$	$-1.1 \cdot 10^8$	$-1.7 \cdot 10^{10}$
z^5	36.0	0	41.0	$-6.7 \cdot 10^{-14}$	37.0	$-9.8 \cdot 10^{-3}$	37.0	-0.046
z^4	0	0	$8.0 \cdot 10^{-14}$	$-3.5 \cdot 10^{-13}$	$-3.4 \cdot 10^7$	$6.9 \cdot 10^9$	$-2.0 \cdot 10^8$	$-3.1 \cdot 10^{10}$
z^3	27.0	0	33.0	$-1.8 \cdot 10^{-14}$	28.0	-0.014	28.0	-0.051
z^2	0	0	$1.6 \cdot 10^{-13}$	$-8.0 \cdot 10^{-14}$	$-1.0 \cdot 10^7$	$2.1 \cdot 10^9$	$-6.0 \cdot 10^7$	$-9.3 \cdot 10^9$
z	2.9	0	3.7	$3.2 \cdot 10^{-14}$	3.0	$-2.4 \cdot 10^{-3}$	3.1	$-7.0 \cdot 10^{-3}$
1.0	0	0	$1.0 \cdot 10^{-14}$	$1.0 \cdot 10^{-14}$	$-2.8 \cdot 10^5$	$5.5 \cdot 10^7$	$-1.6 \cdot 10^6$	$-2.5 \cdot 10^8$
z^8	1.0	0	1.0	0	1.0	0	1.0	0
z^7	0	0	$-1.4 \cdot 10^{-14}$	$-3.8 \cdot 10^{-14}$	$-6.0 \cdot 10^6$	$1.2 \cdot 10^9$	$-3.5 \cdot 10^7$	$-5.4 \cdot 10^9$
z^6	21.0	0	23.0	$-5.5 \cdot 10^{-14}$	21.0	$-3.4 \cdot 10^{-3}$	21.0	-0.019
z^5	0	0	$-5.0 \cdot 10^{-14}$	$-2.4 \cdot 10^{-13}$	$-3.2 \cdot 10^7$	$6.3 \cdot 10^9$	$-1.9 \cdot 10^8$	$-2.9 \cdot 10^{10}$
z^4	39.0	0	46.0	$-6.0 \cdot 10^{-14}$	40.0	-0.015	40.0	-0.062
z^3	0	0	$2.0 \cdot 10^{-13}$	$-2.4 \cdot 10^{-13}$	$-2.4 \cdot 10^7$	$4.7 \cdot 10^9$	$-1.4 \cdot 10^8$	$-2.1 \cdot 10^{10}$
z^2	12.0	0	15.0	$3.2 \cdot 10^{-14}$	12.0	$-8.0 \cdot 10^{-3}$	12.0	-0.025
z	0	0	$6.2 \cdot 10^{-14}$	$1.8 \cdot 10^{-14}$	$-2.5 \cdot 10^6$	$5.1 \cdot 10^8$	$-1.5 \cdot 10^7$	$-2.3 \cdot 10^9$
1.0	0.32	0	0.42	$8.3 \cdot 10^{-15}$	0.33	$-3.2 \cdot 10^{-4}$	0.33	$-8.5 \cdot 10^{-4}$

Table 3: Case 1a (cpv MACA64 MR2023b), $r = 8$, Z4: numerator (first lines) and denominator (last lines) coefficients.

4.3 Preliminary conclusions

As highlighted in this simple case (of two circles), both AAA and AAA-Lawson variants produce **complex-valued approximants**, whereas the optimal result suggests a solution with real coefficients only. Alternatively, LF always produces real-valued approximants (up to machine precision), without explicitly enforcing such a property⁷. Second, only the LF seems to be able to recover the structure of the solution, i.e., the real coefficients and the alternate null and non-null coefficients in the numerator and denominator monomial coefficients, thus discovering the exact complexity and structure of the optimal approximant. Moreover, in AAA and AAA-Lawson, as r increases, the spurious imaginary part of the coefficients becomes no longer negligible and even predominant. Finally, inspecting the quality of the approximation evaluated through the lens of σ_r , it appears that AAA is not accurate. Its conjunction with the Lawson iteration, however, seems to lead to significantly better results, i.e., close to the Zolotarev optimal, and better (in optimal value) than those obtained with LF. The analysis of the poles and zeros of the Z4 approximations (and comparing with the optimal ones) showed, however, an inaccurate restitution of the AAA-Lawson, while the LF provides an accurate matching.

4.4 Some (open) questions

Along with these first comments, we believe it is legitimate to seek the significance in how spurious these complex terms are, and how they can provide a better σ_r approximation. Indeed, while from a practical point of view, a better approximation (lower σ_r) may be satisfactory, from a theoretical point of view, improving accuracy by adding spurious coefficients is not.

Finally, it is to be mentioned that all experiments were run on four different computational environments (consisting of various operating systems and different versions of MATLAB). While the LF led to the same results (on each machine), the AAA consistently led to significantly different results, both concerning the distribution of poles and zeros and the magnitude of the monomial coefficients appearing in the numerator and denominator. We report here only the results obtained using the computational environment of the third author of this work, for reasons of brevity.

4.5 More complete and accurate approximations

By continuing with this two-circle case, let us now compare the approximation features for significantly higher orders. In fig. 1 we display the rational approximation for Z3 with contours.

⁷Notice that it is also possible to enforce parameters' realness, as explained in [6, §2.4.5].

Then, both the poles and the zeros for Z3 and Z4 are displayed. Each frame illustrates the rational approximation reached by each method. In fig. 1, the approximation order r has been selected such that the normalized singular values of the LF are below 10^{-14} , and then the same objective is used for the three other methods.

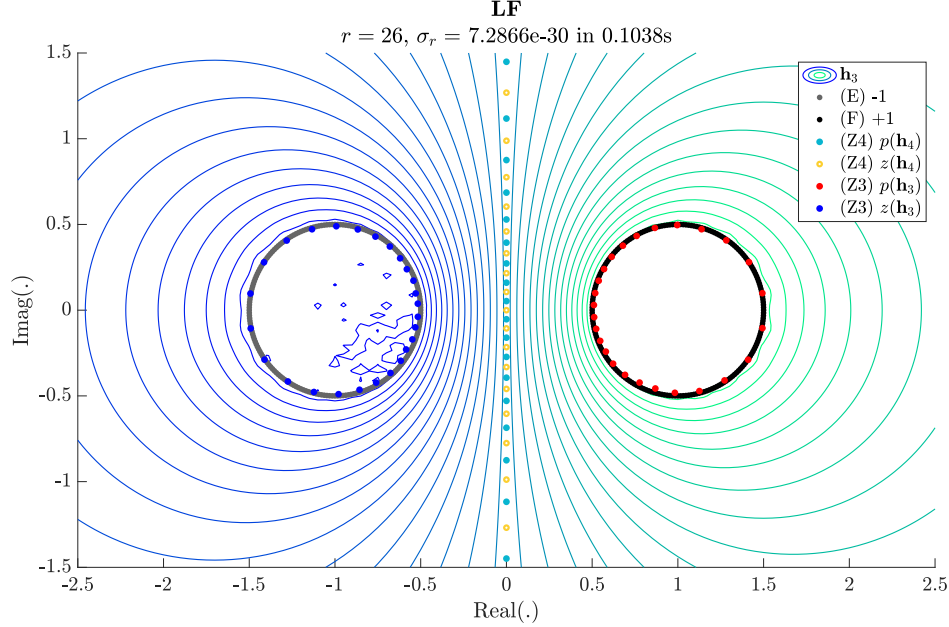


Figure 1: Case '1a', with $r = 26$: different approximation methods: LF (top left), AAA (top right), AAA with damping (bottom left), and AAA-Lawson with damping and 200 iterations (bottom right). Each frame shows: the Zolotarev rational functions of degree r for Z3 ($\mathbf{h}_3$) and Z4 ($\mathbf{h}_4$) defined by connected sets E (on the left) and F (on the right); the Z3 function approximate contours levels $\log_{10} |\mathbf{h}_3(z)| = 0, -1, -2, \dots, -30$ between the two domains; black circles and dots respectively mark zeros and poles of the sign function $\mathbf{h}_4$; blue and red dots respectively denote the zeros and poles of $\mathbf{h}_3$. The minimum values σ_r and computation time are given in the title, as well as the options of each AAA method, and give the number of singularities targeted (and obtained).

What is remarkable in fig. 1, for the LF case, is the clean distribution of the poles and zeros of both Z3 and Z4 problems along the ± 1 circles and vertical axis, with a perfect symmetry. This feature is guaranteed by the almost perfect alternate null and non-null coefficients and the null (machine precision) imaginary part. All other methods provide a slightly irregular distribution with imaginary part in the monomial coefficients. Moreover, in this setup, the optimal $\sigma_r \approx 1.8144 \cdot 10^{-30}$ eq. (13) while the LF reaches the best approximation with $\sigma_n \approx 7.2866 \cdot 10^{-30}$, very close to the optimal one. Notice also that the computation time is much lower for the LF with respect to the AAA versions (and even more with respect to the AAA-Lawson).

For this very same topology, let us now evaluate the Zolotarev ratio and computation time for different rational approximation orders. They are reported in ???. By inspecting the latter, one observes that the LF requires a constant computation time while the AAA variants are dependent

on the approximation order r . Indeed, LF benefits from the fact that it is not iterative.

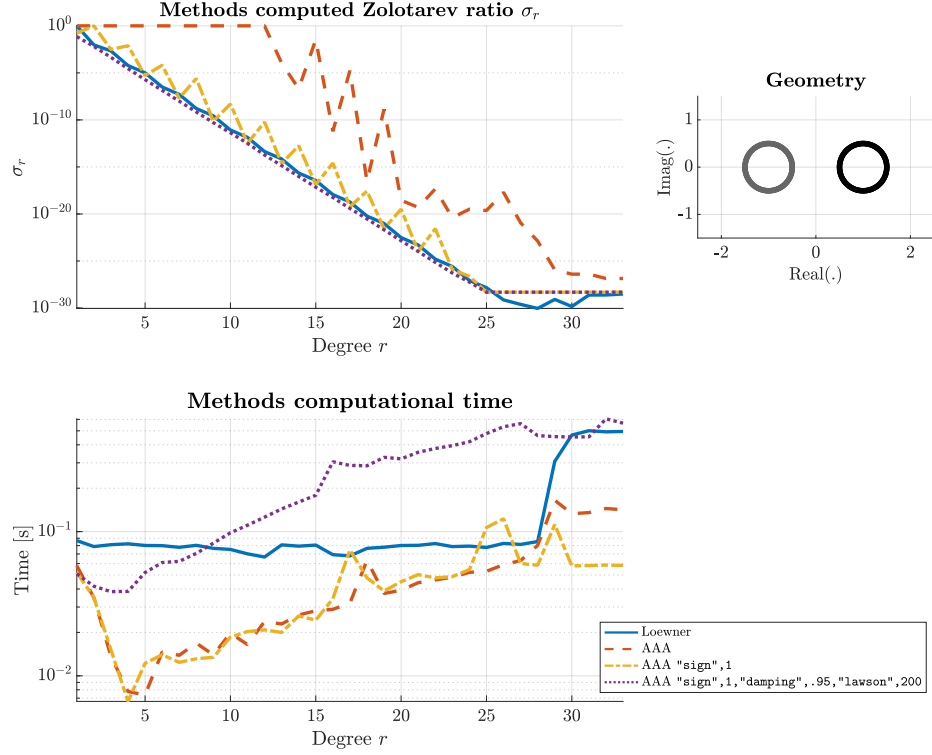


Figure 2: Case '1a': computation time and accuracy comparison for different approximation settings; we compare the standard LF with different AAA and AAA-Lawson settings.

4.6 More experiments

5 Description of the report and overview of the results

5.1 Methodology

5.2 Overview of the results

5.3 Preliminary conclusions

6 Conclusion

In this paper, we have shown that the Loewner framework is a viable candidate for approximating the Z3 and Z4 Zolotarev problems. In particular, we have shown that the LF (i) is fast and accurate, (ii) approximates by compression with controlled complexity and accuracy, without either user intervention or iteration, and (iii) is able to discover the structure of the optimal solution (in symmetric cases). We have also provided a comprehensive comparison study of LF with different versions of the AAA algorithm. In all configurations tested, the standard AAA variant was not able to accurately approximate the Z3 and Z4 problems. This pitfall was corrected thanks to the use of the sign constraint and the Lawson iteration (as in [24]). However, this was achieved at the price of a dedicated data treatment, leading to a higher computational cost. In

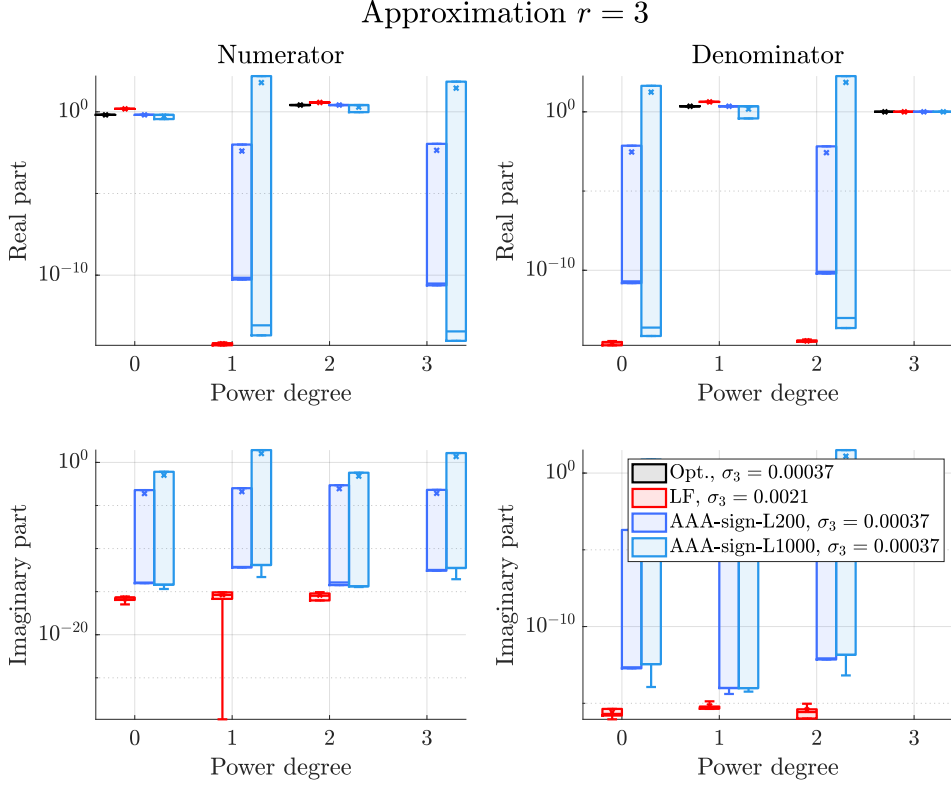


Figure 3: Case 1a, $r = 3$, Z4. Numerator and denominator dispersion over the different platforms.

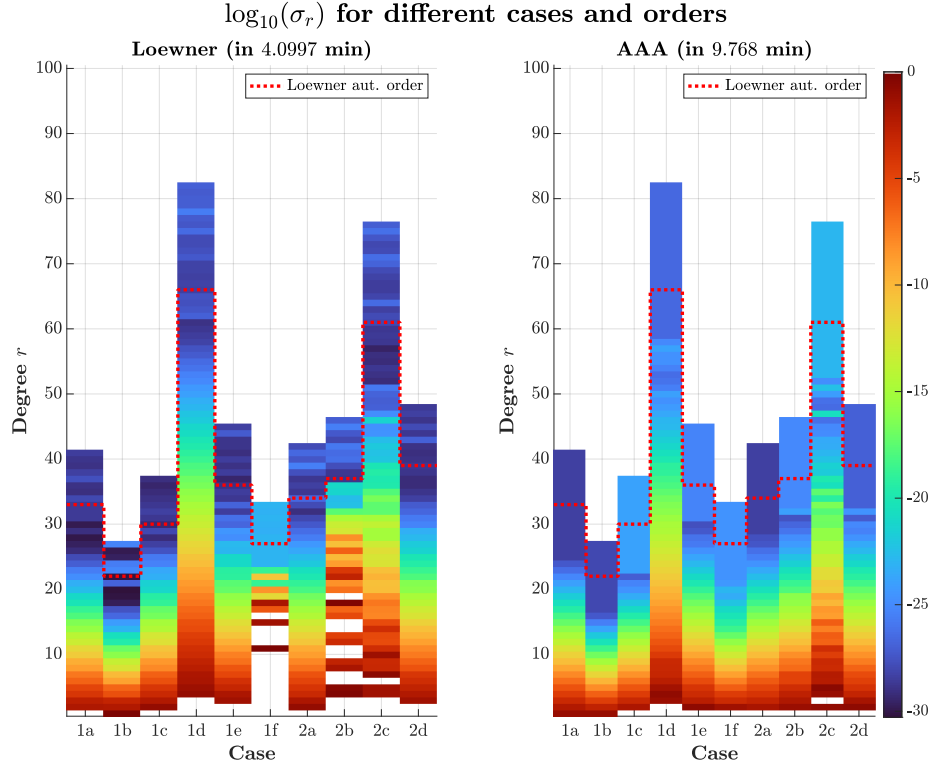
addition, we observed that the Lawson iteration is not monotonically converging, which may result in a worse approximation than for AAA with sign only option (and without any iteration). Finally, we showed that in all configurations, AAA (and variants) do not capture the inherent **symmetries** present in the topology of the configuration. Indeed, this indicates (i) an addition of poles and zeros and/or (ii) polynomial coefficients' imaginary parts, hard to interpret and not in the correct optimal solution space. The code used for generating all results reported here (figures and tables) is made available at the following address:

<https://github.com/cpoussot/zolotarev>

Acknowledgments

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A Detailed results on the 1a case



A.1 Approximation objective $r = 3$

A.1.1 Numerator and denominator coefficients

The numerator and denominator coefficients (both real and complex parts) are given in the following tables. Each table gives the results obtained on different platform.

Basis	Opt., $\sigma_3 = 0.00037$		LF, $\sigma_3 = 0.0021$		AAA-sign-L200, $\sigma_3 = 0.00044$		AAA-sign-L1000, $\sigma_3 = 0.0005$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^3	0	0	0	0	-0.011	$6.5 \cdot 10^{-4}$	-70.0	12.0
z^2	2.6	0	3.7	$3.3 \cdot 10^{-16}$	2.6	$-2.2 \cdot 10^{-3}$	0.95	0.064
z	0	0	$-6.9 \cdot 10^{-15}$	$-4.1 \cdot 10^{-16}$	$9.9 \cdot 10^{-3}$	$1.0 \cdot 10^{-3}$	-150.0	27.0
1.0	0.65	0	1.5	$2.7 \cdot 10^{-16}$	0.65	$-6.1 \cdot 10^{-4}$	-0.36	0.082
z^3	1.0	0	1.0	0	1.0	0	1.0	0
z^2	0	0	$-4.3 \cdot 10^{-15}$	$-9.4 \cdot 10^{-16}$	$-6.6 \cdot 10^{-3}$	$2.1 \cdot 10^{-3}$	-180.0	31.0
z	2.2	0	4.2	$1.3 \cdot 10^{-15}$	2.3	$-1.9 \cdot 10^{-3}$	-0.38	0.084
1.0	0	0	$-2.8 \cdot 10^{-15}$	$-2.0 \cdot 10^{-16}$	$7.3 \cdot 10^{-3}$	$2.0 \cdot 10^{-4}$	-44.0	7.6

Table 4: Case 1a (cpv MACA64 MR2023b), $r = 3$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_3 = 0.00037$		LF, $\sigma_3 = 0.0021$		AAA-sign-L200, $\sigma_3 = 0.00044$		AAA-sign-L1000, $\sigma_3 = 0.0005$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^3	0	0	0	0	-0.011	$6.5 \cdot 10^{-4}$	-70.0	12.0
z^2	2.6	0	3.7	$6.2 \cdot 10^{-16}$	2.6	$-2.2 \cdot 10^{-3}$	0.95	0.064
z	0	0	$-7.6 \cdot 10^{-15}$	$1.4 \cdot 10^{-16}$	$9.9 \cdot 10^{-3}$	$1.0 \cdot 10^{-3}$	-150.0	27.0
1.0	0.65	0	1.5	$3.3 \cdot 10^{-17}$	0.65	$-6.1 \cdot 10^{-4}$	-0.36	0.082
z^3	1.0	0	1.0	0	1.0	0	1.0	0
z^2	0	0	$-3.8 \cdot 10^{-15}$	$2.7 \cdot 10^{-16}$	$-6.6 \cdot 10^{-3}$	$2.1 \cdot 10^{-3}$	-180.0	31.0
z	2.2	0	4.2	$6.2 \cdot 10^{-16}$	2.3	$-1.9 \cdot 10^{-3}$	-0.38	0.084
1.0	0	0	$-3.4 \cdot 10^{-15}$	$-9.1 \cdot 10^{-17}$	$7.3 \cdot 10^{-3}$	$2.0 \cdot 10^{-4}$	-44.0	7.6

Table 5: Case 1a (cpv MACA64 MR2022b), $r = 3$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_3 = 0.00037$		LF, $\sigma_3 = 0.0021$		AAA-sign-L200, $\sigma_3 = 0.00037$		AAA-sign-L1000, $\sigma_3 = 0.00037$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^3	0	0	0	0	$-2.4 \cdot 10^{-11}$	$2.7 \cdot 10^{-13}$	$-9.5 \cdot 10^{-15}$	$5.6 \cdot 10^{-13}$
z^2	2.6	0	3.7	$-9.5 \cdot 10^{-17}$	2.6	$-5.8 \cdot 10^{-15}$	2.6	$-4.2 \cdot 10^{-15}$
z	0	0	$-5.1 \cdot 10^{-15}$	$8.3 \cdot 10^{-16}$	$-5.4 \cdot 10^{-11}$	$6.3 \cdot 10^{-13}$	$-2.1 \cdot 10^{-14}$	$1.2 \cdot 10^{-12}$
1.0	0.65	0	1.5	$-2.1 \cdot 10^{-16}$	0.65	$-9.7 \cdot 10^{-15}$	0.65	$-6.4 \cdot 10^{-15}$
z^3	1.0	0	1.0	0	1.0	0	1.0	0
z^2	0	0	$-3.2 \cdot 10^{-15}$	$1.0 \cdot 10^{-16}$	$-6.2 \cdot 10^{-11}$	$7.1 \cdot 10^{-13}$	$-2.3 \cdot 10^{-14}$	$1.4 \cdot 10^{-12}$
z	2.2	0	4.2	$-4.5 \cdot 10^{-16}$	2.2	$-9.9 \cdot 10^{-15}$	2.2	$-9.8 \cdot 10^{-15}$
1.0	0	0	$-1.8 \cdot 10^{-15}$	$4.3 \cdot 10^{-16}$	$-1.6 \cdot 10^{-11}$	$1.9 \cdot 10^{-13}$	$-7.1 \cdot 10^{-15}$	$3.6 \cdot 10^{-13}$

Table 6: Case 1a (cpv PCWIN64 MR2025b), $r = 3$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_3 = 0.00037$		LF, $\sigma_3 = 0.0021$		AAA-sign-L200, $\sigma_3 = 0.00037$		AAA-sign-L1000, $\sigma_3 = 0.00037$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^3	0	0	0	0	$-2.4 \cdot 10^{-11}$	$2.7 \cdot 10^{-13}$	$-9.5 \cdot 10^{-15}$	$5.6 \cdot 10^{-13}$
z^2	2.6	0	3.7	$-9.5 \cdot 10^{-17}$	2.6	$-5.8 \cdot 10^{-15}$	2.6	$-4.2 \cdot 10^{-15}$
z	0	0	$-5.1 \cdot 10^{-15}$	$8.3 \cdot 10^{-16}$	$-5.4 \cdot 10^{-11}$	$6.3 \cdot 10^{-13}$	$-2.1 \cdot 10^{-14}$	$1.2 \cdot 10^{-12}$
1.0	0.65	0	1.5	$-2.1 \cdot 10^{-16}$	0.65	$-9.7 \cdot 10^{-15}$	0.65	$-6.4 \cdot 10^{-15}$
z^3	1.0	0	1.0	0	1.0	0	1.0	0
z^2	0	0	$-3.2 \cdot 10^{-15}$	$1.0 \cdot 10^{-16}$	$-6.2 \cdot 10^{-11}$	$7.1 \cdot 10^{-13}$	$-2.3 \cdot 10^{-14}$	$1.4 \cdot 10^{-12}$
z	2.2	0	4.2	$-4.5 \cdot 10^{-16}$	2.2	$-9.9 \cdot 10^{-15}$	2.2	$-9.8 \cdot 10^{-15}$
1.0	0	0	$-1.8 \cdot 10^{-15}$	$4.3 \cdot 10^{-16}$	$-1.6 \cdot 10^{-11}$	$1.9 \cdot 10^{-13}$	$-7.1 \cdot 10^{-15}$	$3.6 \cdot 10^{-13}$

Table 7: Case 1a (AlexReis PCWIN64 MR2025b), $r = 3$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_3 = 0.00037$		LF, $\sigma_3 = 0.0021$		AAA-sign-L200, $\sigma_3 = 0.00037$		AAA-sign-L1000, $\sigma_3 = 0.00037$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^3	0	0	0	0	$-3.1 \cdot 10^{-11}$	$-3.3 \cdot 10^{-13}$	$-3.6 \cdot 10^{-14}$	$-2.8 \cdot 10^{-14}$
z^2	2.6	0	3.7	$-8.7 \cdot 10^{-16}$	2.6	$-1.2 \cdot 10^{-14}$	2.6	$3.4 \cdot 10^{-15}$
z	0	0	$-6.5 \cdot 10^{-15}$	$1.5 \cdot 10^{-30}$	$-6.9 \cdot 10^{-11}$	$-7.4 \cdot 10^{-13}$	$-8.5 \cdot 10^{-14}$	$-5.0 \cdot 10^{-14}$
1.0	0.65	0	1.5	$1.1 \cdot 10^{-16}$	0.65	$1.1 \cdot 10^{-14}$	0.65	$2.0 \cdot 10^{-15}$
z^3	1.0	0	1.0	0	1.0	0	1.0	0
z^2	0	0	$-3.0 \cdot 10^{-15}$	$-4.1 \cdot 10^{-16}$	$-8.0 \cdot 10^{-11}$	$-8.4 \cdot 10^{-13}$	$-9.8 \cdot 10^{-14}$	$-6.6 \cdot 10^{-14}$
z	2.2	0	4.2	$-4.3 \cdot 10^{-16}$	2.2	$4.1 \cdot 10^{-15}$	2.2	$5.8 \cdot 10^{-15}$
1.0	0	0	$-3.0 \cdot 10^{-15}$	$1.6 \cdot 10^{-16}$	$-2.0 \cdot 10^{-11}$	$-2.2 \cdot 10^{-13}$	$-2.4 \cdot 10^{-14}$	$-1.2 \cdot 10^{-14}$

Table 8: Case 1a (AlexReis PCWIN64perso MR2017a), $r = 3$, Z4: numerator (first lines) and denominator (last lines) coefficients.

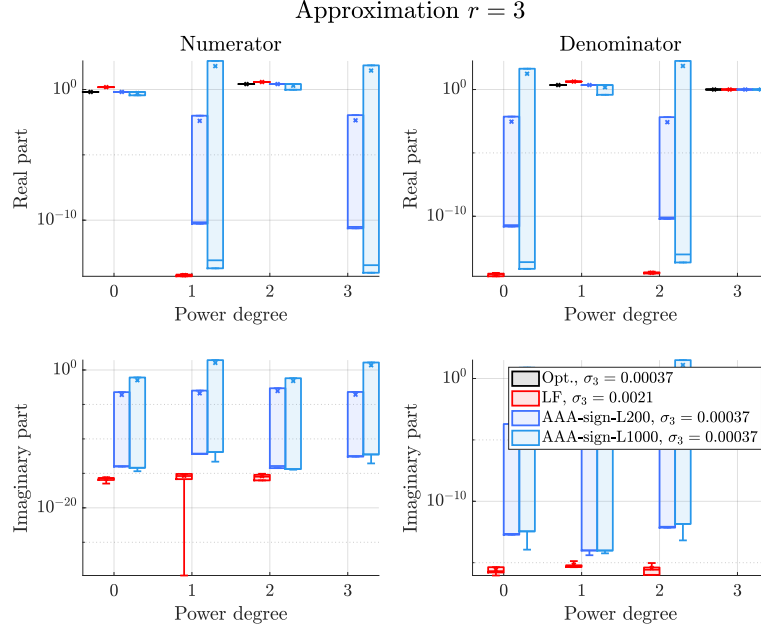


Figure 5: Case 1a, $r = 3$, Z4. Numerator and denominator dispersion over the different platforms.

A.1.2 Poles

The poles are given in the following tables. Each table gives the results obtained on different platform.

	Opt., $\sigma_3 = 0.00037$	LF, $\sigma_3 = 0.0021$	AAA-sign-L200, $\sigma_3 = 0.00044$	AAA-sign-L1000, $\sigma_3 = 0.0005$
1.0	0	$6.58 \cdot 10^{-16} + 4.81 \cdot 10^{-17}i$	$-0.00323 - 9.0 \cdot 10^{-5}i$	$-0.00181 + 0.493i$
2.0	$1.5i$	$2.14 \cdot 10^{-15} - 2.06i$	$0.00557 + 1.5i$	$-0.00165 - 0.494i$
3.0	$-1.5i$	$1.49 \cdot 10^{-15} + 2.06i$	$0.00428 - 1.5i$	$180.0 - 31.3i$

Table 9: Case 1a (cpv MACA64 MR2023b), $r = 3$, Z4: poles.

	Opt., $\sigma_3 = 0.00037$	LF, $\sigma_3 = 0.0021$	AAA-sign-L200, $\sigma_3 = 0.00044$	AAA-sign-L1000, $\sigma_3 = 0.0005$
1.0	0	$8.09 \cdot 10^{-16} + 2.14 \cdot 10^{-17}i$	$-0.00323 - 9.0 \cdot 10^{-5}i$	$-0.00181 + 0.493i$
2.0	$1.5i$	$1.75 \cdot 10^{-15} - 2.06i$	$0.00557 + 1.5i$	$-0.00165 - 0.494i$
3.0	$-1.5i$	$1.29 \cdot 10^{-15} + 2.06i$	$0.00428 - 1.5i$	$180.0 - 31.3i$

Table 10: Case 1a (cpv MACA64 MR2022b), $r = 3$, Z4: poles.

	Opt., $\sigma_3 = 0.00037$	LF, $\sigma_3 = 0.0021$	AAA-sign-L200, $\sigma_3 = 0.00037$	AAA-sign-L1000, $\sigma_3 = 0.00037$
1.0	0	$4.35 \cdot 10^{-16} - 1.02 \cdot 10^{-16}i$	$6.9 \cdot 10^{-12} - 8.47 \cdot 10^{-14}i$	$3.42 \cdot 10^{-15} - 1.59 \cdot 10^{-13}i$
2.0	$1.5i$	$1.25 \cdot 10^{-15} - 2.06i$	$2.76 \cdot 10^{-11} + 1.5i$	$1.22 \cdot 10^{-14} + 1.5i$
3.0	$-1.5i$	$1.47 \cdot 10^{-15} + 2.06i$	$2.76 \cdot 10^{-11} - 1.5i$	$7.73 \cdot 10^{-15} - 1.5i$

Table 11: Case 1a (cpv PCWIN64 MR2025b), $r = 3$, Z4: poles.

	Opt., $\sigma_3 = 0.00037$	LF, $\sigma_3 = 0.0021$	AAA-sign-L200, $\sigma_3 = 0.00037$	AAA-sign-L1000, $\sigma_3 = 0.00037$
1.0	0	$4.35 \cdot 10^{-16} - 1.02 \cdot 10^{-16}i$	$6.9 \cdot 10^{-12} - 8.47 \cdot 10^{-14}i$	$3.42 \cdot 10^{-15} - 1.59 \cdot 10^{-13}i$
2.0	$1.5i$	$1.25 \cdot 10^{-15} - 2.06i$	$2.76 \cdot 10^{-11} + 1.5i$	$1.22 \cdot 10^{-14} + 1.5i$
3.0	$-1.5i$	$1.47 \cdot 10^{-15} + 2.06i$	$2.76 \cdot 10^{-11} - 1.5i$	$7.73 \cdot 10^{-15} - 1.5i$

Table 12: Case 1a (AlexReis PCWIN64 MR2025b), $r = 3$, Z4: poles.

	Opt., $\sigma_3 = 0.00037$	LF, $\sigma_3 = 0.0021$	AAA-sign-L200, $\sigma_3 = 0.00037$	AAA-sign-L1000, $\sigma_3 = 0.00037$
1.0	0	$7.04 \cdot 10^{-16} - 3.76 \cdot 10^{-17}i$	$8.87 \cdot 10^{-12} + 9.81 \cdot 10^{-14}i$	$9.48 \cdot 10^{-15} + 5.12 \cdot 10^{-15}i$
2.0	$1.5i$	$1.05 \cdot 10^{-15} - 2.06i$	$3.54 \cdot 10^{-11} - 1.5i$	$4.65 \cdot 10^{-14} - 1.5i$
3.0	$-1.5i$	$1.26 \cdot 10^{-15} + 2.06i$	$3.54 \cdot 10^{-11} + 1.5i$	$4.04 \cdot 10^{-14} + 1.5i$

Table 13: Case 1a (AlexReis PCWIN64perso MR2017a), $r = 3$, Z4: poles.

A.1.3 Zeros

The zeros are given in the following tables. Each table gives the results obtained on different platform.

	Opt., $\sigma_3 = 0.00037$	LF, $\sigma_3 = 0.0021$	AAA-sign-L200, $\sigma_3 = 0.00044$	AAA-sign-L1000, $\sigma_3 = 0.0005$
1.0	NaN	NaN	$-0.00241 + 0.5i$	$-0.00236 + 1.2 \cdot 10^{-4}i$
2.0	$0.5i$	$9.48 \cdot 10^{-16} - 0.637i$	$-0.00245 - 0.501i$	$0.00667 - 1.48i$
3.0	$-0.5i$	$8.89 \cdot 10^{-16} + 0.637i$	$237.0 + 13.9i$	$0.00871 + 1.48i$

Table 14: Case 1a (cpv MACA64 MR2023b), $r = 3$, Z4: zeros.

	Opt., $\sigma_3 = 0.00037$	LF, $\sigma_3 = 0.0021$	AAA-sign-L200, $\sigma_3 = 0.00044$	AAA-sign-L1000, $\sigma_3 = 0.0005$
1.0	NaN	NaN	$-0.00241 + 0.5i$	$-0.00236 + 1.2 \cdot 10^{-4}i$
2.0	$0.5i$	$9.48 \cdot 10^{-16} - 0.637i$	$-0.00245 - 0.501i$	$0.00667 - 1.48i$
3.0	$-0.5i$	$1.1 \cdot 10^{-15} + 0.637i$	$237.0 + 13.9i$	$0.00871 + 1.48i$

Table 15: Case 1a (cpv MACA64 MR2022b), $r = 3$, Z4: zeros.

	Opt., $\sigma_3 = 0.00037$	LF, $\sigma_3 = 0.0021$	AAA-sign-L200, $\sigma_3 = 0.00037$	AAA-sign-L1000, $\sigma_3 = 0.00037$
1.0	NaN	NaN	$9.21 \cdot 10^{-12} + 0.5i$	$4.98 \cdot 10^{-15} + 0.5i$
2.0	$0.5i$	$7.16 \cdot 10^{-16} + 0.637i$	$9.2 \cdot 10^{-12} - 0.5i$	$2.0 \cdot 10^{-15} - 0.5i$
3.0	$-0.5i$	$6.45 \cdot 10^{-16} - 0.637i$	$1.09 \cdot 10^{11} + 1.22 \cdot 10^9i$	$7.82 \cdot 10^{10} + 4.66 \cdot 10^{12}i$

Table 16: Case 1a (cpv PCWIN64 MR2025b), $r = 3$, Z4: zeros.

	Opt., $\sigma_3 = 0.00037$	LF, $\sigma_3 = 0.0021$	AAA-sign-L200, $\sigma_3 = 0.00037$	AAA-sign-L1000, $\sigma_3 = 0.00037$
1.0	NaN	NaN	$9.21 \cdot 10^{-12} + 0.5i$	$4.98 \cdot 10^{-15} + 0.5i$
2.0	$0.5i$	$7.16 \cdot 10^{-16} + 0.637i$	$9.2 \cdot 10^{-12} - 0.5i$	$2.0 \cdot 10^{-15} - 0.5i$
3.0	$-0.5i$	$6.45 \cdot 10^{-16} - 0.637i$	$1.09 \cdot 10^{11} + 1.22 \cdot 10^9i$	$7.82 \cdot 10^{10} + 4.66 \cdot 10^{12}i$

Table 17: Case 1a (AlexReis PCWIN64 MR2025b), $r = 3$, Z4: zeros.

	Opt., $\sigma_3 = 0.00037$	LF, $\sigma_3 = 0.0021$	AAA-sign-L200, $\sigma_3 = 0.00037$	AAA-sign-L1000, $\sigma_3 = 0.00037$
1.0	NaN	NaN	$1.18 \cdot 10^{-11} - 0.5i$	$1.46 \cdot 10^{-14} - 0.5i$
2.0	$0.5i$	$7.77 \cdot 10^{-16} + 0.637i$	$1.18 \cdot 10^{-11} + 0.5i$	$1.39 \cdot 10^{-14} + 0.5i$
3.0	$-0.5i$	$9.71 \cdot 10^{-16} - 0.637i$	$8.46 \cdot 10^{10} - 9.11 \cdot 10^8i$	$4.45 \cdot 10^{13} - 3.54 \cdot 10^{13}i$

Table 18: Case 1a (AlexReis PCWIN64perso MR2017a), $r = 3$, Z4: zeros.

A.1.4 Poles and zeros visualisation (case cpv MACA64 MR2023b)

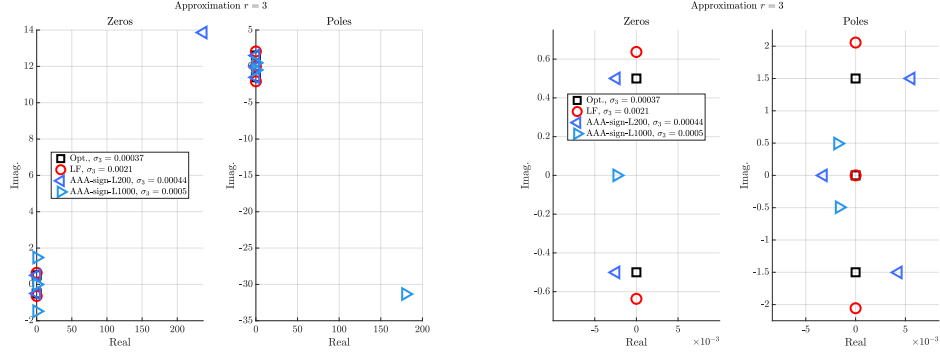


Figure 6: Case 1a (cpv MACA64 MR2023b), $r = 3$, Z4. Left frame: zeros (left) and poles (right). Right frame: same but discarding the largest real entry.

A.1.5 Approximation visualization

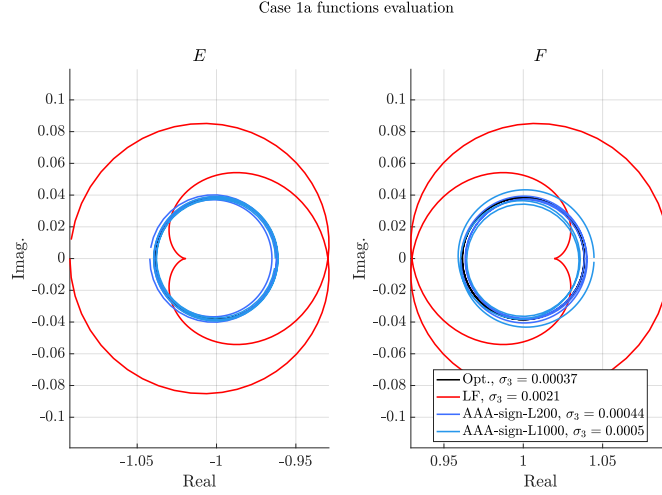


Figure 7: Case 1a (cpv MACA64 MR2023b), $r = 3$, Z4. E and F evaluations.

A.2 Approximation objective $r = 4$

A.2.1 Numerator and denominator coefficients

The numerator and denominator coefficients (both real and complex parts) are given in the following tables. Each table gives the results obtained on different platform.

Basis	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$		LF, $\sigma_4 = 6 \cdot 10^{-05}$		AAA-sign-L200, $\sigma_4 = 2.8 \cdot 10^{-05}$		AAA-sign-L1000, $\sigma_4 = 2.8 \cdot 10^{-05}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^4	0	0	0	0	-140.0	16.0	-150.0	7.3
z^3	3.5	0	3.7	$3.3 \cdot 10^{-16}$	4.7	-0.29	4.8	-0.22
z^2	0	0	$-4.6 \cdot 10^{-15}$	$3.3 \cdot 10^{-15}$	-650.0	70.0	-660.0	33.0
z	2.6	0	3.2	$5.6 \cdot 10^{-16}$	5.5	-0.52	5.6	-0.34
1.0	0	0	$-1.3 \cdot 10^{-15}$	$-1.9 \cdot 10^{-15}$	-80.0	8.8	-82.0	4.1
z^4	1.0	0	1.0	0	1.0	0	1.0	0
z^3	0	0	$-2.7 \cdot 10^{-15}$	$-1.3 \cdot 10^{-15}$	-500.0	54.0	-510.0	25.0
z^2	4.5	0	5.2	$4.8 \cdot 10^{-16}$	7.9	-0.7	7.9	-0.49
z	0	0	$-3.5 \cdot 10^{-15}$	$-2.2 \cdot 10^{-15}$	-370.0	41.0	-380.0	19.0
1.0	0.56	0	0.75	$9.5 \cdot 10^{-17}$	1.4	-0.13	1.4	-0.078

Table 19: Case 1a (cpv MACA64 MR2023b), $r = 4$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$		LF, $\sigma_4 = 6 \cdot 10^{-05}$		AAA-sign-L200, $\sigma_4 = 2.8 \cdot 10^{-05}$		AAA-sign-L1000, $\sigma_4 = 2.8 \cdot 10^{-05}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^4	0	0	0	0	-140.0	16.0	-150.0	7.3
z^3	3.5	0	3.7	$6.4 \cdot 10^{-16}$	4.7	-0.29	4.8	-0.22
z^2	0	0	$-4.8 \cdot 10^{-15}$	$3.7 \cdot 10^{-16}$	-650.0	70.0	-660.0	33.0
z	2.6	0	3.2	$1.8 \cdot 10^{-15}$	5.5	-0.52	5.6	-0.34
1.0	0	0	$-1.0 \cdot 10^{-15}$	$2.6 \cdot 10^{-15}$	-80.0	8.8	-82.0	4.1
z^4	1.0	0	1.0	0	1.0	0	1.0	0
z^3	0	0	$-9.8 \cdot 10^{-16}$	$8.7 \cdot 10^{-16}$	-500.0	54.0	-510.0	25.0
z^2	4.5	0	5.2	$1.9 \cdot 10^{-15}$	7.9	-0.7	7.9	-0.49
z	0	0	$-4.0 \cdot 10^{-15}$	$2.8 \cdot 10^{-15}$	-370.0	41.0	-380.0	19.0
1.0	0.56	0	0.75	$5.9 \cdot 10^{-16}$	1.4	-0.13	1.4	-0.078

Table 20: Case 1a (cpv MACA64 MR2022b), $r = 4$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$		LF, $\sigma_4 = 6 \cdot 10^{-05}$		AAA-sign-L200, $\sigma_4 = 3.3 \cdot 10^{-05}$		AAA-sign-L1000, $\sigma_4 = 3.4 \cdot 10^{-05}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^4	0	0	0	0	$-5.4 \cdot 10^{-3}$	$-3.1 \cdot 10^{-3}$	$-4.8 \cdot 10^{-3}$	$-2.8 \cdot 10^{-3}$
z^3	3.5	0	3.7	$1.4 \cdot 10^{-16}$	3.5	$-3.7 \cdot 10^{-3}$	3.5	$-3.8 \cdot 10^{-3}$
z^2	0	0	$-2.8 \cdot 10^{-15}$	$-2.9 \cdot 10^{-15}$	-0.07	$-2.6 \cdot 10^{-3}$	-0.078	$-2.5 \cdot 10^{-3}$
z	2.6	0	3.2	$-4.7 \cdot 10^{-16}$	2.6	-0.011	2.6	-0.011
1.0	0	0	$-1.8 \cdot 10^{-15}$	$-7.3 \cdot 10^{-16}$	-0.015	$1.3 \cdot 10^{-3}$	-0.017	$1.4 \cdot 10^{-3}$
z^4	1.0	0	1.0	0	1.0	0	1.0	0
z^3	0	0	$-4.6 \cdot 10^{-16}$	$-3.3 \cdot 10^{-16}$	-0.036	$-6.9 \cdot 10^{-3}$	-0.038	$-6.5 \cdot 10^{-3}$
z^2	4.5	0	5.2	$2.4 \cdot 10^{-16}$	4.5	-0.011	4.5	-0.011
z	0	0	$-4.2 \cdot 10^{-15}$	$4.4 \cdot 10^{-16}$	-0.054	$2.2 \cdot 10^{-3}$	-0.062	$2.5 \cdot 10^{-3}$
1.0	0.56	0	0.75	$-5.3 \cdot 10^{-16}$	0.57	$-3.3 \cdot 10^{-3}$	0.57	$-3.3 \cdot 10^{-3}$

Table 21: Case 1a (cpv PCWIN64 MR2025b), $r = 4$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$		LF, $\sigma_4 = 6 \cdot 10^{-05}$		AAA-sign-L200, $\sigma_4 = 3.3 \cdot 10^{-05}$		AAA-sign-L1000, $\sigma_4 = 3.4 \cdot 10^{-05}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^4	0	0	0	0	$-5.4 \cdot 10^{-3}$	$-3.1 \cdot 10^{-3}$	$-4.8 \cdot 10^{-3}$	$-2.8 \cdot 10^{-3}$
z^3	3.5	0	3.7	$1.4 \cdot 10^{-16}$	3.5	$-3.7 \cdot 10^{-3}$	3.5	$-3.8 \cdot 10^{-3}$
z^2	0	0	$-2.8 \cdot 10^{-15}$	$-2.9 \cdot 10^{-15}$	-0.07	$-2.6 \cdot 10^{-3}$	-0.078	$-2.5 \cdot 10^{-3}$
z	2.6	0	3.2	$-4.7 \cdot 10^{-16}$	2.6	-0.011	2.6	-0.011
1.0	0	0	$-1.8 \cdot 10^{-15}$	$-7.3 \cdot 10^{-16}$	-0.015	$1.3 \cdot 10^{-3}$	-0.017	$1.4 \cdot 10^{-3}$
z^4	1.0	0	1.0	0	1.0	0	1.0	0
z^3	0	0	$-4.6 \cdot 10^{-16}$	$-3.3 \cdot 10^{-16}$	-0.036	$-6.9 \cdot 10^{-3}$	-0.038	$-6.5 \cdot 10^{-3}$
z^2	4.5	0	5.2	$2.4 \cdot 10^{-16}$	4.5	-0.011	4.5	-0.011
z	0	0	$-4.2 \cdot 10^{-15}$	$4.4 \cdot 10^{-16}$	-0.054	$2.2 \cdot 10^{-3}$	-0.062	$2.5 \cdot 10^{-3}$
1.0	0.56	0	0.75	$-5.3 \cdot 10^{-16}$	0.57	$-3.3 \cdot 10^{-3}$	0.57	$-3.3 \cdot 10^{-3}$

Table 22: Case 1a (AlexReis PCWIN64 MR2025b), $r = 4$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$		LF, $\sigma_4 = 6 \cdot 10^{-05}$		AAA-sign-L200, $\sigma_4 = 2.7 \cdot 10^{-05}$		AAA-sign-L1000, $\sigma_4 = 2.7 \cdot 10^{-05}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^4	0	0	0	0	$-4.6 \cdot 10^8$	$-2.6 \cdot 10^6$	$-3.6 \cdot 10^9$	$1.9 \cdot 10^{11}$
z^3	3.5	0	3.7	$-8.3 \cdot 10^{-16}$	3.5	$8.7 \cdot 10^{-5}$	3.5	$8.4 \cdot 10^{-4}$
z^2	0	0	$-5.0 \cdot 10^{-15}$	$-2.1 \cdot 10^{-15}$	$-2.1 \cdot 10^9$	$-1.1 \cdot 10^7$	$-1.6 \cdot 10^{10}$	$8.6 \cdot 10^{11}$
z	2.6	0	3.2	$7.3 \cdot 10^{-16}$	2.6	$2.1 \cdot 10^{-4}$	2.6	$2.6 \cdot 10^{-3}$
1.0	0	0	$-7.8 \cdot 10^{-16}$	$1.0 \cdot 10^{-16}$	$-2.6 \cdot 10^8$	$-1.4 \cdot 10^6$	$-2.0 \cdot 10^9$	$1.1 \cdot 10^{11}$
z^4	1.0	0	1.0	0	1.0	0	1.0	0
z^3	0	0	$-1.7 \cdot 10^{-15}$	$-1.5 \cdot 10^{-15}$	$-1.6 \cdot 10^9$	$-8.8 \cdot 10^6$	$-1.3 \cdot 10^{10}$	$6.6 \cdot 10^{11}$
z^2	4.5	0	5.2	$-9.5 \cdot 10^{-16}$	4.5	$2.5 \cdot 10^{-4}$	4.5	$2.9 \cdot 10^{-3}$
z	0	0	$-4.3 \cdot 10^{-15}$	$-1.3 \cdot 10^{-15}$	$-1.2 \cdot 10^9$	$-6.6 \cdot 10^6$	$-9.4 \cdot 10^9$	$5.0 \cdot 10^{11}$
1.0	0.56	0	0.75	$4.5 \cdot 10^{-16}$	0.56	$6.3 \cdot 10^{-5}$	0.57	$1.2 \cdot 10^{-3}$

Table 23: Case 1a (AlexReis PCWIN64perso MR2017a), $r = 4$, Z4: numerator (first lines) and denominator (last lines) coefficients.

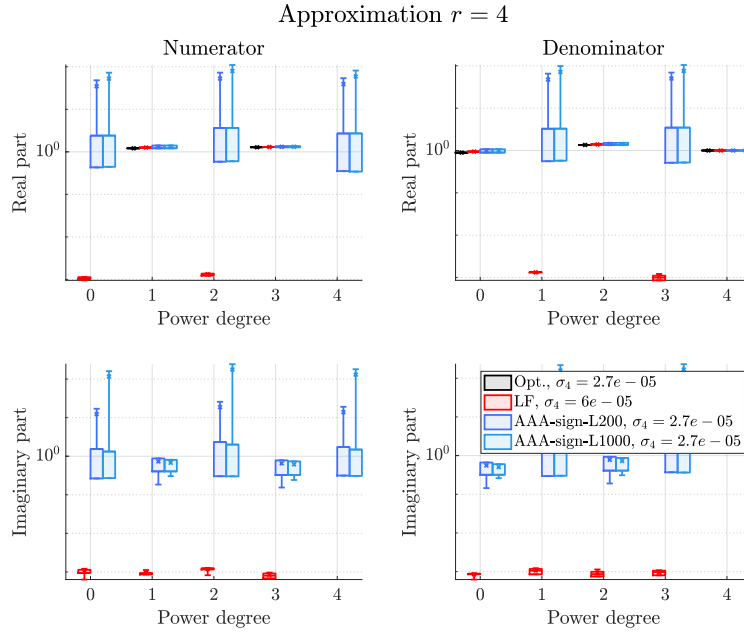


Figure 8: Case 1a, $r = 4$, Z4. Numerator and denominator dispersion over the different platforms.

A.2.2 Poles

The poles are given in the following tables. Each table gives the results obtained on different platform.

	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$	LF, $\sigma_4 = 6 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_4 = 2.8 \cdot 10^{-05}$	AAA-sign-L1000, $\sigma_4 = 2.8 \cdot 10^{-05}$
1.0	$-1.94 \cdot 10^{-16} + 2.09i$	$3.21 \cdot 10^{-16} - 0.384i$	$0.00377 + 6.65 \cdot 10^{-5}i$	$0.00375 - 2.07 \cdot 10^{-5}i$
2.0	$-1.94 \cdot 10^{-16} - 2.09i$	$3.09 \cdot 10^{-16} + 0.384i$	$0.00535 - 0.864i$	$0.00513 + 0.864i$
3.0	$-1.39 \cdot 10^{-17} + 0.359i$	$1.12 \cdot 10^{-15} - 2.25i$	$0.00516 + 0.864i$	$0.00532 - 0.864i$
4.0	$-1.39 \cdot 10^{-17} - 0.359i$	$9.09 \cdot 10^{-16} + 2.25i$	$499.0 - 54.4i$	$507.0 - 25.2i$

Table 24: Case 1a (cpv MACA64 MR2023b), $r = 4$, Z4: poles.

	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$	LF, $\sigma_4 = 6 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_4 = 2.8 \cdot 10^{-05}$	AAA-sign-L1000, $\sigma_4 = 2.8 \cdot 10^{-05}$
1.0	$-1.94 \cdot 10^{-16} + 2.09i$	$3.19 \cdot 10^{-16} + 0.384i$	$0.00377 + 6.65 \cdot 10^{-5}i$	$0.00375 - 2.07 \cdot 10^{-5}i$
2.0	$-1.94 \cdot 10^{-16} - 2.09i$	$4.66 \cdot 10^{-16} - 0.384i$	$0.00535 - 0.864i$	$0.00513 + 0.864i$
3.0	$-1.39 \cdot 10^{-17} + 0.359i$	$4.91 \cdot 10^{-16} - 2.25i$	$0.00516 + 0.864i$	$0.00532 - 0.864i$
4.0	$-1.39 \cdot 10^{-17} - 0.359i$	$-3.02 \cdot 10^{-16} + 2.25i$	$499.0 - 54.4i$	$507.0 - 25.2i$

Table 25: Case 1a (cpv MACA64 MR2022b), $r = 4$, Z4: poles.

	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$	LF, $\sigma_4 = 6 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_4 = 3.3 \cdot 10^{-05}$	AAA-sign-L1000, $\sigma_4 = 3.4 \cdot 10^{-05}$
1.0	$-1.94 \cdot 10^{-16} + 2.09i$	$5.75 \cdot 10^{-16} + 0.384i$	$0.00636 + 0.36i$	$0.00731 + 0.36i$
2.0	$-1.94 \cdot 10^{-16} - 2.09i$	$2.75 \cdot 10^{-16} - 0.384i$	$0.0052 - 0.361i$	$0.00612 - 0.361i$
3.0	$-1.39 \cdot 10^{-17} + 0.359i$	$-1.15 \cdot 10^{-16} - 2.25i$	$0.00935 - 2.1i$	$0.00961 - 2.09i$
4.0	$-1.39 \cdot 10^{-17} - 0.359i$	$-2.71 \cdot 10^{-16} + 2.25i$	$0.0147 + 2.1i$	$0.0149 + 2.1i$

Table 26: Case 1a (cpv PCWIN64 MR2025b), $r = 4$, Z4: poles.

	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$	LF, $\sigma_4 = 6 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_4 = 3.3 \cdot 10^{-05}$	AAA-sign-L1000, $\sigma_4 = 3.4 \cdot 10^{-05}$
1.0	$-1.94 \cdot 10^{-16} + 2.09i$	$5.75 \cdot 10^{-16} + 0.384i$	$0.00636 + 0.36i$	$0.00731 + 0.36i$
2.0	$-1.94 \cdot 10^{-16} - 2.09i$	$2.75 \cdot 10^{-16} - 0.384i$	$0.0052 - 0.361i$	$0.00612 - 0.361i$
3.0	$-1.39 \cdot 10^{-17} + 0.359i$	$-1.15 \cdot 10^{-16} - 2.25i$	$0.00935 - 2.1i$	$0.00961 - 2.09i$
4.0	$-1.39 \cdot 10^{-17} - 0.359i$	$-2.71 \cdot 10^{-16} + 2.25i$	$0.0147 + 2.1i$	$0.0149 + 2.1i$

Table 27: Case 1a (AlexReis PCWIN64 MR2025b), $r = 4$, Z4: poles.

	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$	LF, $\sigma_4 = 6 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_4 = 2.7 \cdot 10^{-05}$	AAA-sign-L1000, $\sigma_4 = 2.7 \cdot 10^{-05}$
1.0	$-1.94 \cdot 10^{-16} + 2.09i$	$5.68 \cdot 10^{-16} - 0.384i$	$4.73 \cdot 10^{-10} - 2.59 \cdot 10^{-12}i$	$1.93 \cdot 10^{-14} + 1.16 \cdot 10^{-12}i$
2.0	$-1.94 \cdot 10^{-16} - 2.09i$	$2.52 \cdot 10^{-16} + 0.384i$	$9.46 \cdot 10^{-10} + 0.866i$	$3.13 \cdot 10^{-14} - 0.866i$
3.0	$-1.39 \cdot 10^{-17} + 0.359i$	$1.77 \cdot 10^{-16} - 2.25i$	$9.46 \cdot 10^{-10} - 0.866i$	$5.34 \cdot 10^{-14} + 0.866i$
4.0	$-1.39 \cdot 10^{-17} - 0.359i$	$6.55 \cdot 10^{-16} + 2.25i$	$1.58 \cdot 10^9 + 8.84 \cdot 10^6i$	$1.26 \cdot 10^{10} - 6.6 \cdot 10^{11}i$

Table 28: Case 1a (AlexReis PCWIN64perso MR2017a), $r = 4$, Z4: poles.

A.2.3 Zeros

The zeros are given in the following tables. Each table gives the results obtained on different platform.

	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$	LF, $\sigma_4 = 6 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_4 = 2.8 \cdot 10^{-05}$	AAA-sign-L1000, $\sigma_4 = 2.8 \cdot 10^{-05}$
1.0	NaN	NaN	0.00403 - 0.358 _z	0.00404 + 0.358 _z
2.0	0	$4.1 \cdot 10^{-16} + 5.92 \cdot 10^{-16} z$	0.00406 + 0.358 _z	0.00402 - 0.358 _z
3.0	0.866 _z	$3.69 \cdot 10^{-16} + 0.929 z$	0.0128 - 2.08 _z	0.0127 - 2.09 _z
4.0	-0.866 _z	$4.46 \cdot 10^{-16} - 0.929 z$	0.0117 + 2.09 _z	0.0116 + 2.09 _z

Table 29: Case 1a (cpv MACA64 MR2023b), $r = 4$, Z4: zeros.

	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$	LF, $\sigma_4 = 6 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_4 = 2.8 \cdot 10^{-05}$	AAA-sign-L1000, $\sigma_4 = 2.8 \cdot 10^{-05}$
1.0	NaN	NaN	0.00403 - 0.358 _z	0.00404 + 0.358 _z
2.0	0	$3.12 \cdot 10^{-16} - 7.29 \cdot 10^{-16} z$	0.00406 + 0.358 _z	0.00402 - 0.358 _z
3.0	0.866 _z	$6.57 \cdot 10^{-16} - 0.929 z$	0.0128 - 2.08 _z	0.0127 - 2.09 _z
4.0	-0.866 _z	$3.16 \cdot 10^{-16} + 0.929 z$	0.0117 + 2.09 _z	0.0116 + 2.09 _z

Table 30: Case 1a (cpv MACA64 MR2022b), $r = 4$, Z4: zeros.

	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$	LF, $\sigma_4 = 6 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_4 = 3.3 \cdot 10^{-05}$	AAA-sign-L1000, $\sigma_4 = 3.4 \cdot 10^{-05}$
1.0	NaN	NaN	$0.00562 - 4.73 \cdot 10^{-4} z$	$0.00656 - 5.2 \cdot 10^{-4} z$
2.0	0	$5.68 \cdot 10^{-16} + 2.27 \cdot 10^{-16} z$	0.00528 - 0.87 _z	0.00611 - 0.869 _z
3.0	0.866 _z	$3.38 \cdot 10^{-18} - 0.929 z$	0.00792 + 0.871 _z	0.00875 + 0.87 _z
4.0	-0.866 _z	$1.73 \cdot 10^{-16} + 0.929 z$	486.0 - 282.0 _z	539.0 - 319.0 _z

Table 31: Case 1a (cpv PCWIN64 MR2025b), $r = 4$, Z4: zeros.

	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$	LF, $\sigma_4 = 6 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_4 = 3.3 \cdot 10^{-05}$	AAA-sign-L1000, $\sigma_4 = 3.4 \cdot 10^{-05}$
1.0	NaN	NaN	$0.00562 - 4.73 \cdot 10^{-4} z$	$0.00656 - 5.2 \cdot 10^{-4} z$
2.0	0	$5.68 \cdot 10^{-16} + 2.27 \cdot 10^{-16} z$	0.00528 - 0.87 _z	0.00611 - 0.869 _z
3.0	0.866 _z	$3.38 \cdot 10^{-18} - 0.929 z$	0.00792 + 0.871 _z	0.00875 + 0.87 _z
4.0	-0.866 _z	$1.73 \cdot 10^{-16} + 0.929 z$	486.0 - 282.0 _z	539.0 - 319.0 _z

Table 32: Case 1a (AlexReis PCWIN64 MR2025b), $r = 4$, Z4: zeros.

	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$	LF, $\sigma_4 = 6 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_4 = 2.7 \cdot 10^{-05}$	AAA-sign-L1000, $\sigma_4 = 2.7 \cdot 10^{-05}$
1.0	NaN	NaN	$5.54 \cdot 10^{-10} + 0.359 z$	$2.3 \cdot 10^{-14} - 0.359 z$
2.0	0	$2.43 \cdot 10^{-16} - 3.21 \cdot 10^{-17} z$	$5.54 \cdot 10^{-10} - 0.359 z$	$2.5 \cdot 10^{-14} + 0.359 z$
3.0	0.866 _z	$7.52 \cdot 10^{-16} - 0.929 z$	$3.23 \cdot 10^{-9} + 2.09 z$	$1.03 \cdot 10^{-13} - 2.09 z$
4.0	-0.866 _z	$3.33 \cdot 10^{-16} + 0.929 z$	$3.23 \cdot 10^{-9} - 2.09 z$	$1.92 \cdot 10^{-13} + 2.09 z$

Table 33: Case 1a (AlexReis PCWIN64perso MR2017a), $r = 4$, Z4: zeros.

A.2.4 Poles and zeros visualisation (case cpv MACA64 MR2023b)

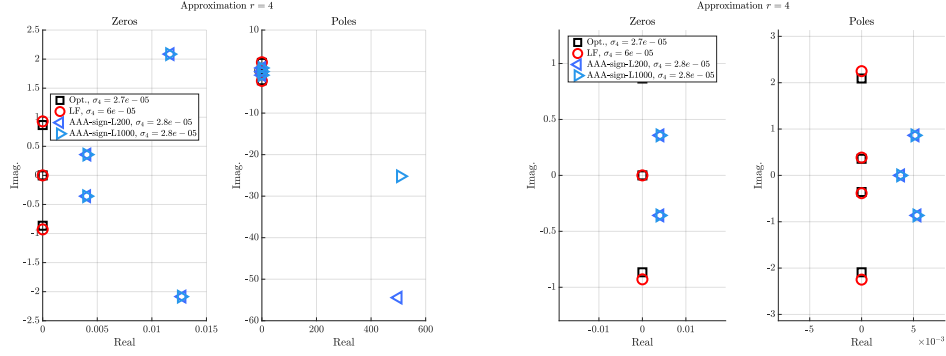


Figure 9: Case 1a (cpv MACA64 MR2023b), $r = 4$, Z4. Left frame: zeros (left) and poles (right). Right frame: same but discarding the largest real entry.

A.2.5 Approximation visualization

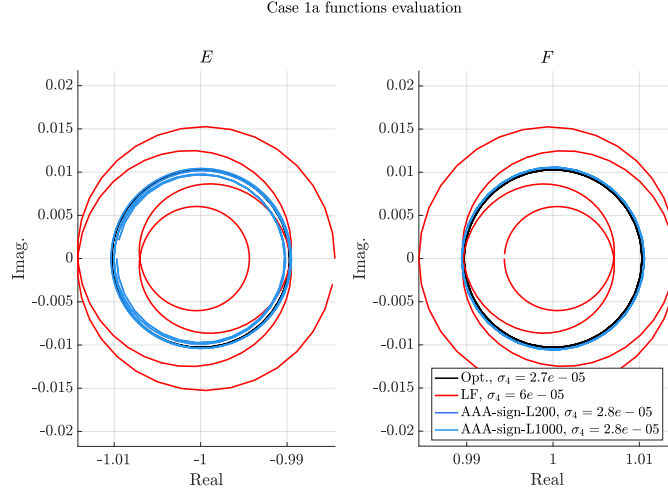


Figure 10: Case 1a (cpv MACA64 MR2023b), $r = 4$, Z4. E and F evaluations.

A.3 Approximation objective $r = 5$

A.3.1 Numerator and denominator coefficients

The numerator and denominator coefficients (both real and complex parts) are given in the following tables. Each table gives the results obtained on different platform.

Basis	Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$		LF, $\sigma_5 = 1 \cdot 10^{-05}$		AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$		AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^5	0	0	0	0	$-1.1 \cdot 10^{-11}$	$-4.8 \cdot 10^{-11}$	$2.1 \cdot 10^{-13}$	$4.5 \cdot 10^{-11}$
z^4	4.3	0	5.5	$-5.6 \cdot 10^{-15}$	4.3	$-6.4 \cdot 10^{-13}$	4.3	$1.2 \cdot 10^{-12}$
z^3	0	0	$-4.3 \cdot 10^{-15}$	$-2.7 \cdot 10^{-15}$	$-8.3 \cdot 10^{-11}$	$-3.6 \cdot 10^{-10}$	$1.5 \cdot 10^{-12}$	$3.4 \cdot 10^{-10}$
z^2	6.5	0	12.0	$-1.8 \cdot 10^{-14}$	6.5	$-2.9 \cdot 10^{-12}$	6.5	$5.2 \cdot 10^{-12}$
z	0	0	$-4.2 \cdot 10^{-15}$	$-1.0 \cdot 10^{-14}$	$-3.1 \cdot 10^{-11}$	$-1.4 \cdot 10^{-10}$	$5.5 \cdot 10^{-13}$	$1.3 \cdot 10^{-10}$
1.0	0.49	0	1.1	$-1.4 \cdot 10^{-15}$	0.49	$-3.6 \cdot 10^{-13}$	0.49	$6.3 \cdot 10^{-13}$
z^5	1.0	0	1.0	0	1.0	0	1.0	0
z^4	0	0	$-3.8 \cdot 10^{-15}$	$-1.4 \cdot 10^{-15}$	$-4.8 \cdot 10^{-11}$	$-2.1 \cdot 10^{-10}$	$8.8 \cdot 10^{-13}$	$2.0 \cdot 10^{-10}$
z^3	7.5	0	11.0	$-1.9 \cdot 10^{-14}$	7.5	$-2.2 \cdot 10^{-12}$	7.5	$4.0 \cdot 10^{-12}$
z^2	0	0	$-4.8 \cdot 10^{-15}$	$-1.1 \cdot 10^{-14}$	$-7.2 \cdot 10^{-11}$	$-3.1 \cdot 10^{-10}$	$1.3 \cdot 10^{-12}$	$3.0 \cdot 10^{-10}$
z	2.8	0	5.8	$-7.1 \cdot 10^{-15}$	2.8	$-1.7 \cdot 10^{-12}$	2.8	$3.0 \cdot 10^{-12}$
1.0	0	0	$-9.8 \cdot 10^{-16}$	$-1.1 \cdot 10^{-15}$	$-5.4 \cdot 10^{-12}$	$-2.4 \cdot 10^{-11}$	$9.5 \cdot 10^{-14}$	$2.3 \cdot 10^{-11}$

Table 34: Case 1a (cpv MACA64 MR2023b), $r = 5$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$		LF, $\sigma_5 = 1 \cdot 10^{-05}$		AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$		AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^5	0	0	0	0	$-4.1 \cdot 10^{-11}$	$3.9 \cdot 10^{-11}$	$-2.9 \cdot 10^{-14}$	$1.2 \cdot 10^{-11}$
z^4	4.3	0	5.5	$-6.5 \cdot 10^{-16}$	4.3	$-5.4 \cdot 10^{-13}$	4.3	$1.3 \cdot 10^{-13}$
z^3	0	0	$-2.1 \cdot 10^{-14}$	$4.3 \cdot 10^{-15}$	$-3.1 \cdot 10^{-10}$	$2.9 \cdot 10^{-10}$	$-2.9 \cdot 10^{-13}$	$9.0 \cdot 10^{-11}$
z^2	6.5	0	12.0	$7.5 \cdot 10^{-15}$	6.5	$-2.4 \cdot 10^{-12}$	6.5	$5.9 \cdot 10^{-13}$
z	0	0	$-1.3 \cdot 10^{-14}$	$1.6 \cdot 10^{-14}$	$-1.2 \cdot 10^{-10}$	$1.1 \cdot 10^{-10}$	$-6.8 \cdot 10^{-14}$	$3.4 \cdot 10^{-11}$
1.0	0.49	0	1.1	$2.0 \cdot 10^{-16}$	0.49	$-3.0 \cdot 10^{-13}$	0.49	$8.0 \cdot 10^{-14}$
z^5	1.0	0	1.0	0	1.0	0	1.0	0
z^4	0	0	$-5.8 \cdot 10^{-15}$	$1.8 \cdot 10^{-15}$	$-1.8 \cdot 10^{-10}$	$1.7 \cdot 10^{-10}$	$-1.7 \cdot 10^{-13}$	$5.2 \cdot 10^{-11}$
z^3	7.5	0	11.0	$3.2 \cdot 10^{-15}$	7.5	$-1.8 \cdot 10^{-12}$	7.5	$4.5 \cdot 10^{-13}$
z^2	0	0	$-2.3 \cdot 10^{-14}$	$1.8 \cdot 10^{-14}$	$-2.7 \cdot 10^{-10}$	$2.5 \cdot 10^{-10}$	$-2.2 \cdot 10^{-13}$	$7.8 \cdot 10^{-11}$
z	2.8	0	5.8	$4.1 \cdot 10^{-15}$	2.8	$-1.4 \cdot 10^{-12}$	2.8	$3.5 \cdot 10^{-13}$
1.0	0	0	$-3.4 \cdot 10^{-15}$	$2.4 \cdot 10^{-15}$	$-2.0 \cdot 10^{-11}$	$1.9 \cdot 10^{-11}$	$-2.3 \cdot 10^{-15}$	$5.9 \cdot 10^{-12}$

Table 35: Case 1a (cpv MACA64 MR2022b), $r = 5$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$		LF, $\sigma_5 = 1 \cdot 10^{-05}$		AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$		AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^5	0	0	0	0	$1.1 \cdot 10^3$	-31.0	$1.1 \cdot 10^3$	-31.0
z^4	4.3	0	5.5	$-3.9 \cdot 10^{-15}$	3.8	0.021	3.8	0.021
z^3	0	0	$-6.8 \cdot 10^{-15}$	$7.6 \cdot 10^{-15}$	$8.1 \cdot 10^3$	-240.0	$8.1 \cdot 10^3$	-230.0
z^2	6.5	0	12.0	$-1.5 \cdot 10^{-14}$	4.1	0.086	4.1	0.085
z	0	0	$-1.4 \cdot 10^{-14}$	$-1.2 \cdot 10^{-15}$	$3.1 \cdot 10^3$	-89.0	$3.1 \cdot 10^3$	-87.0
1.0	0.49	0	1.1	$-2.1 \cdot 10^{-15}$	0.19	$9.6 \cdot 10^{-3}$	0.19	$9.5 \cdot 10^{-3}$
z^5	1.0	0	1.0	0	1.0	0	1.0	$-5.8 \cdot 10^{-17}$
z^4	0	0	$-1.4 \cdot 10^{-16}$	$-5.8 \cdot 10^{-16}$	$4.7 \cdot 10^3$	-140.0	$4.7 \cdot 10^3$	-130.0
z^3	7.5	0	11.0	$-1.3 \cdot 10^{-14}$	5.7	0.071	5.7	0.07
z^2	0	0	$-1.7 \cdot 10^{-14}$	$1.2 \cdot 10^{-14}$	$7.1 \cdot 10^3$	-200.0	$7.1 \cdot 10^3$	-200.0
z	2.8	0	5.8	$-1.1 \cdot 10^{-14}$	1.4	0.047	1.4	0.046
1.0	0	0	$-3.3 \cdot 10^{-15}$	$5.1 \cdot 10^{-16}$	530.0	-15.0	530.0	-15.0

Table 36: Case 1a (cpv PCWIN64 MR2025b), $r = 5$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$		LF, $\sigma_5 = 1 \cdot 10^{-05}$		AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$		AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^5	0	0	0	0	$1.1 \cdot 10^3$	-31.0	$1.1 \cdot 10^3$	-31.0
z^4	4.3	0	5.5	$-3.9 \cdot 10^{-15}$	3.8	0.021	3.8	0.021
z^3	0	0	$-6.8 \cdot 10^{-15}$	$7.6 \cdot 10^{-15}$	$8.1 \cdot 10^3$	-240.0	$8.1 \cdot 10^3$	-230.0
z^2	6.5	0	12.0	$-1.5 \cdot 10^{-14}$	4.1	0.086	4.1	0.085
z	0	0	$-1.4 \cdot 10^{-14}$	$-1.2 \cdot 10^{-15}$	$3.1 \cdot 10^3$	-89.0	$3.1 \cdot 10^3$	-87.0
1.0	0.49	0	1.1	$-2.1 \cdot 10^{-15}$	0.19	$9.6 \cdot 10^{-3}$	0.19	$9.5 \cdot 10^{-3}$
z^5	1.0	0	1.0	0	1.0	0	1.0	$-5.8 \cdot 10^{-17}$
z^4	0	0	$-1.4 \cdot 10^{-16}$	$-5.8 \cdot 10^{-16}$	$4.7 \cdot 10^3$	-140.0	$4.7 \cdot 10^3$	-130.0
z^3	7.5	0	11.0	$-1.3 \cdot 10^{-14}$	5.7	0.071	5.7	0.07
z^2	0	0	$-1.7 \cdot 10^{-14}$	$1.2 \cdot 10^{-14}$	$7.1 \cdot 10^3$	-200.0	$7.1 \cdot 10^3$	-200.0
z	2.8	0	5.8	$-1.1 \cdot 10^{-14}$	1.4	0.047	1.4	0.046
1.0	0	0	$-3.3 \cdot 10^{-15}$	$5.1 \cdot 10^{-16}$	530.0	-15.0	530.0	-15.0

Table 37: Case 1a (AlexReis PCWIN64 MR2025b), $r = 5$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$		LF, $\sigma_5 = 1 \cdot 10^{-05}$		AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$		AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^5	0	0	0	0	$-4.2 \cdot 10^{-11}$	$-2.1 \cdot 10^{-11}$	$1.3 \cdot 10^{-13}$	$-2.0 \cdot 10^{-11}$
z^4	4.3	0	5.5	$-3.8 \cdot 10^{-15}$	4.3	$1.4 \cdot 10^{-14}$	4.3	$-3.7 \cdot 10^{-14}$
z^3	0	0	$-2.2 \cdot 10^{-14}$	$-2.1 \cdot 10^{-15}$	$-3.1 \cdot 10^{-10}$	$-1.6 \cdot 10^{-10}$	$8.9 \cdot 10^{-13}$	$-1.5 \cdot 10^{-10}$
z^2	6.5	0	12.0	$-1.3 \cdot 10^{-14}$	6.5	$-3.9 \cdot 10^{-14}$	6.5	$-1.2 \cdot 10^{-13}$
z	0	0	$-1.4 \cdot 10^{-14}$	$-1.8 \cdot 10^{-15}$	$-1.2 \cdot 10^{-10}$	$-5.9 \cdot 10^{-11}$	$4.0 \cdot 10^{-13}$	$-5.5 \cdot 10^{-11}$
1.0	0.49	0	1.1	$-5.7 \cdot 10^{-16}$	0.49	$-5.3 \cdot 10^{-15}$	0.49	$2.6 \cdot 10^{-15}$
z^5	1.0	0	1.0	0	1.0	0	1.0	0
z^4	0	0	$-4.3 \cdot 10^{-15}$	$1.6 \cdot 10^{-15}$	$-1.8 \cdot 10^{-10}$	$-9.2 \cdot 10^{-11}$	$5.0 \cdot 10^{-13}$	$-8.5 \cdot 10^{-11}$
z^3	7.5	0	11.0	$-9.9 \cdot 10^{-15}$	7.5	$2.2 \cdot 10^{-15}$	7.5	$-1.2 \cdot 10^{-13}$
z^2	0	0	$-2.3 \cdot 10^{-14}$	$3.7 \cdot 10^{-15}$	$-2.7 \cdot 10^{-10}$	$-1.4 \cdot 10^{-10}$	$8.4 \cdot 10^{-13}$	$-1.3 \cdot 10^{-10}$
z	2.8	0	5.8	$-3.7 \cdot 10^{-15}$	2.8	$-2.7 \cdot 10^{-14}$	2.8	$-3.2 \cdot 10^{-14}$
1.0	0	0	$-2.5 \cdot 10^{-15}$	$-3.5 \cdot 10^{-16}$	$-2.0 \cdot 10^{-11}$	$-1.0 \cdot 10^{-11}$	$7.4 \cdot 10^{-14}$	$-9.5 \cdot 10^{-12}$

Table 38: Case 1a (AlexReis PCWIN64perso MR2017a), $r = 5$, Z4: numerator (first lines) and denominator (last lines) coefficients.

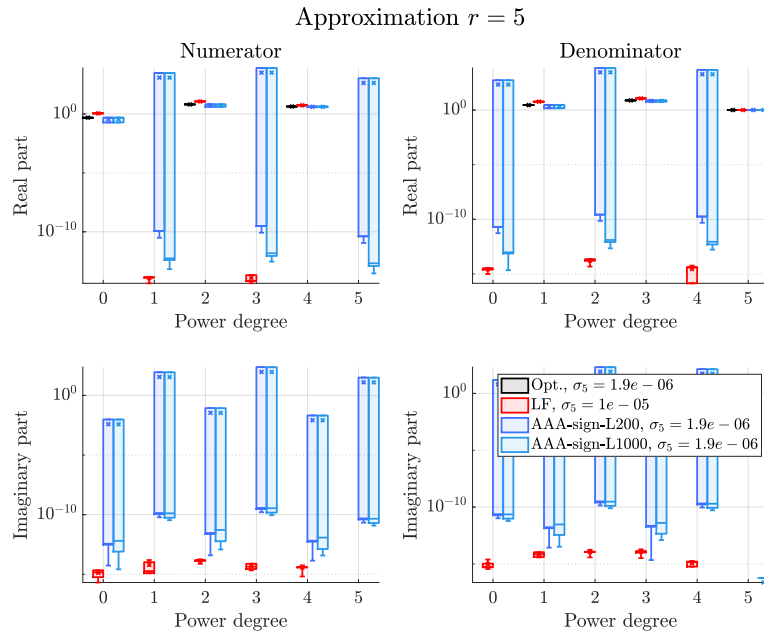


Figure 11: Case 1a, $r = 5$, Z4. Numerator and denominator dispersion over the different platforms.

A.3.2 Poles

The poles are given in the following tables. Each table gives the results obtained on different platform.

	Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$	LF, $\sigma_5 = 1 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$	AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$
1.0	0	$1.69 \cdot 10^{-16} + 1.84 \cdot 10^{-16}i$	$1.92 \cdot 10^{-12} + 8.44 \cdot 10^{-12}i$	$-3.41 \cdot 10^{-14} - 8.04 \cdot 10^{-12}i$
2.0	$3.89 \cdot 10^{-16} + 2.67i$	$2.45 \cdot 10^{-16} - 0.729i$	$2.84 \cdot 10^{-12} - 0.629i$	$-2.12 \cdot 10^{-13} + 0.629i$
3.0	$3.89 \cdot 10^{-16} - 2.67i$	$-1.57 \cdot 10^{-16} + 0.729i$	$3.03 \cdot 10^{-12} + 0.629i$	$1.08 \cdot 10^{-13} - 0.629i$
4.0	$-5.55 \cdot 10^{-17} + 0.629i$	$-1.18 \cdot 10^{-15} - 3.3i$	$1.98 \cdot 10^{-11} - 2.67i$	$-1.09 \cdot 10^{-12} + 2.67i$
5.0	$-5.55 \cdot 10^{-17} - 0.629i$	$4.72 \cdot 10^{-15} + 3.3i$	$2.05 \cdot 10^{-11} + 2.67i$	$3.47 \cdot 10^{-13} - 2.67i$

Table 39: Case 1a (cpv MACA64 MR2023b), $r = 5$, Z4: poles.

	Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$	LF, $\sigma_5 = 1 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$	AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$
1.0	0	$5.89 \cdot 10^{-16} - 4.22 \cdot 10^{-16}i$	$7.1 \cdot 10^{-12} - 6.72 \cdot 10^{-12}i$	$-8.34 \cdot 10^{-16} - 2.1 \cdot 10^{-12}i$
2.0	$3.89 \cdot 10^{-16} + 2.67i$	$4.48 \cdot 10^{-16} + 0.729i$	$1.09 \cdot 10^{-11} + 0.629i$	$-8.47 \cdot 10^{-15} + 0.629i$
3.0	$3.89 \cdot 10^{-16} - 2.67i$	$8.9 \cdot 10^{-16} - 0.729i$	$1.08 \cdot 10^{-11} - 0.629i$	$3.14 \cdot 10^{-14} - 0.629i$
4.0	$-5.55 \cdot 10^{-17} + 0.629i$	$1.51 \cdot 10^{-15} + 3.3i$	$7.46 \cdot 10^{-11} + 2.67i$	$-7.54 \cdot 10^{-15} + 2.67i$
5.0	$-5.55 \cdot 10^{-17} - 0.629i$	$2.43 \cdot 10^{-15} - 3.3i$	$7.4 \cdot 10^{-11} - 2.67i$	$1.54 \cdot 10^{-13} - 2.67i$

Table 40: Case 1a (cpv MACA64 MR2022b), $r = 5$, Z4: poles.

	Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$	LF, $\sigma_5 = 1 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$	AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$
1.0	0	$5.62 \cdot 10^{-16} - 8.81 \cdot 10^{-17}i$	$-7.91 \cdot 10^{-5} + 0.281i$	$-7.91 \cdot 10^{-5} + 0.281i$
2.0	$3.89 \cdot 10^{-16} + 2.67i$	$7.72 \cdot 10^{-16} + 0.729i$	$-7.96 \cdot 10^{-5} - 0.281i$	$-7.95 \cdot 10^{-5} - 0.281i$
3.0	$3.89 \cdot 10^{-16} - 2.67i$	$2.77 \cdot 10^{-16} - 0.729i$	$-3.61 \cdot 10^{-4} + 1.19i$	$-3.61 \cdot 10^{-4} + 1.19i$
4.0	$-5.55 \cdot 10^{-17} + 0.629i$	$-2.67 \cdot 10^{-15} - 3.3i$	$-3.65 \cdot 10^{-4} - 1.19i$	$-3.65 \cdot 10^{-4} - 1.19i$
5.0	$-5.55 \cdot 10^{-17} - 0.629i$	$1.2 \cdot 10^{-15} + 3.3i$	$-4700.0 + 136.0i$	$-4700.0 + 134.0i$

Table 41: Case 1a (cpv PCWIN64 MR2025b), $r = 5$, Z4: poles.

	Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$	LF, $\sigma_5 = 1 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$	AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$
1.0	0	$5.62 \cdot 10^{-16} - 8.81 \cdot 10^{-17}i$	$-7.91 \cdot 10^{-5} + 0.281i$	$-7.91 \cdot 10^{-5} + 0.281i$
2.0	$3.89 \cdot 10^{-16} + 2.67i$	$7.72 \cdot 10^{-16} + 0.729i$	$-7.96 \cdot 10^{-5} - 0.281i$	$-7.95 \cdot 10^{-5} - 0.281i$
3.0	$3.89 \cdot 10^{-16} - 2.67i$	$2.77 \cdot 10^{-16} - 0.729i$	$-3.61 \cdot 10^{-4} + 1.19i$	$-3.61 \cdot 10^{-4} + 1.19i$
4.0	$-5.55 \cdot 10^{-17} + 0.629i$	$-2.67 \cdot 10^{-15} - 3.3i$	$-3.65 \cdot 10^{-4} - 1.19i$	$-3.65 \cdot 10^{-4} - 1.19i$
5.0	$-5.55 \cdot 10^{-17} - 0.629i$	$1.2 \cdot 10^{-15} + 3.3i$	$-4700.0 + 136.0i$	$-4700.0 + 134.0i$

Table 42: Case 1a (AlexReis PCWIN64 MR2025b), $r = 5$, Z4: poles.

	Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$	LF, $\sigma_5 = 1 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$	AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$
1.0	0	$4.33 \cdot 10^{-16} + 6.11 \cdot 10^{-17}i$	$7.24 \cdot 10^{-12} + 3.66 \cdot 10^{-12}i$	$-2.63 \cdot 10^{-14} + 3.39 \cdot 10^{-12}i$
2.0	$3.89 \cdot 10^{-16} + 2.67i$	$6.49 \cdot 10^{-16} + 0.729i$	$1.11 \cdot 10^{-11} - 0.629i$	$-3.24 \cdot 10^{-14} - 0.629i$
3.0	$3.89 \cdot 10^{-16} - 2.67i$	$8.51 \cdot 10^{-16} - 0.729i$	$1.11 \cdot 10^{-11} + 0.629i$	$-3.71 \cdot 10^{-14} + 0.629i$
4.0	$-5.55 \cdot 10^{-17} + 0.629i$	$2.72 \cdot 10^{-15} + 3.3i$	$7.58 \cdot 10^{-11} - 2.67i$	$-2.25 \cdot 10^{-13} - 2.67i$
5.0	$-5.55 \cdot 10^{-17} - 0.629i$	$-3.32 \cdot 10^{-16} - 3.3i$	$7.58 \cdot 10^{-11} + 2.67i$	$-1.79 \cdot 10^{-13} + 2.67i$

Table 43: Case 1a (AlexReis PCWIN64perso MR2017a), $r = 5$, Z4: poles.

A.3.3 Zeros

The zeros are given in the following tables. Each table gives the results obtained on different platform.

	Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$	LF, $\sigma_5 = 1 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$	AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$
1.0	NaN	NaN	$2.08 \cdot 10^{-12} - 0.281_z$	$-1.06 \cdot 10^{-13} + 0.281_z$
2.0	$5.55 \cdot 10^{-17} + 1.19_z$	$2.26 \cdot 10^{-16} - 0.32_z$	$2.16 \cdot 10^{-12} + 0.281_z$	$3.17 \cdot 10^{-14} - 0.281_z$
3.0	$5.55 \cdot 10^{-17} - 1.19_z$	$1.36 \cdot 10^{-16} + 0.32_z$	$5.38 \cdot 10^{-12} - 1.19_z$	$-4.15 \cdot 10^{-13} + 1.19_z$
4.0	$6.94 \cdot 10^{-17} + 0.281_z$	$5.92 \cdot 10^{-16} + 1.42_z$	$5.74 \cdot 10^{-12} + 1.19_z$	$2.15 \cdot 10^{-13} - 1.19_z$
5.0	$6.94 \cdot 10^{-17} - 0.281_z$	$-1.68 \cdot 10^{-16} - 1.42_z$	$1.99 \cdot 10^{10} - 8.58 \cdot 10^{10}_z$	$-4.5 \cdot 10^8 + 9.58 \cdot 10^{10}_z$

Table 44: Case 1a (cpv MACA64 MR2023b), $r = 5$, Z4: zeros.

	Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$	LF, $\sigma_5 = 1 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$	AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$
1.0	NaN	NaN	$7.88 \cdot 10^{-12} + 0.281_z$	$-7.88 \cdot 10^{-15} + 0.281_z$
2.0	$5.55 \cdot 10^{-17} + 1.19_z$	$5.94 \cdot 10^{-16} + 0.32_z$	$7.81 \cdot 10^{-12} - 0.281_z$	$1.46 \cdot 10^{-14} - 0.281_z$
3.0	$5.55 \cdot 10^{-17} - 1.19_z$	$4.39 \cdot 10^{-16} - 0.32_z$	$2.07 \cdot 10^{-11} + 1.19_z$	$-9.47 \cdot 10^{-15} + 1.19_z$
4.0	$6.94 \cdot 10^{-17} + 0.281_z$	$1.88 \cdot 10^{-15} - 1.42_z$	$2.04 \cdot 10^{-11} - 1.19_z$	$6.07 \cdot 10^{-14} - 1.19_z$
5.0	$6.94 \cdot 10^{-17} - 0.281_z$	$9.89 \cdot 10^{-16} + 1.42_z$	$5.49 \cdot 10^{10} + 5.28 \cdot 10^{10}_z$	$8.85 \cdot 10^8 + 3.6 \cdot 10^{11}_z$

Table 45: Case 1a (cpv MACA64 MR2022b), $r = 5$, Z4: zeros.

	Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$	LF, $\sigma_5 = 1 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$	AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$
1.0	NaN	NaN	$-6.3 \cdot 10^{-5} - 4.98 \cdot 10^{-6}_z$	$-6.3 \cdot 10^{-5} - 4.9 \cdot 10^{-6}_z$
2.0	$5.55 \cdot 10^{-17} + 1.19_z$	$5.2 \cdot 10^{-16} - 0.32_z$	$-1.45 \cdot 10^{-4} + 0.629_z$	$-1.45 \cdot 10^{-4} + 0.629_z$
3.0	$5.55 \cdot 10^{-17} - 1.19_z$	$7.03 \cdot 10^{-16} + 0.32_z$	$-1.46 \cdot 10^{-4} - 0.629_z$	$-1.46 \cdot 10^{-4} - 0.629_z$
4.0	$6.94 \cdot 10^{-17} + 0.281_z$	$4.37 \cdot 10^{-16} + 1.42_z$	$-0.00156 + 2.66_z$	$-0.00156 + 2.66_z$
5.0	$6.94 \cdot 10^{-17} - 0.281_z$	$-4.16 \cdot 10^{-16} - 1.42_z$	$-0.00158 - 2.67_z$	$-0.00158 - 2.67_z$

Table 46: Case 1a (cpv PCWIN64 MR2025b), $r = 5$, Z4: zeros.

	Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$	LF, $\sigma_5 = 1 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$	AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$
1.0	NaN	NaN	$-6.3 \cdot 10^{-5} - 4.98 \cdot 10^{-6}_z$	$-6.3 \cdot 10^{-5} - 4.9 \cdot 10^{-6}_z$
2.0	$5.55 \cdot 10^{-17} + 1.19_z$	$5.2 \cdot 10^{-16} - 0.32_z$	$-1.45 \cdot 10^{-4} + 0.629_z$	$-1.45 \cdot 10^{-4} + 0.629_z$
3.0	$5.55 \cdot 10^{-17} - 1.19_z$	$7.03 \cdot 10^{-16} + 0.32_z$	$-1.46 \cdot 10^{-4} - 0.629_z$	$-1.46 \cdot 10^{-4} - 0.629_z$
4.0	$6.94 \cdot 10^{-17} + 0.281_z$	$4.37 \cdot 10^{-16} + 1.42_z$	$-0.00156 + 2.66_z$	$-0.00156 + 2.66_z$
5.0	$6.94 \cdot 10^{-17} - 0.281_z$	$-4.16 \cdot 10^{-16} - 1.42_z$	$-0.00158 - 2.67_z$	$-0.00158 - 2.67_z$

Table 47: Case 1a (AlexReis PCWIN64 MR2025b), $r = 5$, Z4: zeros.

	Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$	LF, $\sigma_5 = 1 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$	AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$
1.0	NaN	NaN	$8.01 \cdot 10^{-12} - 0.281_z$	$-2.07 \cdot 10^{-14} - 0.281_z$
2.0	$5.55 \cdot 10^{-17} + 1.19_z$	$6.68 \cdot 10^{-16} - 0.32_z$	$8.0 \cdot 10^{-12} + 0.281_z$	$-3.44 \cdot 10^{-14} + 0.281_z$
3.0	$5.55 \cdot 10^{-17} - 1.19_z$	$4.52 \cdot 10^{-16} + 0.32_z$	$2.09 \cdot 10^{-11} - 1.19_z$	$-6.2 \cdot 10^{-14} - 1.19_z$
4.0	$6.94 \cdot 10^{-17} + 0.281_z$	$1.09 \cdot 10^{-15} - 1.42_z$	$2.09 \cdot 10^{-11} + 1.19_z$	$-4.62 \cdot 10^{-14} + 1.19_z$
5.0	$6.94 \cdot 10^{-17} - 0.281_z$	$1.78 \cdot 10^{-15} + 1.42_z$	$8.25 \cdot 10^{10} - 4.18 \cdot 10^{10}_z$	$-1.39 \cdot 10^9 - 2.19 \cdot 10^{11}_z$

Table 48: Case 1a (AlexReis PCWIN64perso MR2017a), $r = 5$, Z4: zeros.

A.3.4 Poles and zeros visualisation (case cpv MACA64 MR2023b)

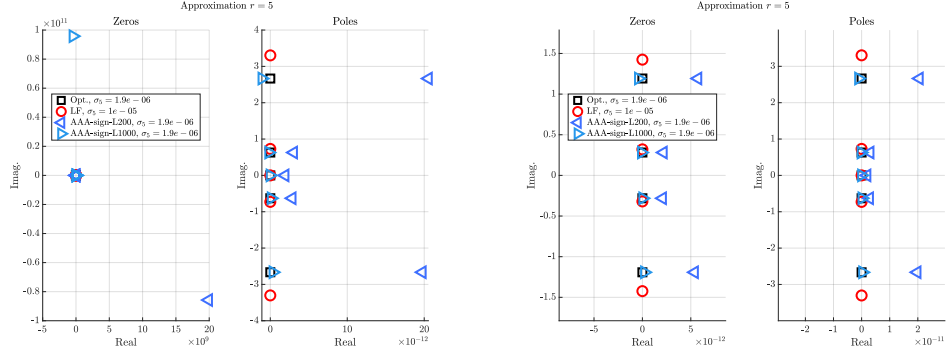


Figure 12: Case 1a (cpv MACA64 MR2023b), $r = 5$, Z4. Left frame: zeros (left) and poles (right). Right frame: same but discarding the largest real entry.

A.3.5 Approximation visualization

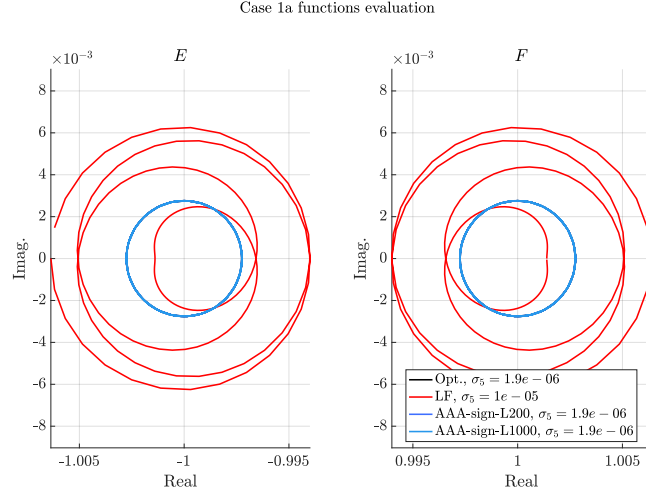


Figure 13: Case 1a (cpv MACA64 MR2023b), $r = 5$, Z4. E and F evaluations.

A.4 Approximation objective $r = 6$

A.4.1 Numerator and denominator coefficients

The numerator and denominator coefficients (both real and complex parts) are given in the following tables. Each table gives the results obtained on different platform.

Basis	Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$		LF, $\sigma_6 = 3.2 \cdot 10^{-07}$		AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$		AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^6	0	0	0	0	$-1.2 \cdot 10^{-10}$	$-2.0 \cdot 10^{-10}$	$1.9 \cdot 10^{-12}$	$-8.0 \cdot 10^{-11}$
z^5	5.2	0	5.5	$-5.7 \cdot 10^{-15}$	5.2	$-6.3 \cdot 10^{-12}$	5.2	$-4.6 \cdot 10^{-12}$
z^4	0	0	$-2.3 \cdot 10^{-14}$	$-1.3 \cdot 10^{-14}$	$-1.4 \cdot 10^{-9}$	$-2.3 \cdot 10^{-9}$	$2.1 \cdot 10^{-11}$	$-9.2 \cdot 10^{-10}$
z^3	13.0	0	15.0	$-1.1 \cdot 10^{-14}$	13.0	$-4.7 \cdot 10^{-11}$	13.0	$-3.5 \cdot 10^{-11}$
z^2	0	0	$-4.2 \cdot 10^{-14}$	$-3.3 \cdot 10^{-14}$	$-1.1 \cdot 10^{-9}$	$-1.7 \cdot 10^{-9}$	$1.5 \cdot 10^{-11}$	$-7.0 \cdot 10^{-10}$
z	2.9	0	3.7	$6.4 \cdot 10^{-16}$	2.9	$-1.7 \cdot 10^{-11}$	2.9	$-1.3 \cdot 10^{-11}$
1.0	0	0	$-4.1 \cdot 10^{-15}$	$-2.9 \cdot 10^{-15}$	$-5.3 \cdot 10^{-11}$	$-8.7 \cdot 10^{-11}$	$7.8 \cdot 10^{-13}$	$-3.6 \cdot 10^{-11}$
z^6	1.0	0	1.0	0	1.0	0	1.0	0
z^5	0	0	$-7.0 \cdot 10^{-15}$	$-4.7 \cdot 10^{-15}$	$-6.5 \cdot 10^{-10}$	$-1.0 \cdot 10^{-9}$	$9.8 \cdot 10^{-12}$	$-4.2 \cdot 10^{-10}$
z^4	11.0	0	12.0	$-1.4 \cdot 10^{-14}$	11.0	$-2.7 \cdot 10^{-11}$	11.0	$-2.0 \cdot 10^{-11}$
z^3	0	0	$-4.5 \cdot 10^{-14}$	$-3.7 \cdot 10^{-14}$	$-1.6 \cdot 10^{-9}$	$-2.6 \cdot 10^{-9}$	$2.4 \cdot 10^{-11}$	$-1.1 \cdot 10^{-9}$
z^2	8.4	0	10.0	$-1.8 \cdot 10^{-16}$	8.4	$-4.0 \cdot 10^{-11}$	8.4	$-3.0 \cdot 10^{-11}$
z	0	0	$-2.0 \cdot 10^{-14}$	$-9.8 \cdot 10^{-15}$	$-3.6 \cdot 10^{-10}$	$-6.0 \cdot 10^{-10}$	$5.3 \cdot 10^{-12}$	$-2.5 \cdot 10^{-10}$
1.0	0.42	0	0.56	$-5.6 \cdot 10^{-16}$	0.42	$-3.0 \cdot 10^{-12}$	0.42	$-2.3 \cdot 10^{-12}$

Table 49: Case 1a (cpv MACA64 MR2023b), $r = 6$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$		LF, $\sigma_6 = 3.2 \cdot 10^{-07}$		AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$		AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^6	0	0	0	0	$-1.2 \cdot 10^{-10}$	$-6.5 \cdot 10^{-11}$	$-1.8 \cdot 10^{-12}$	$3.7 \cdot 10^{-10}$
z^5	5.2	0	5.5	$-8.5 \cdot 10^{-16}$	5.2	$4.5 \cdot 10^{-12}$	5.2	$5.7 \cdot 10^{-12}$
z^4	0	0	$-2.2 \cdot 10^{-14}$	$-7.0 \cdot 10^{-15}$	$-1.4 \cdot 10^{-9}$	$-7.1 \cdot 10^{-10}$	$-2.0 \cdot 10^{-11}$	$4.2 \cdot 10^{-9}$
z^3	13.0	0	15.0	$-2.3 \cdot 10^{-15}$	13.0	$3.3 \cdot 10^{-11}$	13.0	$4.2 \cdot 10^{-11}$
z^2	0	0	$-3.0 \cdot 10^{-14}$	$-6.1 \cdot 10^{-15}$	$-1.0 \cdot 10^{-9}$	$-5.2 \cdot 10^{-10}$	$-1.5 \cdot 10^{-11}$	$3.2 \cdot 10^{-9}$
z	2.9	0	3.7	$-4.8 \cdot 10^{-15}$	2.9	$1.2 \cdot 10^{-11}$	2.9	$1.6 \cdot 10^{-11}$
1.0	0	0	$-8.2 \cdot 10^{-16}$	$-2.6 \cdot 10^{-16}$	$-5.2 \cdot 10^{-11}$	$-2.5 \cdot 10^{-11}$	$-7.2 \cdot 10^{-13}$	$1.6 \cdot 10^{-10}$
z^6	1.0	0	1.0	$3.5 \cdot 10^{-17}$	1.0	$-3.6 \cdot 10^{-18}$	1.0	0
z^5	0	0	$-4.6 \cdot 10^{-15}$	$-1.1 \cdot 10^{-15}$	$-6.4 \cdot 10^{-10}$	$-3.3 \cdot 10^{-10}$	$-9.4 \cdot 10^{-12}$	$1.9 \cdot 10^{-9}$
z^4	11.0	0	12.0	$5.7 \cdot 10^{-16}$	11.0	$1.9 \cdot 10^{-11}$	11.0	$2.4 \cdot 10^{-11}$
z^3	0	0	$-3.3 \cdot 10^{-14}$	$-3.1 \cdot 10^{-15}$	$-1.6 \cdot 10^{-9}$	$-8.1 \cdot 10^{-10}$	$-2.3 \cdot 10^{-11}$	$4.9 \cdot 10^{-9}$
z^2	8.4	0	10.0	$-9.1 \cdot 10^{-15}$	8.4	$2.9 \cdot 10^{-11}$	8.4	$3.6 \cdot 10^{-11}$
z	0	0	$-1.2 \cdot 10^{-14}$	$-6.9 \cdot 10^{-15}$	$-3.6 \cdot 10^{-10}$	$-1.8 \cdot 10^{-10}$	$-5.0 \cdot 10^{-12}$	$1.1 \cdot 10^{-9}$
1.0	0.42	0	0.56	$3.5 \cdot 10^{-17}$	0.42	$2.1 \cdot 10^{-12}$	0.42	$2.7 \cdot 10^{-12}$

Table 50: Case 1a (cpv MACA64 MR2022b), $r = 6$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$		LF, $\sigma_6 = 3.2 \cdot 10^{-07}$		AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$		AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^6	0	0	0	0	$-4.4 \cdot 10^8$	$-2.2 \cdot 10^9$	$-6.7 \cdot 10^7$	$-6.4 \cdot 10^9$
z^5	5.2	0	5.5	$-3.6 \cdot 10^{-15}$	5.2	$-4.6 \cdot 10^{-4}$	5.2	$5.3 \cdot 10^{-4}$
z^4	0	0	$-2.6 \cdot 10^{-16}$	$1.4 \cdot 10^{-14}$	$-5.0 \cdot 10^9$	$-2.5 \cdot 10^{10}$	$-7.5 \cdot 10^8$	$-7.2 \cdot 10^{10}$
z^3	13.0	0	15.0	$-1.8 \cdot 10^{-14}$	13.0	$-3.2 \cdot 10^{-3}$	13.0	$3.8 \cdot 10^{-3}$
z^2	0	0	$-1.9 \cdot 10^{-15}$	$9.7 \cdot 10^{-15}$	$-3.7 \cdot 10^9$	$-1.9 \cdot 10^{10}$	$-5.6 \cdot 10^8$	$-5.4 \cdot 10^{10}$
z	2.9	0	3.7	$-5.0 \cdot 10^{-15}$	2.9	$-1.3 \cdot 10^{-3}$	2.9	$1.7 \cdot 10^{-3}$
1.0	0	0	$-4.4 \cdot 10^{-16}$	$2.2 \cdot 10^{-16}$	$-1.9 \cdot 10^8$	$-9.4 \cdot 10^8$	$-2.8 \cdot 10^7$	$-2.7 \cdot 10^9$
z^6	1.0	0	1.0	0	1.0	0	1.0	0
z^5	0	0	$1.8 \cdot 10^{-15}$	$2.9 \cdot 10^{-15}$	$-2.3 \cdot 10^9$	$-1.2 \cdot 10^{10}$	$-3.5 \cdot 10^8$	$-3.3 \cdot 10^{10}$
z^4	11.0	0	12.0	$-9.1 \cdot 10^{-15}$	11.0	$-1.9 \cdot 10^{-3}$	11.0	$2.1 \cdot 10^{-3}$
z^3	0	0	$-4.6 \cdot 10^{-15}$	$1.5 \cdot 10^{-14}$	$-5.8 \cdot 10^9$	$-2.9 \cdot 10^{10}$	$-8.7 \cdot 10^8$	$-8.3 \cdot 10^{10}$
z^2	8.4	0	10.0	$-1.5 \cdot 10^{-14}$	8.5	$-2.8 \cdot 10^{-3}$	8.4	$3.6 \cdot 10^{-3}$
z	0	0	$-9.5 \cdot 10^{-18}$	$2.7 \cdot 10^{-15}$	$-1.3 \cdot 10^9$	$-6.5 \cdot 10^9$	$-2.0 \cdot 10^8$	$-1.9 \cdot 10^{10}$
1.0	0.42	0	0.56	$1.9 \cdot 10^{-16}$	0.42	$-2.4 \cdot 10^{-4}$	0.42	$2.7 \cdot 10^{-4}$

Table 51: Case 1a (cpv PCWIN64 MR2025b), $r = 6$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$		LF, $\sigma_6 = 3.2 \cdot 10^{-07}$		AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$		AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^6	0	0	0	0	$-4.4 \cdot 10^8$	$-2.2 \cdot 10^9$	$-6.7 \cdot 10^7$	$-6.4 \cdot 10^9$
z^5	5.2	0	5.5	$-3.6 \cdot 10^{-15}$	5.2	$-4.6 \cdot 10^{-4}$	5.2	$5.3 \cdot 10^{-4}$
z^4	0	0	$-2.6 \cdot 10^{-16}$	$1.4 \cdot 10^{-14}$	$-5.0 \cdot 10^9$	$-2.5 \cdot 10^{10}$	$-7.5 \cdot 10^8$	$-7.2 \cdot 10^{10}$
z^3	13.0	0	15.0	$-1.8 \cdot 10^{-14}$	13.0	$-3.2 \cdot 10^{-3}$	13.0	$3.8 \cdot 10^{-3}$
z^2	0	0	$-1.9 \cdot 10^{-15}$	$9.7 \cdot 10^{-15}$	$-3.7 \cdot 10^9$	$-1.9 \cdot 10^{10}$	$-5.6 \cdot 10^8$	$-5.4 \cdot 10^{10}$
z	2.9	0	3.7	$-5.0 \cdot 10^{-15}$	2.9	$-1.3 \cdot 10^{-3}$	2.9	$1.7 \cdot 10^{-3}$
1.0	0	0	$-4.4 \cdot 10^{-16}$	$2.2 \cdot 10^{-16}$	$-1.9 \cdot 10^8$	$-9.4 \cdot 10^8$	$-2.8 \cdot 10^7$	$-2.7 \cdot 10^9$
z^6	1.0	0	1.0	0	1.0	0	1.0	0
z^5	0	0	$1.8 \cdot 10^{-15}$	$2.9 \cdot 10^{-15}$	$-2.3 \cdot 10^9$	$-1.2 \cdot 10^{10}$	$-3.5 \cdot 10^8$	$-3.3 \cdot 10^{10}$
z^4	11.0	0	12.0	$-9.1 \cdot 10^{-15}$	11.0	$-1.9 \cdot 10^{-3}$	11.0	$2.1 \cdot 10^{-3}$
z^3	0	0	$-4.6 \cdot 10^{-15}$	$1.5 \cdot 10^{-14}$	$-5.8 \cdot 10^9$	$-2.9 \cdot 10^{10}$	$-8.7 \cdot 10^8$	$-8.3 \cdot 10^{10}$
z^2	8.4	0	10.0	$-1.5 \cdot 10^{-14}$	8.5	$-2.8 \cdot 10^{-3}$	8.4	$3.6 \cdot 10^{-3}$
z	0	0	$-9.5 \cdot 10^{-18}$	$2.7 \cdot 10^{-15}$	$-1.3 \cdot 10^9$	$-6.5 \cdot 10^9$	$-2.0 \cdot 10^8$	$-1.9 \cdot 10^{10}$
1.0	0.42	0	0.56	$1.9 \cdot 10^{-16}$	0.42	$-2.4 \cdot 10^{-4}$	0.42	$2.7 \cdot 10^{-4}$

Table 52: Case 1a (AlexReis PCWIN64 MR2025b), $r = 6$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$		LF, $\sigma_6 = 3.2 \cdot 10^{-07}$		AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$		AAA-sign-L1000, $\sigma_6 = 1.8 \cdot 10^{-07}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^6	0	0	0	0	-150.0	-7.0	74.0	4.9
z^5	5.2	0	5.5	$-2.9 \cdot 10^{-15}$	4.2	-0.04	2.7	-0.089
z^4	0	0	$-2.3 \cdot 10^{-15}$	$2.6 \cdot 10^{-15}$	$-1.6 \cdot 10^3$	-79.0	840.0	56.0
z^3	13.0	0	15.0	$-2.3 \cdot 10^{-14}$	5.8	-0.27	-6.0	-0.66
z^2	0	0	$1.2 \cdot 10^{-14}$	$4.9 \cdot 10^{-15}$	$-1.2 \cdot 10^3$	-59.0	640.0	42.0
z	2.9	0	3.7	$-8.8 \cdot 10^{-15}$	0.31	-0.088	-4.3	-0.24
1.0	0	0	$1.4 \cdot 10^{-16}$	$3.0 \cdot 10^{-15}$	-62.0	-3.0	32.0	2.1
z^6	1.0	0	1.0	0	1.0	0	1.0	$4.3 \cdot 10^{-17}$
z^5	0	0	$-3.3 \cdot 10^{-16}$	$-2.3 \cdot 10^{-15}$	-760.0	-36.0	390.0	26.0
z^4	11.0	0	12.0	$-1.1 \cdot 10^{-14}$	7.0	-0.16	0.35	-0.38
z^3	0	0	$1.4 \cdot 10^{-14}$	$-2.7 \cdot 10^{-15}$	$-1.9 \cdot 10^3$	-91.0	980.0	64.0
z^2	8.4	0	10.0	$-2.0 \cdot 10^{-14}$	2.3	-0.22	-8.1	-0.56
z	0	0	$4.7 \cdot 10^{-15}$	$5.8 \cdot 10^{-15}$	-430.0	-20.0	220.0	15.0
1.0	0.42	0	0.56	$-7.4 \cdot 10^{-16}$	-0.023	-0.014	-0.83	-0.041

Table 53: Case 1a (AlexReis PCWIN64perso MR2017a), $r = 6$, Z4: numerator (first lines) and denominator (last lines) coefficients.

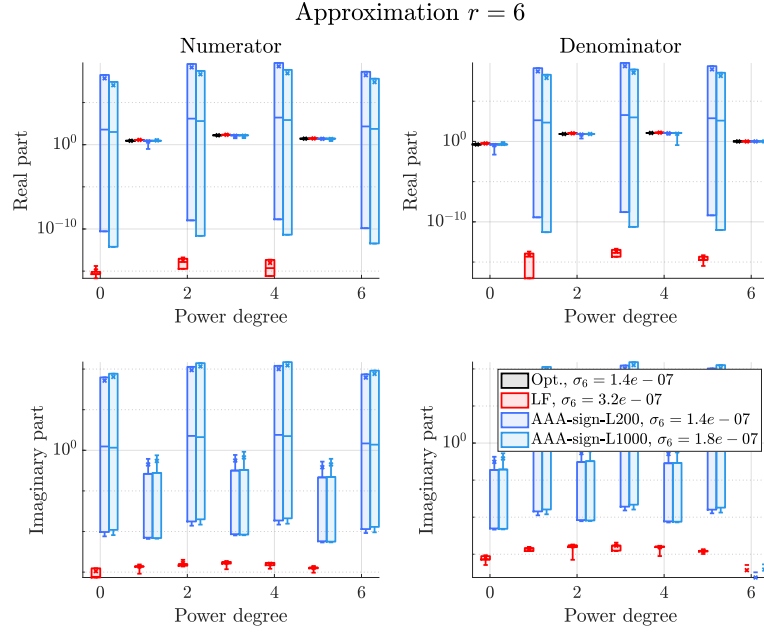


Figure 14: Case 1a, $r = 6$, Z4. Numerator and denominator dispersion over the different platforms.

A.4.2 Poles

The poles are given in the following tables. Each table gives the results obtained on different platform.

	Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$	LF, $\sigma_6 = 3.2 \cdot 10^{-07}$	AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$	AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$
1.0	$5.55 \cdot 10^{-17} + 3.23i$	$8.54 \cdot 10^{-16} - 0.243i$	$1.9 \cdot 10^{-11} - 0.232i$	$-4.84 \cdot 10^{-13} - 0.232i$
2.0	$5.55 \cdot 10^{-17} - 3.23i$	$1.13 \cdot 10^{-15} + 0.243i$	$1.96 \cdot 10^{-11} + 0.232i$	$-7.24 \cdot 10^{-14} + 0.232i$
3.0	$-1.39 \cdot 10^{-16} + 0.866i$	$1.4 \cdot 10^{-15} - 0.908i$	$3.5 \cdot 10^{-11} - 0.866i$	$-1.3 \cdot 10^{-12} - 0.866i$
4.0	$-1.39 \cdot 10^{-16} - 0.866i$	$1.03 \cdot 10^{-16} + 0.908i$	$3.7 \cdot 10^{-11} + 0.866i$	$2.56 \cdot 10^{-13} + 0.866i$
5.0	$1.21 \cdot 10^{-17} + 0.232i$	$-4.06 \cdot 10^{-16} - 3.4i$	$2.65 \cdot 10^{-10} - 3.23i$	$-7.0 \cdot 10^{-12} - 3.23i$
6.0	$1.21 \cdot 10^{-17} - 0.232i$	$3.94 \cdot 10^{-15} + 3.4i$	$2.72 \cdot 10^{-10} + 3.23i$	$-1.24 \cdot 10^{-12} + 3.23i$

Table 54: Case 1a (cpv MACA64 MR2023b), $r = 6$, Z4: poles.

	Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$	LF, $\sigma_6 = 3.2 \cdot 10^{-07}$	AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$	AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$
1.0	$1.39 \cdot 10^{-16} + 3.23i$	$7.01 \cdot 10^{-16} - 0.243i$	$1.93 \cdot 10^{-11} - 0.232i$	$2.13 \cdot 10^{-14} + 0.232i$
2.0	$1.39 \cdot 10^{-16} - 3.23i$	$3.33 \cdot 10^{-16} + 0.243i$	$1.89 \cdot 10^{-11} + 0.232i$	$5.06 \cdot 10^{-13} - 0.232i$
3.0	$-2.78 \cdot 10^{-17} + 0.866i$	$1.39 \cdot 10^{-16} - 0.908i$	$3.64 \cdot 10^{-11} - 0.866i$	$-3.95 \cdot 10^{-13} + 0.866i$
4.0	$-2.78 \cdot 10^{-17} - 0.866i$	$1.34 \cdot 10^{-15} + 0.908i$	$3.49 \cdot 10^{-11} + 0.866i$	$1.43 \cdot 10^{-12} - 0.866i$
5.0	$1.73 \cdot 10^{-18} + 0.232i$	$1.83 \cdot 10^{-15} - 3.4i$	$2.69 \cdot 10^{-10} - 3.23i$	$4.22 \cdot 10^{-13} + 3.23i$
6.0	$1.73 \cdot 10^{-18} - 0.232i$	$3.22 \cdot 10^{-16} + 3.4i$	$2.63 \cdot 10^{-10} + 3.23i$	$7.46 \cdot 10^{-12} - 3.23i$

Table 55: Case 1a (cpv MACA64 MR2022b), $r = 6$, Z4: poles.

	Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$	LF, $\sigma_6 = 3.2 \cdot 10^{-07}$	AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$	AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$
1.0	$5.55 \cdot 10^{-17} + 3.23_z$	$-2.65 \cdot 10^{-16} + 0.243_z$	$1.25 \cdot 10^{-11} - 6.27 \cdot 10^{-11}_z$	$2.52 \cdot 10^{-13} - 2.27 \cdot 10^{-11}_z$
2.0	$5.55 \cdot 10^{-17} - 3.23_z$	$2.35 \cdot 10^{-16} - 0.243_z$	$1.66 \cdot 10^{-11} + 0.5_z$	$3.42 \cdot 10^{-13} + 0.5_z$
3.0	$-1.39 \cdot 10^{-16} + 0.866_z$	$7.55 \cdot 10^{-16} + 0.908_z$	$1.68 \cdot 10^{-11} - 0.5_z$	$3.23 \cdot 10^{-13} - 0.5_z$
4.0	$-1.39 \cdot 10^{-16} - 0.866_z$	$-1.43 \cdot 10^{-16} - 0.908_z$	$4.96 \cdot 10^{-11} + 1.5_z$	$9.76 \cdot 10^{-13} + 1.5_z$
5.0	$1.21 \cdot 10^{-17} + 0.232_z$	$3.11 \cdot 10^{-17} + 3.4_z$	$5.05 \cdot 10^{-11} - 1.5_z$	$9.48 \cdot 10^{-13} - 1.5_z$
6.0	$1.21 \cdot 10^{-17} - 0.232_z$	$-2.45 \cdot 10^{-15} - 3.4_z$	$2.31 \cdot 10^9 + 1.15 \cdot 10^{10}_z$	$3.47 \cdot 10^8 + 3.3 \cdot 10^{10}_z$

Table 56: Case 1a (cpv PCWIN64 MR2025b), $r = 6$, Z4: poles.

	Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$	LF, $\sigma_6 = 3.2 \cdot 10^{-07}$	AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$	AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$
1.0	$5.55 \cdot 10^{-17} + 3.23_z$	$-2.65 \cdot 10^{-16} + 0.243_z$	$1.25 \cdot 10^{-11} - 6.27 \cdot 10^{-11}_z$	$2.52 \cdot 10^{-13} - 2.27 \cdot 10^{-11}_z$
2.0	$5.55 \cdot 10^{-17} - 3.23_z$	$2.35 \cdot 10^{-16} - 0.243_z$	$1.66 \cdot 10^{-11} + 0.5_z$	$3.42 \cdot 10^{-13} + 0.5_z$
3.0	$-1.39 \cdot 10^{-16} + 0.866_z$	$7.55 \cdot 10^{-16} + 0.908_z$	$1.68 \cdot 10^{-11} - 0.5_z$	$3.23 \cdot 10^{-13} - 0.5_z$
4.0	$-1.39 \cdot 10^{-16} - 0.866_z$	$-1.43 \cdot 10^{-16} - 0.908_z$	$4.96 \cdot 10^{-11} + 1.5_z$	$9.76 \cdot 10^{-13} + 1.5_z$
5.0	$1.21 \cdot 10^{-17} + 0.232_z$	$3.11 \cdot 10^{-17} + 3.4_z$	$5.05 \cdot 10^{-11} - 1.5_z$	$9.48 \cdot 10^{-13} - 1.5_z$
6.0	$1.21 \cdot 10^{-17} - 0.232_z$	$-2.45 \cdot 10^{-15} - 3.4_z$	$2.31 \cdot 10^9 + 1.15 \cdot 10^{10}_z$	$3.47 \cdot 10^8 + 3.3 \cdot 10^{10}_z$

Table 57: Case 1a (AlexReis PCWIN64 MR2025b), $r = 6$, Z4: poles.

	Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$	LF, $\sigma_6 = 3.2 \cdot 10^{-07}$	AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$	AAA-sign-L1000, $\sigma_6 = 1.8 \cdot 10^{-07}$
1.0	$1.39 \cdot 10^{-16} + 3.23_z$	$-3.2 \cdot 10^{-16} + 0.243_z$	$-5.48 \cdot 10^{-5} - 3.03 \cdot 10^{-5}_z$	$0.00375 - 5.94 \cdot 10^{-5}_z$
2.0	$1.39 \cdot 10^{-16} - 3.23_z$	$-1.23 \cdot 10^{-16} - 0.243_z$	$2.15 \cdot 10^{-4} + 0.5_z$	$0.0032 + 0.502_z$
3.0	$-2.78 \cdot 10^{-17} + 0.866_z$	$-9.82 \cdot 10^{-16} - 0.908_z$	$2.38 \cdot 10^{-4} - 0.5_z$	$0.00297 - 0.502_z$
4.0	$-2.78 \cdot 10^{-17} - 0.866_z$	$1.25 \cdot 10^{-16} + 0.908_z$	$0.00276 + 1.5_z$	$-0.00243 - 1.51_z$
5.0	$1.73 \cdot 10^{-18} + 0.232_z$	$-7.29 \cdot 10^{-16} - 3.4_z$	$0.00277 - 1.5_z$	$-0.00183 + 1.51_z$
6.0	$1.73 \cdot 10^{-18} - 0.232_z$	$2.36 \cdot 10^{-15} + 3.4_z$	$760.0 + 36.2_z$	$-387.0 - 25.7_z$

Table 58: Case 1a (AlexReis PCWIN64perso MR2017a), $r = 6$, Z4: poles.

A.4.3 Zeros

The zeros are given in the following tables. Each table gives the results obtained on different platform.

	Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$	LF, $\sigma_6 = 3.2 \cdot 10^{-07}$	AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$	AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$
1.0	NaN	NaN	$1.8 \cdot 10^{-11} + 2.97 \cdot 10^{-11}_z$	$-2.66 \cdot 10^{-13} + 1.23 \cdot 10^{-11}_z$
2.0	0	$1.1 \cdot 10^{-15} + 7.9 \cdot 10^{-16}_z$	$2.34 \cdot 10^{-11} - 0.5_z$	$-7.85 \cdot 10^{-13} - 0.5_z$
3.0	1.5_z	$1.13 \cdot 10^{-15} - 0.523_z$	$2.46 \cdot 10^{-11} + 0.5_z$	$1.08 \cdot 10^{-13} + 0.5_z$
4.0	-1.5_z	$6.11 \cdot 10^{-16} + 0.523_z$	$7.02 \cdot 10^{-11} - 1.5_z$	$-2.43 \cdot 10^{-12} - 1.5_z$
5.0	0.5_z	$8.46 \cdot 10^{-16} - 1.57_z$	$7.38 \cdot 10^{-11} + 1.5_z$	$2.62 \cdot 10^{-13} + 1.5_z$
6.0	-0.5_z	$5.09 \cdot 10^{-16} + 1.57_z$	$1.18 \cdot 10^{10} - 1.88 \cdot 10^{10}_z$	$-1.58 \cdot 10^9 - 6.51 \cdot 10^{10}_z$

Table 59: Case 1a (cpv MACA64 MR2023b), $r = 6$, Z4: zeros.

	Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$	LF, $\sigma_6 = 3.2 \cdot 10^{-07}$	AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$	AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$
1.0	NaN	NaN	$1.78 \cdot 10^{-11} + 8.65 \cdot 10^{-12}_z$	$2.47 \cdot 10^{-13} - 5.44 \cdot 10^{-11}_z$
2.0	0	$2.1 \cdot 10^{-16} + 1.53 \cdot 10^{-16}_z$	$2.42 \cdot 10^{-11} - 0.5_z$	$-1.87 \cdot 10^{-13} + 0.5_z$
3.0	1.5_z	$5.51 \cdot 10^{-16} - 0.523_z$	$2.33 \cdot 10^{-11} + 0.5_z$	$8.55 \cdot 10^{-13} - 0.5_z$
4.0	-1.5_z	$1.27 \cdot 10^{-15} + 0.523_z$	$7.25 \cdot 10^{-11} - 1.5_z$	$-5.55 \cdot 10^{-13} + 1.5_z$
5.0	0.5_z	$9.91 \cdot 10^{-16} - 1.57_z$	$7.0 \cdot 10^{-11} + 1.5_z$	$2.66 \cdot 10^{-12} - 1.5_z$
6.0	-0.5_z	$8.9 \cdot 10^{-16} + 1.57_z$	$3.3 \cdot 10^{10} - 1.73 \cdot 10^{10}_z$	$6.91 \cdot 10^7 + 1.4 \cdot 10^{10}_z$

Table 60: Case 1a (cpv MACA64 MR2022b), $r = 6$, Z4: zeros.

	Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$	LF, $\sigma_6 = 3.2 \cdot 10^{-07}$	AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$	AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$
1.0	NaN	NaN	$1.34 \cdot 10^{-11} + 0.232z$	$2.75 \cdot 10^{-13} + 0.232z$
2.0	0	$1.18 \cdot 10^{-16} - 5.87 \cdot 10^{-17}z$	$1.35 \cdot 10^{-11} - 0.232z$	$2.66 \cdot 10^{-13} - 0.232z$
3.0	$1.5z$	$3.18 \cdot 10^{-17} + 0.523z$	$2.48 \cdot 10^{-11} + 0.866z$	$5.0 \cdot 10^{-13} + 0.866z$
4.0	$-1.5z$	$-1.45 \cdot 10^{-17} - 0.523z$	$2.53 \cdot 10^{-11} - 0.866z$	$4.74 \cdot 10^{-13} - 0.866z$
5.0	$0.5z$	$4.28 \cdot 10^{-16} + 1.57z$	$1.86 \cdot 10^{-10} + 3.23z$	$3.6 \cdot 10^{-12} + 3.23z$
6.0	$-0.5z$	$-5.15 \cdot 10^{-16} - 1.57z$	$1.88 \cdot 10^{-10} - 3.23z$	$3.54 \cdot 10^{-12} - 3.23z$

Table 61: Case 1a (cpv PCWIN64 MR2025b), $r = 6$, Z4: zeros.

	Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$	LF, $\sigma_6 = 3.2 \cdot 10^{-07}$	AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$	AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$
1.0	NaN	NaN	$1.34 \cdot 10^{-11} + 0.232z$	$2.75 \cdot 10^{-13} + 0.232z$
2.0	0	$1.18 \cdot 10^{-16} - 5.87 \cdot 10^{-17}z$	$1.35 \cdot 10^{-11} - 0.232z$	$2.66 \cdot 10^{-13} - 0.232z$
3.0	$1.5z$	$3.18 \cdot 10^{-17} + 0.523z$	$2.48 \cdot 10^{-11} + 0.866z$	$5.0 \cdot 10^{-13} + 0.866z$
4.0	$-1.5z$	$-1.45 \cdot 10^{-17} - 0.523z$	$2.53 \cdot 10^{-11} - 0.866z$	$4.74 \cdot 10^{-13} - 0.866z$
5.0	$0.5z$	$4.28 \cdot 10^{-16} + 1.57z$	$1.86 \cdot 10^{-10} + 3.23z$	$3.6 \cdot 10^{-12} + 3.23z$
6.0	$-0.5z$	$-5.15 \cdot 10^{-16} - 1.57z$	$1.88 \cdot 10^{-10} - 3.23z$	$3.54 \cdot 10^{-12} - 3.23z$

Table 62: Case 1a (AlexReis PCWIN64 MR2025b), $r = 6$, Z4: zeros.

	Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$	LF, $\sigma_6 = 3.2 \cdot 10^{-07}$	AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$	AAA-sign-L1000, $\sigma_6 = 1.8 \cdot 10^{-07}$
1.0	NaN	NaN	$-4.25 \cdot 10^{-6} + 0.232z$	$0.00366 + 0.233z$
2.0	0	$-3.79 \cdot 10^{-17} - 8.1 \cdot 10^{-16}z$	$1.06 \cdot 10^{-5} - 0.232z$	$0.00355 - 0.233z$
3.0	$1.5z$	$-7.15 \cdot 10^{-16} - 0.523z$	$8.28 \cdot 10^{-4} + 0.866z$	$0.00158 - 0.869z$
4.0	$-1.5z$	$-3.03 \cdot 10^{-16} + 0.523z$	$8.5 \cdot 10^{-4} - 0.866z$	$0.00196 + 0.87z$
5.0	$0.5z$	$1.55 \cdot 10^{-15} + 1.57z$	$0.0135 + 3.23z$	$-0.024 - 3.24z$
6.0	$-0.5z$	$-8.33 \cdot 10^{-17} - 1.57z$	$0.0135 - 3.23z$	$-0.0228 + 3.25z$

Table 63: Case 1a (AlexReis PCWIN64perso MR2017a), $r = 6$, Z4: zeros.

A.4.4 Poles and zeros visualisation (case cpv MACA64 MR2023b)

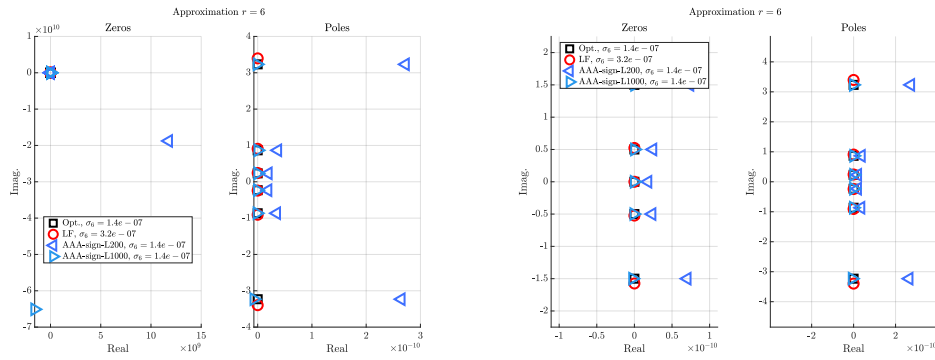


Figure 15: Case 1a (cpv MACA64 MR2023b), $r = 6$, Z4. Left frame: zeros (left) and poles (right). Right frame: same but discarding the largest real entry.

A.4.5 Approximation visualization

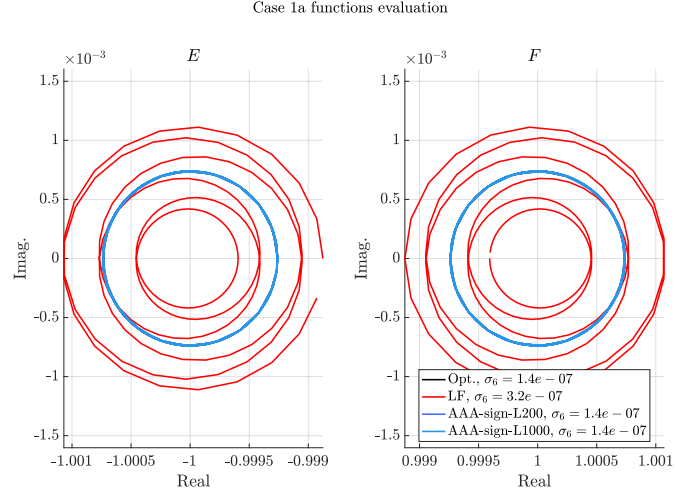


Figure 16: Case 1a (cpv MACA64 MR2023b), $r = 6$, Z4. E and F evaluations.

A.5 Approximation objective $r = 7$

A.5.1 Numerator and denominator coefficients

The numerator and denominator coefficients (both real and complex parts) are given in the following tables. Each table gives the results obtained on different platform.

Basis	Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$		LF, $\sigma_7 = 5.2 \cdot 10^{-08}$		AAA-sign-L200, $\sigma_7 = 9.8 \cdot 10^{-09}$		AAA-sign-L1000, $\sigma_7 = 9.8 \cdot 10^{-09}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^7	0	0	0	0	$-2.5 \cdot 10^6$	$-1.8 \cdot 10^8$	$-5.6 \cdot 10^5$	$-2.6 \cdot 10^8$
z^6	6.1	0	7.2	$-2.2 \cdot 10^{-14}$	6.0	$1.1 \cdot 10^{-3}$	6.0	$7.4 \cdot 10^{-4}$
z^5	0	0	$-1.5 \cdot 10^{-13}$	$-9.8 \cdot 10^{-15}$	$-4.0 \cdot 10^7$	$-2.8 \cdot 10^9$	$-8.8 \cdot 10^6$	$-4.0 \cdot 10^9$
z^4	23.0	0	36.0	$-9.6 \cdot 10^{-14}$	22.0	0.012	22.0	$8.5 \cdot 10^{-3}$
z^3	0	0	$-3.3 \cdot 10^{-13}$	$-7.5 \cdot 10^{-14}$	$-5.0 \cdot 10^7$	$-3.5 \cdot 10^9$	$-1.1 \cdot 10^7$	$-5.0 \cdot 10^9$
z^2	10.0	0	20.0	$-1.3 \cdot 10^{-13}$	10.0	$8.7 \cdot 10^{-3}$	10.0	$6.6 \cdot 10^{-3}$
z	0	0	$-5.4 \cdot 10^{-14}$	$-3.7 \cdot 10^{-14}$	$-7.5 \cdot 10^6$	$-5.2 \cdot 10^8$	$-1.6 \cdot 10^6$	$-7.6 \cdot 10^8$
1.0	0.37	0	0.85	$-1.3 \cdot 10^{-14}$	0.35	$4.3 \cdot 10^{-4}$	0.35	$3.4 \cdot 10^{-4}$
z^7	1.0	0	1.0	0	1.0	$1.7 \cdot 10^{-17}$	1.0	0
z^6	0	0	$-3.0 \cdot 10^{-14}$	$-2.0 \cdot 10^{-14}$	$-1.5 \cdot 10^7$	$-1.1 \cdot 10^9$	$-3.4 \cdot 10^6$	$-1.6 \cdot 10^9$
z^5	16.0	0	22.0	$-5.9 \cdot 10^{-14}$	16.0	$5.5 \cdot 10^{-3}$	16.0	$3.9 \cdot 10^{-3}$
z^4	0	0	$-3.0 \cdot 10^{-13}$	$-8.4 \cdot 10^{-15}$	$-5.8 \cdot 10^7$	$-4.0 \cdot 10^9$	$-1.3 \cdot 10^7$	$-5.8 \cdot 10^9$
z^3	20.0	0	35.0	$-8.8 \cdot 10^{-14}$	19.0	0.014	19.0	0.01
z^2	0	0	$-1.8 \cdot 10^{-13}$	$-7.1 \cdot 10^{-14}$	$-2.6 \cdot 10^7$	$-1.8 \cdot 10^9$	$-5.7 \cdot 10^6$	$-2.6 \cdot 10^9$
z	3.0	0	6.3	$-5.7 \cdot 10^{-14}$	2.9	$3.0 \cdot 10^{-3}$	2.9	$2.3 \cdot 10^{-3}$
1.0	0	0	$-6.8 \cdot 10^{-15}$	$-3.7 \cdot 10^{-16}$	$-9.3 \cdot 10^5$	$-6.5 \cdot 10^7$	$-2.0 \cdot 10^5$	$-9.4 \cdot 10^7$

Table 64: Case 1a (cpv MACA64 MR2023b), $r = 7$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$		LF, $\sigma_7 = 5.2 \cdot 10^{-08}$		AAA-sign-L200, $\sigma_7 = 1.1 \cdot 10^{-08}$		AAA-sign-L1000, $\sigma_7 = 9.9 \cdot 10^{-09}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^7	0	0	0	0	-93.0	0.99	$1.5 \cdot 10^{-4}$	$-7.1 \cdot 10^{-5}$
z^6	6.1	0	7.2	$1.3 \cdot 10^{-14}$	4.5	$8.6 \cdot 10^{-3}$	6.1	$-1.5 \cdot 10^{-5}$
z^5	0	0	$-2.3 \cdot 10^{-15}$	$5.2 \cdot 10^{-14}$	$-1.5 \cdot 10^3$	16.0	$-3.9 \cdot 10^{-4}$	$-1.0 \cdot 10^{-3}$
z^4	23.0	0	36.0	$9.9 \cdot 10^{-14}$	5.9	0.097	23.0	$-1.1 \cdot 10^{-4}$
z^3	0	0	$6.8 \cdot 10^{-15}$	$1.2 \cdot 10^{-13}$	$-1.8 \cdot 10^3$	19.0	$-3.7 \cdot 10^{-3}$	$-1.3 \cdot 10^{-3}$
z^2	10.0	0	20.0	$1.1 \cdot 10^{-13}$	-2.2	0.074	10.0	$-4.2 \cdot 10^{-5}$
z	0	0	$1.1 \cdot 10^{-15}$	$3.6 \cdot 10^{-14}$	-270.0	2.9	$-9.8 \cdot 10^{-4}$	$-1.9 \cdot 10^{-4}$
1.0	0.37	0	0.85	$3.8 \cdot 10^{-15}$	-0.25	$3.8 \cdot 10^{-3}$	0.37	$5.3 \cdot 10^{-7}$
z^7	1.0	0	1.0	$5.2 \cdot 10^{-17}$	1.0	0	1.0	0
z^6	0	0	$-4.1 \cdot 10^{-15}$	$9.4 \cdot 10^{-15}$	-560.0	6.0	$3.7 \cdot 10^{-4}$	$-4.1 \cdot 10^{-4}$
z^5	16.0	0	22.0	$5.5 \cdot 10^{-14}$	7.9	0.044	16.0	$-6.1 \cdot 10^{-5}$
z^4	0	0	$1.6 \cdot 10^{-14}$	$1.3 \cdot 10^{-13}$	$-2.1 \cdot 10^3$	22.0	$-2.4 \cdot 10^{-3}$	$-1.5 \cdot 10^{-3}$
z^3	20.0	0	35.0	$1.3 \cdot 10^{-13}$	0.34	0.11	20.0	$-9.6 \cdot 10^{-5}$
z^2	0	0	$1.6 \cdot 10^{-15}$	$7.5 \cdot 10^{-14}$	-940.0	10.0	$-2.7 \cdot 10^{-3}$	$-6.5 \cdot 10^{-4}$
z	3.0	0	6.3	$4.2 \cdot 10^{-14}$	-1.3	0.026	3.0	$-5.5 \cdot 10^{-6}$
1.0	0	0	$-4.0 \cdot 10^{-16}$	$4.8 \cdot 10^{-15}$	-33.0	0.36	$-1.4 \cdot 10^{-4}$	$-2.5 \cdot 10^{-5}$

Table 65: Case 1a (cpv MACA64 MR2022b), $r = 7$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$		LF, $\sigma_7 = 5.2 \cdot 10^{-08}$		AAA-sign-L200, $\sigma_7 = 9.8 \cdot 10^{-09}$		AAA-sign-L1000, $\sigma_7 = 9.8 \cdot 10^{-09}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^7	0	0	0	0	$-2.0 \cdot 10^6$	$-1.5 \cdot 10^8$	$6.3 \cdot 10^4$	$-1.7 \cdot 10^8$
z^6	6.1	0	7.2	$-1.6 \cdot 10^{-14}$	6.1	$-3.2 \cdot 10^{-5}$	6.1	$4.4 \cdot 10^{-4}$
z^5	0	0	$-4.3 \cdot 10^{-14}$	$8.7 \cdot 10^{-14}$	$-3.1 \cdot 10^7$	$-2.3 \cdot 10^9$	$9.9 \cdot 10^5$	$-2.7 \cdot 10^9$
z^4	23.0	0	36.0	$-1.1 \cdot 10^{-13}$	23.0	$-3.5 \cdot 10^{-4}$	23.0	$4.4 \cdot 10^{-3}$
z^3	0	0	$-4.7 \cdot 10^{-14}$	$2.0 \cdot 10^{-13}$	$-3.9 \cdot 10^7$	$-2.9 \cdot 10^9$	$1.2 \cdot 10^6$	$-3.4 \cdot 10^9$
z^2	10.0	0	20.0	$-7.3 \cdot 10^{-14}$	10.0	$-3.1 \cdot 10^{-4}$	10.0	$3.0 \cdot 10^{-3}$
z	0	0	$-2.4 \cdot 10^{-15}$	$4.0 \cdot 10^{-14}$	$-5.8 \cdot 10^6$	$-4.4 \cdot 10^8$	$1.9 \cdot 10^5$	$-5.1 \cdot 10^8$
1.0	0.37	0	0.85	$-9.5 \cdot 10^{-15}$	0.37	$-1.6 \cdot 10^{-5}$	0.37	$1.3 \cdot 10^{-4}$
z^7	1.0	0	1.0	0	1.0	0	1.0	0
z^6	0	0	$-1.0 \cdot 10^{-14}$	$1.4 \cdot 10^{-14}$	$-1.2 \cdot 10^7$	$-9.0 \cdot 10^8$	$3.8 \cdot 10^5$	$-1.0 \cdot 10^9$
z^5	16.0	0	22.0	$-4.1 \cdot 10^{-14}$	16.0	$-1.7 \cdot 10^{-4}$	16.0	$2.1 \cdot 10^{-3}$
z^4	0	0	$-7.1 \cdot 10^{-14}$	$2.0 \cdot 10^{-13}$	$-4.5 \cdot 10^7$	$-3.4 \cdot 10^9$	$1.4 \cdot 10^6$	$-3.9 \cdot 10^9$
z^3	20.0	0	35.0	$-7.1 \cdot 10^{-14}$	20.0	$-4.2 \cdot 10^{-4}$	20.0	$4.9 \cdot 10^{-3}$
z^2	0	0	$-1.4 \cdot 10^{-14}$	$1.2 \cdot 10^{-13}$	$-2.0 \cdot 10^7$	$-1.5 \cdot 10^9$	$6.4 \cdot 10^5$	$-1.7 \cdot 10^9$
z	3.0	0	6.3	$-2.6 \cdot 10^{-14}$	3.0	$-1.1 \cdot 10^{-4}$	3.0	$9.7 \cdot 10^{-4}$
1.0	0	0	$1.0 \cdot 10^{-15}$	$7.7 \cdot 10^{-15}$	$-7.2 \cdot 10^5$	$-5.4 \cdot 10^7$	$2.3 \cdot 10^4$	$-6.2 \cdot 10^7$

Table 66: Case 1a (cpv PCWIN64 MR2025b), $r = 7$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$		LF, $\sigma_7 = 5.2 \cdot 10^{-08}$		AAA-sign-L200, $\sigma_7 = 9.8 \cdot 10^{-09}$		AAA-sign-L1000, $\sigma_7 = 9.8 \cdot 10^{-09}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^7	0	0	0	0	$-2.0 \cdot 10^6$	$-1.5 \cdot 10^8$	$6.3 \cdot 10^4$	$-1.7 \cdot 10^8$
z^6	6.1	0	7.2	$-1.6 \cdot 10^{-14}$	6.1	$-3.2 \cdot 10^{-5}$	6.1	$4.4 \cdot 10^{-4}$
z^5	0	0	$-4.3 \cdot 10^{-14}$	$8.7 \cdot 10^{-14}$	$-3.1 \cdot 10^7$	$-2.3 \cdot 10^9$	$9.9 \cdot 10^5$	$-2.7 \cdot 10^9$
z^4	23.0	0	36.0	$-1.1 \cdot 10^{-13}$	23.0	$-3.5 \cdot 10^{-4}$	23.0	$4.4 \cdot 10^{-3}$
z^3	0	0	$-4.7 \cdot 10^{-14}$	$2.0 \cdot 10^{-13}$	$-3.9 \cdot 10^7$	$-2.9 \cdot 10^9$	$1.2 \cdot 10^6$	$-3.4 \cdot 10^9$
z^2	10.0	0	20.0	$-7.3 \cdot 10^{-14}$	10.0	$-3.1 \cdot 10^{-4}$	10.0	$3.0 \cdot 10^{-3}$
z	0	0	$-2.4 \cdot 10^{-15}$	$4.0 \cdot 10^{-14}$	$-5.8 \cdot 10^6$	$-4.4 \cdot 10^8$	$1.9 \cdot 10^5$	$-5.1 \cdot 10^8$
1.0	0.37	0	0.85	$-9.5 \cdot 10^{-15}$	0.37	$-1.6 \cdot 10^{-5}$	0.37	$1.3 \cdot 10^{-4}$
z^7	1.0	0	1.0	0	1.0	0	1.0	0
z^6	0	0	$-1.0 \cdot 10^{-14}$	$1.4 \cdot 10^{-14}$	$-1.2 \cdot 10^7$	$-9.0 \cdot 10^8$	$3.8 \cdot 10^5$	$-1.0 \cdot 10^9$
z^5	16.0	0	22.0	$-4.1 \cdot 10^{-14}$	16.0	$-1.7 \cdot 10^{-4}$	16.0	$2.1 \cdot 10^{-3}$
z^4	0	0	$-7.1 \cdot 10^{-14}$	$2.0 \cdot 10^{-13}$	$-4.5 \cdot 10^7$	$-3.4 \cdot 10^9$	$1.4 \cdot 10^6$	$-3.9 \cdot 10^9$
z^3	20.0	0	35.0	$-7.1 \cdot 10^{-14}$	20.0	$-4.2 \cdot 10^{-4}$	20.0	$4.9 \cdot 10^{-3}$
z^2	0	0	$-1.4 \cdot 10^{-14}$	$1.2 \cdot 10^{-13}$	$-2.0 \cdot 10^7$	$-1.5 \cdot 10^9$	$6.4 \cdot 10^5$	$-1.7 \cdot 10^9$
z	3.0	0	6.3	$-2.6 \cdot 10^{-14}$	3.0	$-1.1 \cdot 10^{-4}$	3.0	$9.7 \cdot 10^{-4}$
1.0	0	0	$1.0 \cdot 10^{-15}$	$7.7 \cdot 10^{-15}$	$-7.2 \cdot 10^5$	$-5.4 \cdot 10^7$	$2.3 \cdot 10^4$	$-6.2 \cdot 10^7$

Table 67: Case 1a (AlexReis PCWIN64 MR2025b), $r = 7$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$		LF, $\sigma_7 = 5.2 \cdot 10^{-08}$		AAA-sign-L200, $\sigma_7 = 9.8 \cdot 10^{-09}$		AAA-sign-L1000, $\sigma_7 = 9.8 \cdot 10^{-09}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^7	0	0	0	0	$-1.9 \cdot 10^7$	$-4.9 \cdot 10^8$	$2.1 \cdot 10^5$	$-1.6 \cdot 10^8$
z^6	6.1	0	7.2	$-2.2 \cdot 10^{-15}$	6.1	$5.2 \cdot 10^{-5}$	6.1	$2.4 \cdot 10^{-4}$
z^5	0	0	$-8.6 \cdot 10^{-14}$	$-4.8 \cdot 10^{-14}$	$-2.9 \cdot 10^8$	$-7.7 \cdot 10^9$	$3.3 \cdot 10^6$	$-2.5 \cdot 10^9$
z^4	23.0	0	36.0	$-1.9 \cdot 10^{-14}$	23.0	$-3.0 \cdot 10^{-4}$	23.0	$3.2 \cdot 10^{-3}$
z^3	0	0	$-1.5 \cdot 10^{-13}$	$-4.2 \cdot 10^{-14}$	$-3.7 \cdot 10^8$	$-9.7 \cdot 10^9$	$4.1 \cdot 10^6$	$-3.1 \cdot 10^9$
z^2	10.0	0	20.0	$1.1 \cdot 10^{-14}$	10.0	$-3.5 \cdot 10^{-4}$	10.0	$2.6 \cdot 10^{-3}$
z	0	0	$-3.5 \cdot 10^{-14}$	$-2.4 \cdot 10^{-15}$	$-5.5 \cdot 10^7$	$-1.5 \cdot 10^9$	$6.1 \cdot 10^5$	$-4.7 \cdot 10^8$
1.0	0.37	0	0.85	$5.8 \cdot 10^{-16}$	0.37	$-1.1 \cdot 10^{-5}$	0.37	$1.3 \cdot 10^{-4}$
z^7	1.0	0	1.0	0	1.0	0	1.0	0
z^6	0	0	$-2.0 \cdot 10^{-14}$	$-9.7 \cdot 10^{-15}$	$-1.1 \cdot 10^8$	$-3.0 \cdot 10^9$	$1.3 \cdot 10^6$	$-9.7 \cdot 10^8$
z^5	16.0	0	22.0	$-1.6 \cdot 10^{-15}$	16.0	$-5.8 \cdot 10^{-6}$	16.0	$1.4 \cdot 10^{-3}$
z^4	0	0	$-1.4 \cdot 10^{-13}$	$-6.1 \cdot 10^{-14}$	$-4.2 \cdot 10^8$	$-1.1 \cdot 10^{10}$	$4.7 \cdot 10^6$	$-3.6 \cdot 10^9$
z^3	20.0	0	35.0	$4.0 \cdot 10^{-14}$	20.0	$-4.9 \cdot 10^{-4}$	20.0	$3.8 \cdot 10^{-3}$
z^2	0	0	$-8.4 \cdot 10^{-14}$	$-1.6 \cdot 10^{-14}$	$-1.9 \cdot 10^8$	$-5.0 \cdot 10^9$	$2.1 \cdot 10^6$	$-1.6 \cdot 10^9$
z	3.0	0	6.3	$1.8 \cdot 10^{-14}$	3.0	$-1.1 \cdot 10^{-4}$	3.0	$9.0 \cdot 10^{-4}$
1.0	0	0	$-3.7 \cdot 10^{-15}$	$-1.7 \cdot 10^{-15}$	$-6.8 \cdot 10^6$	$-1.8 \cdot 10^8$	$7.6 \cdot 10^4$	$-5.8 \cdot 10^7$

Table 68: Case 1a (AlexReis PCWIN64perso MR2017a), $r = 7$, Z4: numerator (first lines) and denominator (last lines) coefficients.

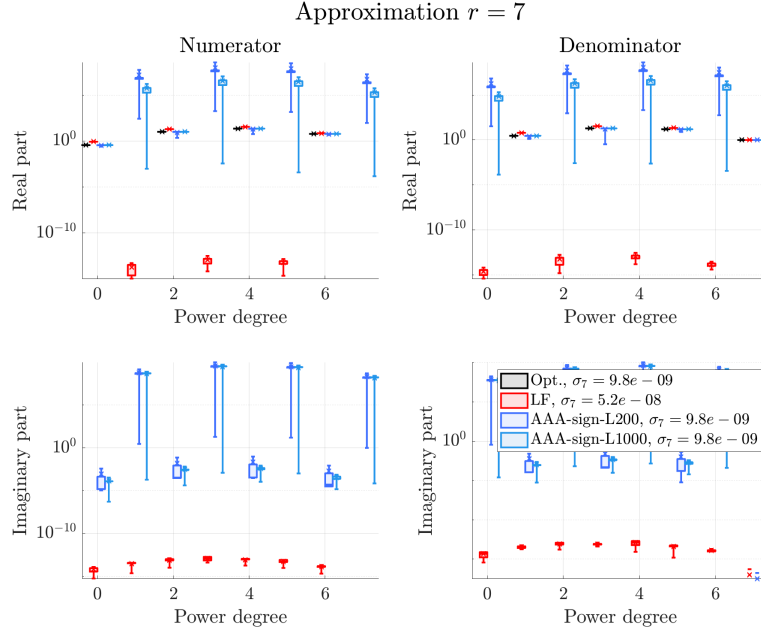


Figure 17: Case 1a, $r = 7$, Z4. Numerator and denominator dispersion over the different platforms.

A.5.2 Poles

The poles are given in the following tables. Each table gives the results obtained on different platform.

	Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$	LF, $\sigma_7 = 5.2 \cdot 10^{-08}$	AAA-sign-L200, $\sigma_7 = 9.8 \cdot 10^{-09}$	AAA-sign-L1000, $\sigma_7 = 9.8 \cdot 10^{-09}$
1.0	0	$1.08 \cdot 10^{-15} + 5.86 \cdot 10^{-17}i$	$1.63 \cdot 10^{-11} + 0.198i$	$5.75 \cdot 10^{-12} + 0.198i$
2.0	$-7.63 \cdot 10^{-17} + 3.79i$	$-1.46 \cdot 10^{-16} - 0.458i$	$5.67 \cdot 10^{-12} - 0.198i$	$-2.73 \cdot 10^{-12} - 0.198i$
3.0	$-7.63 \cdot 10^{-17} - 3.79i$	$3.4 \cdot 10^{-15} + 0.458i$	$3.53 \cdot 10^{-11} + 0.691i$	$1.68 \cdot 10^{-11} + 0.691i$
4.0	$2.78 \cdot 10^{-17} + 1.09i$	$3.32 \cdot 10^{-15} + 1.23i$	$-1.59 \cdot 10^{-12} - 0.691i$	$-1.26 \cdot 10^{-11} - 0.691i$
5.0	$2.78 \cdot 10^{-17} - 1.09i$	$5.36 \cdot 10^{-15} - 1.23i$	$1.01 \cdot 10^{-10} + 1.8i$	$4.36 \cdot 10^{-11} + 1.8i$
6.0	$7.63 \cdot 10^{-17} + 0.417i$	$1.82 \cdot 10^{-15} - 4.47i$	$6.14 \cdot 10^{-12} - 1.8i$	$-3.18 \cdot 10^{-11} - 1.8i$
7.0	$7.63 \cdot 10^{-17} - 0.417i$	$1.53 \cdot 10^{-14} + 4.47i$	$1.54 \cdot 10^7 + 1.07 \cdot 10^9i$	$3.38 \cdot 10^6 + 1.55 \cdot 10^9i$

Table 69: Case 1a (cpv MACA64 MR2023b), $r = 7$, Z4: poles.

	Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$	LF, $\sigma_7 = 5.2 \cdot 10^{-08}$	AAA-sign-L200, $\sigma_7 = 1.1 \cdot 10^{-08}$	AAA-sign-L1000, $\sigma_7 = 9.9 \cdot 10^{-09}$
1.0	0	$6.4 \cdot 10^{-17} - 7.54 \cdot 10^{-16}i$	$-8.57 \cdot 10^{-4} - 0.197i$	$4.89 \cdot 10^{-5} + 8.52 \cdot 10^{-6}i$
2.0	$-1.46 \cdot 10^{-16} + 3.79i$	$-7.57 \cdot 10^{-16} + 0.458i$	$-8.53 \cdot 10^{-4} + 0.197i$	$4.97 \cdot 10^{-5} - 0.417i$
3.0	$-1.46 \cdot 10^{-16} - 3.79i$	$6.74 \cdot 10^{-16} - 0.458i$	$-1.38 \cdot 10^{-4} - 0.689i$	$4.82 \cdot 10^{-5} + 0.417i$
4.0	$2.78 \cdot 10^{-17} + 1.09i$	$-7.92 \cdot 10^{-16} + 1.23i$	$-1.19 \cdot 10^{-4} + 0.689i$	$3.1 \cdot 10^{-5} - 1.09i$
5.0	$2.78 \cdot 10^{-17} - 1.09i$	$-2.55 \cdot 10^{-16} - 1.23i$	$0.00463 - 1.79i$	$3.26 \cdot 10^{-5} + 1.09i$
6.0	$-3.47 \cdot 10^{-17} + 0.417i$	$-4.93 \cdot 10^{-15} + 4.47i$	$0.0047 + 1.79i$	$-2.98 \cdot 10^{-4} - 3.79i$
7.0	$-3.47 \cdot 10^{-17} - 0.417i$	$1.15 \cdot 10^{-14} - 4.47i$	$565.0 - 6.0i$	$-2.83 \cdot 10^{-4} + 3.8i$

Table 70: Case 1a (cpv MACA64 MR2022b), $r = 7$, Z4: poles.

	Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$	LF, $\sigma_7 = 5.2 \cdot 10^{-08}$	AAA-sign-L200, $\sigma_7 = 9.8 \cdot 10^{-09}$	AAA-sign-L1000, $\sigma_7 = 9.8 \cdot 10^{-09}$
1.0	0	$-1.61 \cdot 10^{-16} - 1.23 \cdot 10^{-15}z$	$1.16 \cdot 10^{-11} + 0.198z$	$-2.97 \cdot 10^{-12} + 0.198z$
2.0	$-7.63 \cdot 10^{-17} + 3.79z$	$6.42 \cdot 10^{-16} + 0.458z$	$1.16 \cdot 10^{-11} - 0.198z$	$2.94 \cdot 10^{-12} - 0.198z$
3.0	$-7.63 \cdot 10^{-17} - 3.79z$	$-4.86 \cdot 10^{-16} - 0.458z$	$1.81 \cdot 10^{-11} + 0.691z$	$-1.0 \cdot 10^{-11} + 0.691z$
4.0	$2.78 \cdot 10^{-17} + 1.09z$	$1.24 \cdot 10^{-15} + 1.23z$	$1.81 \cdot 10^{-11} - 0.691z$	$9.84 \cdot 10^{-12} - 0.691z$
5.0	$2.78 \cdot 10^{-17} - 1.09z$	$1.68 \cdot 10^{-15} - 1.23z$	$5.9 \cdot 10^{-11} + 1.8z$	$-2.56 \cdot 10^{-11} + 1.8z$
6.0	$7.63 \cdot 10^{-17} + 0.417z$	$8.32 \cdot 10^{-15} + 4.47z$	$5.86 \cdot 10^{-11} - 1.8z$	$2.36 \cdot 10^{-11} - 1.8z$
7.0	$7.63 \cdot 10^{-17} - 0.417z$	$-9.17 \cdot 10^{-16} - 4.47z$	$1.2 \cdot 10^7 + 9.01 \cdot 10^8z$	$-3.8 \cdot 10^5 + 1.04 \cdot 10^9z$

Table 71: Case 1a (cpv PCWIN64 MR2025b), $r = 7$, Z4: poles.

	Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$	LF, $\sigma_7 = 5.2 \cdot 10^{-08}$	AAA-sign-L200, $\sigma_7 = 9.8 \cdot 10^{-09}$	AAA-sign-L1000, $\sigma_7 = 9.8 \cdot 10^{-09}$
1.0	0	$-1.61 \cdot 10^{-16} - 1.23 \cdot 10^{-15}z$	$1.16 \cdot 10^{-11} + 0.198z$	$-2.97 \cdot 10^{-12} + 0.198z$
2.0	$-7.63 \cdot 10^{-17} + 3.79z$	$6.42 \cdot 10^{-16} + 0.458z$	$1.16 \cdot 10^{-11} - 0.198z$	$2.94 \cdot 10^{-12} - 0.198z$
3.0	$-7.63 \cdot 10^{-17} - 3.79z$	$-4.86 \cdot 10^{-16} - 0.458z$	$1.81 \cdot 10^{-11} + 0.691z$	$-1.0 \cdot 10^{-11} + 0.691z$
4.0	$2.78 \cdot 10^{-17} + 1.09z$	$1.24 \cdot 10^{-15} + 1.23z$	$1.81 \cdot 10^{-11} - 0.691z$	$9.84 \cdot 10^{-12} - 0.691z$
5.0	$2.78 \cdot 10^{-17} - 1.09z$	$1.68 \cdot 10^{-15} - 1.23z$	$5.9 \cdot 10^{-11} + 1.8z$	$-2.56 \cdot 10^{-11} + 1.8z$
6.0	$7.63 \cdot 10^{-17} + 0.417z$	$8.32 \cdot 10^{-15} + 4.47z$	$5.86 \cdot 10^{-11} - 1.8z$	$2.36 \cdot 10^{-11} - 1.8z$
7.0	$7.63 \cdot 10^{-17} - 0.417z$	$-9.17 \cdot 10^{-16} - 4.47z$	$1.2 \cdot 10^7 + 9.01 \cdot 10^8z$	$-3.8 \cdot 10^5 + 1.04 \cdot 10^9z$

Table 72: Case 1a (AlexReis PCWIN64 MR2025b), $r = 7$, Z4: poles.

	Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$	LF, $\sigma_7 = 5.2 \cdot 10^{-08}$	AAA-sign-L200, $\sigma_7 = 9.8 \cdot 10^{-09}$	AAA-sign-L1000, $\sigma_7 = 9.8 \cdot 10^{-09}$
1.0	0	$5.88 \cdot 10^{-16} + 2.7 \cdot 10^{-16}z$	$9.66 \cdot 10^{-12} + 0.198z$	$-3.3 \cdot 10^{-12} + 0.198z$
2.0	$-1.46 \cdot 10^{-16} + 3.79z$	$3.26 \cdot 10^{-16} + 0.458z$	$1.05 \cdot 10^{-11} - 0.198z$	$1.72 \cdot 10^{-12} - 0.198z$
3.0	$-1.46 \cdot 10^{-16} - 3.79z$	$1.1 \cdot 10^{-15} - 0.458z$	$1.43 \cdot 10^{-11} + 0.691z$	$-9.82 \cdot 10^{-12} + 0.691z$
4.0	$2.78 \cdot 10^{-17} + 1.09z$	$2.71 \cdot 10^{-15} - 1.23z$	$1.69 \cdot 10^{-11} - 0.691z$	$7.05 \cdot 10^{-12} - 0.691z$
5.0	$2.78 \cdot 10^{-17} - 1.09z$	$1.12 \cdot 10^{-15} + 1.23z$	$4.72 \cdot 10^{-11} + 1.8z$	$-2.61 \cdot 10^{-11} + 1.8z$
6.0	$-3.47 \cdot 10^{-17} + 0.417z$	$6.39 \cdot 10^{-15} - 4.47z$	$5.39 \cdot 10^{-11} - 1.8z$	$1.58 \cdot 10^{-11} - 1.8z$
7.0	$-3.47 \cdot 10^{-17} - 0.417z$	$7.26 \cdot 10^{-15} + 4.47z$	$1.13 \cdot 10^8 + 2.98 \cdot 10^9z$	$-1.26 \cdot 10^6 + 9.7 \cdot 10^8z$

Table 73: Case 1a (AlexReis PCWIN64perso MR2017a), $r = 7$, Z4: poles.

A.5.3 Zeros

The zeros are given in the following tables. Each table gives the results obtained on different platform.

	Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$	LF, $\sigma_7 = 5.2 \cdot 10^{-08}$	AAA-sign-L200, $\sigma_7 = 9.8 \cdot 10^{-09}$	AAA-sign-L1000, $\sigma_7 = 9.8 \cdot 10^{-09}$
1.0	NaN	NaN	$1.05 \cdot 10^{-11} - 6.74 \cdot 10^{-10}z$	$1.46 \cdot 10^{-12} - 4.66 \cdot 10^{-10}z$
2.0	$1.73 \cdot 10^{-17} + 1.8z$	$2.11 \cdot 10^{-16} - 0.216z$	$2.4 \cdot 10^{-11} + 0.417z$	$1.06 \cdot 10^{-11} + 0.417z$
3.0	$1.73 \cdot 10^{-17} - 1.8z$	$2.13 \cdot 10^{-15} + 0.216z$	$1.61 \cdot 10^{-12} - 0.417z$	$-7.23 \cdot 10^{-12} - 0.417z$
4.0	$5.55 \cdot 10^{-17} + 0.691z$	$1.4 \cdot 10^{-15} - 0.768z$	$5.5 \cdot 10^{-11} + 1.09z$	$2.6 \cdot 10^{-11} + 1.09z$
5.0	$5.55 \cdot 10^{-17} - 0.691z$	$4.64 \cdot 10^{-15} + 0.768z$	$-2.6 \cdot 10^{-12} - 1.09z$	$-1.99 \cdot 10^{-11} - 1.09z$
6.0	$1.21 \cdot 10^{-17} + 0.198z$	$5.1 \cdot 10^{-15} + 2.07z$	$3.02 \cdot 10^{-10} + 3.79z$	$1.01 \cdot 10^{-10} + 3.79z$
7.0	$1.21 \cdot 10^{-17} - 0.198z$	$7.22 \cdot 10^{-15} - 2.07z$	$1.02 \cdot 10^{-10} - 3.79z$	$-5.76 \cdot 10^{-11} - 3.79z$

Table 74: Case 1a (cpv MACA64 MR2023b), $r = 7$, Z4: zeros.

	Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$	LF, $\sigma_7 = 5.2 \cdot 10^{-08}$	AAA-sign-L200, $\sigma_7 = 1.1 \cdot 10^{-08}$	AAA-sign-L1000, $\sigma_7 = 9.9 \cdot 10^{-09}$
1.0	NaN	NaN	$-9.17 \cdot 10^{-4} + 4.3 \cdot 10^{-6}z$	$4.98 \cdot 10^{-5} - 0.198z$
2.0	$-3.12 \cdot 10^{-17} + 1.8z$	$2.07 \cdot 10^{-16} + 0.216z$	$-6.4 \cdot 10^{-4} - 0.416z$	$4.86 \cdot 10^{-5} + 0.198z$
3.0	$-3.12 \cdot 10^{-17} - 1.8z$	$-1.59 \cdot 10^{-16} - 0.216z$	$-6.31 \cdot 10^{-4} + 0.416z$	$4.58 \cdot 10^{-5} - 0.691z$
4.0	$4.16 \cdot 10^{-17} + 0.691z$	$-1.16 \cdot 10^{-15} + 0.768z$	$0.00106 - 1.08z$	$4.52 \cdot 10^{-5} + 0.691z$
5.0	$4.16 \cdot 10^{-17} - 0.691z$	$1.19 \cdot 10^{-15} - 0.768z$	$0.0011 + 1.08z$	$-1.89 \cdot 10^{-5} - 1.8z$
6.0	$5.2 \cdot 10^{-18} + 0.198z$	$-1.59 \cdot 10^{-16} + 2.07z$	$0.0242 - 3.78z$	$-1.35 \cdot 10^{-5} + 1.8z$
7.0	$5.2 \cdot 10^{-18} - 0.198z$	$8.43 \cdot 10^{-16} - 2.07z$	$0.0244 + 3.78z$	$-3.31 \cdot 10^4 - 1.58 \cdot 10^4z$

Table 75: Case 1a (cpv MACA64 MR2022b), $r = 7$, Z4: zeros.

	Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$	LF, $\sigma_7 = 5.2 \cdot 10^{-08}$	AAA-sign-L200, $\sigma_7 = 9.8 \cdot 10^{-09}$	AAA-sign-L1000, $\sigma_7 = 9.8 \cdot 10^{-09}$
1.0	NaN	NaN	$1.11 \cdot 10^{-11} - 8.34 \cdot 10^{-10}i$	$-2.09 \cdot 10^{-14} - 7.35 \cdot 10^{-10}i$
2.0	$1.73 \cdot 10^{-17} + 1.8i$	$8.87 \cdot 10^{-16} + 0.216i$	$1.36 \cdot 10^{-11} + 0.417i$	$-6.16 \cdot 10^{-12} + 0.417i$
3.0	$1.73 \cdot 10^{-17} - 1.8i$	$-8.68 \cdot 10^{-16} - 0.216i$	$1.36 \cdot 10^{-11} - 0.417i$	$6.12 \cdot 10^{-12} - 0.417i$
4.0	$5.55 \cdot 10^{-17} + 0.691i$	$2.39 \cdot 10^{-16} + 0.768i$	$2.86 \cdot 10^{-11} + 1.09i$	$-1.55 \cdot 10^{-11} + 1.09i$
5.0	$5.55 \cdot 10^{-17} - 0.691i$	$4.76 \cdot 10^{-16} - 0.768i$	$2.84 \cdot 10^{-11} - 1.09i$	$1.49 \cdot 10^{-11} - 1.09i$
6.0	$1.21 \cdot 10^{-17} + 0.198i$	$3.76 \cdot 10^{-15} + 2.07i$	$2.24 \cdot 10^{-10} + 3.79i$	$-5.59 \cdot 10^{-11} + 3.79i$
7.0	$1.21 \cdot 10^{-17} - 0.198i$	$1.55 \cdot 10^{-15} - 2.07i$	$2.23 \cdot 10^{-10} - 3.79i$	$4.61 \cdot 10^{-11} - 3.79i$

Table 76: Case 1a (cpv PCWIN64 MR2025b), $r = 7$, Z4: zeros.

	Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$	LF, $\sigma_7 = 5.2 \cdot 10^{-08}$	AAA-sign-L200, $\sigma_7 = 9.8 \cdot 10^{-09}$	AAA-sign-L1000, $\sigma_7 = 9.8 \cdot 10^{-09}$
1.0	NaN	NaN	$1.11 \cdot 10^{-11} - 8.34 \cdot 10^{-10}i$	$-2.09 \cdot 10^{-14} - 7.35 \cdot 10^{-10}i$
2.0	$1.73 \cdot 10^{-17} + 1.8i$	$8.87 \cdot 10^{-16} + 0.216i$	$1.36 \cdot 10^{-11} + 0.417i$	$-6.16 \cdot 10^{-12} + 0.417i$
3.0	$1.73 \cdot 10^{-17} - 1.8i$	$-8.68 \cdot 10^{-16} - 0.216i$	$1.36 \cdot 10^{-11} - 0.417i$	$6.12 \cdot 10^{-12} - 0.417i$
4.0	$5.55 \cdot 10^{-17} + 0.691i$	$2.39 \cdot 10^{-16} + 0.768i$	$2.86 \cdot 10^{-11} + 1.09i$	$-1.55 \cdot 10^{-11} + 1.09i$
5.0	$5.55 \cdot 10^{-17} - 0.691i$	$4.76 \cdot 10^{-16} - 0.768i$	$2.84 \cdot 10^{-11} - 1.09i$	$1.49 \cdot 10^{-11} - 1.09i$
6.0	$1.21 \cdot 10^{-17} + 0.198i$	$3.76 \cdot 10^{-15} + 2.07i$	$2.24 \cdot 10^{-10} + 3.79i$	$-5.59 \cdot 10^{-11} + 3.79i$
7.0	$1.21 \cdot 10^{-17} - 0.198i$	$1.55 \cdot 10^{-15} - 2.07i$	$2.23 \cdot 10^{-10} - 3.79i$	$4.61 \cdot 10^{-11} - 3.79i$

Table 77: Case 1a (AlexReis PCWIN64 MR2025b), $r = 7$, Z4: zeros.

	Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$	LF, $\sigma_7 = 5.2 \cdot 10^{-08}$	AAA-sign-L200, $\sigma_7 = 9.8 \cdot 10^{-09}$	AAA-sign-L1000, $\sigma_7 = 9.8 \cdot 10^{-09}$
1.0	NaN	NaN	$9.56 \cdot 10^{-12} - 2.53 \cdot 10^{-10}i$	$-7.45 \cdot 10^{-13} - 7.84 \cdot 10^{-10}i$
2.0	$-3.12 \cdot 10^{-17} + 1.8i$	$8.38 \cdot 10^{-16} - 0.216i$	$1.09 \cdot 10^{-11} + 0.417i$	$-6.18 \cdot 10^{-12} + 0.417i$
3.0	$-3.12 \cdot 10^{-17} - 1.8i$	$8.24 \cdot 10^{-16} + 0.216i$	$1.26 \cdot 10^{-11} - 0.417i$	$4.24 \cdot 10^{-12} - 0.417i$
4.0	$4.16 \cdot 10^{-17} + 0.691i$	$1.45 \cdot 10^{-15} - 0.768i$	$2.24 \cdot 10^{-11} + 1.09i$	$-1.53 \cdot 10^{-11} + 1.09i$
5.0	$4.16 \cdot 10^{-17} - 0.691i$	$3.88 \cdot 10^{-16} + 0.768i$	$2.65 \cdot 10^{-11} - 1.09i$	$1.06 \cdot 10^{-11} - 1.09i$
6.0	$5.2 \cdot 10^{-18} + 0.198i$	$3.77 \cdot 10^{-15} - 2.07i$	$1.85 \cdot 10^{-10} + 3.79i$	$-6.37 \cdot 10^{-11} + 3.79i$
7.0	$5.2 \cdot 10^{-18} - 0.198i$	$4.72 \cdot 10^{-15} + 2.07i$	$1.99 \cdot 10^{-10} - 3.79i$	$2.35 \cdot 10^{-11} - 3.79i$

Table 78: Case 1a (AlexReis PCWIN64perso MR2017a), $r = 7$, Z4: zeros.

A.5.4 Poles and zeros visualisation (case cpv MACA64 MR2023b)

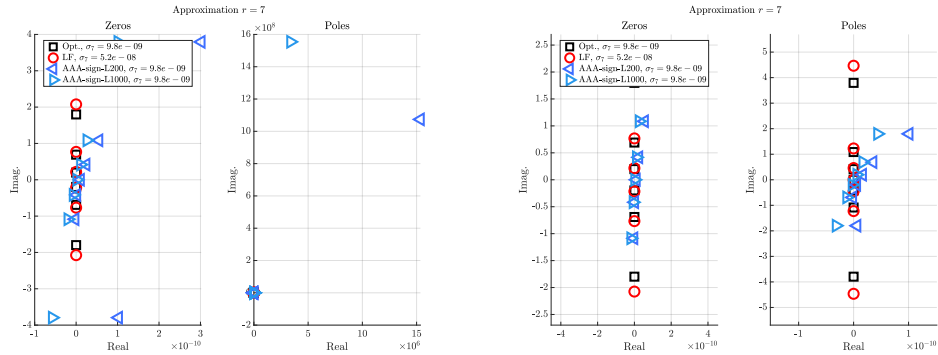


Figure 18: Case 1a (cpv MACA64 MR2023b), $r = 7$, Z4. Left frame: zeros (left) and poles (right). Right frame: same but discarding the largest real entry.

A.5.5 Approximation visualization

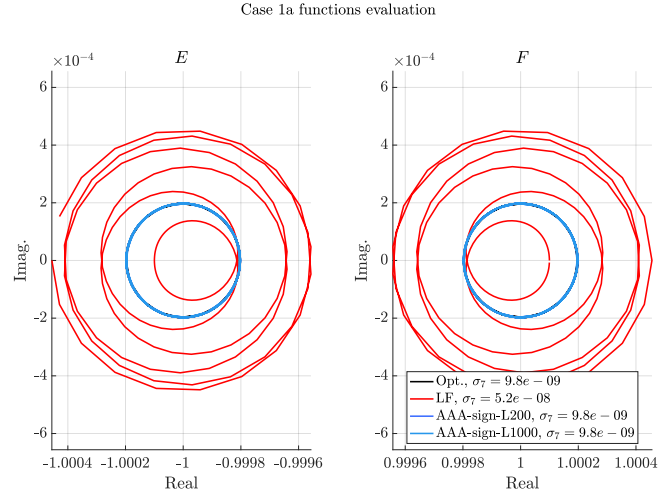


Figure 19: Case 1a (cpv MACA64 MR2023b), $r = 7$, Z4. E and F evaluations.

A.6 Approximation objective $r = 8$

A.6.1 Numerator and denominator coefficients

The numerator and denominator coefficients (both real and complex parts) are given in the following tables. Each table gives the results obtained on different platform.

Basis	Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$		LF, $\sigma_8 = 1.6 \cdot 10^{-09}$		AAA-sign-L200, $\sigma_8 = 7.1 \cdot 10^{-10}$		AAA-sign-L1000, $\sigma_8 = 7.1 \cdot 10^{-10}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^8	0	0	0	0	$-8.7 \cdot 10^5$	$1.7 \cdot 10^8$	$-5.1 \cdot 10^6$	$-7.9 \cdot 10^8$
z^7	6.9	0	7.2	$-1.8 \cdot 10^{-14}$	7.0	$-4.9 \cdot 10^{-4}$	7.0	$-3.4 \cdot 10^{-3}$
z^6	0	0	$-5.1 \cdot 10^{-14}$	$-1.6 \cdot 10^{-13}$	$-1.8 \cdot 10^7$	$3.7 \cdot 10^9$	$-1.1 \cdot 10^8$	$-1.7 \cdot 10^{10}$
z^5	36.0	0	41.0	$-6.7 \cdot 10^{-14}$	37.0	$-9.8 \cdot 10^{-3}$	37.0	-0.046
z^4	0	0	$8.0 \cdot 10^{-14}$	$-3.5 \cdot 10^{-13}$	$-3.4 \cdot 10^7$	$6.9 \cdot 10^9$	$-2.0 \cdot 10^8$	$-3.1 \cdot 10^{10}$
z^3	27.0	0	33.0	$-1.8 \cdot 10^{-14}$	28.0	-0.014	28.0	-0.051
z^2	0	0	$1.6 \cdot 10^{-13}$	$-8.0 \cdot 10^{-14}$	$-1.0 \cdot 10^7$	$2.1 \cdot 10^9$	$-6.0 \cdot 10^7$	$-9.3 \cdot 10^9$
z	2.9	0	3.7	$3.2 \cdot 10^{-14}$	3.0	$-2.4 \cdot 10^{-3}$	3.1	$-7.0 \cdot 10^{-3}$
1.0	0	0	$1.0 \cdot 10^{-14}$	$1.0 \cdot 10^{-14}$	$-2.8 \cdot 10^5$	$5.5 \cdot 10^7$	$-1.6 \cdot 10^6$	$-2.5 \cdot 10^8$
z^8	1.0	0	1.0	0	1.0	0	1.0	0
z^7	0	0	$-1.4 \cdot 10^{-14}$	$-3.8 \cdot 10^{-14}$	$-6.0 \cdot 10^6$	$1.2 \cdot 10^9$	$-3.5 \cdot 10^7$	$-5.4 \cdot 10^9$
z^6	21.0	0	23.0	$-5.5 \cdot 10^{-14}$	21.0	$-3.4 \cdot 10^{-3}$	21.0	-0.019
z^5	0	0	$-5.0 \cdot 10^{-14}$	$-2.4 \cdot 10^{-13}$	$-3.2 \cdot 10^7$	$6.3 \cdot 10^9$	$-1.9 \cdot 10^8$	$-2.9 \cdot 10^{10}$
z^4	39.0	0	46.0	$-6.0 \cdot 10^{-14}$	40.0	-0.015	40.0	-0.062
z^3	0	0	$2.0 \cdot 10^{-13}$	$-2.4 \cdot 10^{-13}$	$-2.4 \cdot 10^7$	$4.7 \cdot 10^9$	$-1.4 \cdot 10^8$	$-2.1 \cdot 10^{10}$
z^2	12.0	0	15.0	$3.2 \cdot 10^{-14}$	12.0	$-8.0 \cdot 10^{-3}$	12.0	-0.025
z	0	0	$6.2 \cdot 10^{-14}$	$1.8 \cdot 10^{-14}$	$-2.5 \cdot 10^6$	$5.1 \cdot 10^8$	$-1.5 \cdot 10^7$	$-2.3 \cdot 10^9$
1.0	0.32	0	0.42	$8.3 \cdot 10^{-15}$	0.33	$-3.2 \cdot 10^{-4}$	0.33	$-8.5 \cdot 10^{-4}$

Table 79: Case 1a (cpv MACA64 MR2023b), $r = 8$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$		LF, $\sigma_8 = 1.6 \cdot 10^{-09}$		AAA-sign-L200, $\sigma_8 = 7.4 \cdot 10^{-10}$		AAA-sign-L1000, $\sigma_8 = 7.4 \cdot 10^{-10}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^8	0	0	0	0	$4.6 \cdot 10^{-3}$	$1.0 \cdot 10^{-4}$	$4.6 \cdot 10^{-3}$	$1.3 \cdot 10^{-4}$
z^7	6.9	0	7.2	$1.3 \cdot 10^{-14}$	6.9	$-8.5 \cdot 10^{-5}$	6.9	$-8.5 \cdot 10^{-5}$
z^6	0	0	$6.3 \cdot 10^{-14}$	$-1.2 \cdot 10^{-13}$	0.12	$1.5 \cdot 10^{-3}$	0.12	$2.0 \cdot 10^{-3}$
z^5	36.0	0	41.0	$1.9 \cdot 10^{-13}$	36.0	$-1.3 \cdot 10^{-3}$	36.0	$-1.3 \cdot 10^{-3}$
z^4	0	0	$3.4 \cdot 10^{-13}$	$-3.0 \cdot 10^{-13}$	0.25	$1.8 \cdot 10^{-3}$	0.25	$2.8 \cdot 10^{-3}$
z^3	27.0	0	33.0	$3.6 \cdot 10^{-13}$	27.0	$-1.5 \cdot 10^{-3}$	27.0	$-1.6 \cdot 10^{-3}$
z^2	0	0	$1.6 \cdot 10^{-13}$	$-1.4 \cdot 10^{-13}$	0.086	$2.6 \cdot 10^{-4}$	0.086	$5.5 \cdot 10^{-4}$
z	2.9	0	3.7	$3.8 \cdot 10^{-14}$	2.9	$-2.2 \cdot 10^{-4}$	2.9	$-2.3 \cdot 10^{-4}$
1.0	0	0	$6.1 \cdot 10^{-15}$	$-6.4 \cdot 10^{-15}$	$2.6 \cdot 10^{-3}$	$-3.5 \cdot 10^{-7}$	$2.6 \cdot 10^{-3}$	$7.3 \cdot 10^{-6}$
z^8	1.0	0	1.0	0	1.0	0	1.0	0
z^7	0	0	$3.0 \cdot 10^{-15}$	$-2.1 \cdot 10^{-14}$	0.035	$6.0 \cdot 10^{-4}$	0.035	$7.7 \cdot 10^{-4}$
z^6	21.0	0	23.0	$7.1 \cdot 10^{-14}$	21.0	$-5.0 \cdot 10^{-4}$	21.0	$-5.1 \cdot 10^{-4}$
z^5	0	0	$2.2 \cdot 10^{-13}$	$-2.3 \cdot 10^{-13}$	0.22	$2.2 \cdot 10^{-3}$	0.22	$3.0 \cdot 10^{-3}$
z^4	39.0	0	46.0	$3.4 \cdot 10^{-13}$	39.0	$-1.8 \cdot 10^{-3}$	39.0	$-1.8 \cdot 10^{-3}$
z^3	0	0	$3.1 \cdot 10^{-13}$	$-2.5 \cdot 10^{-13}$	0.19	$9.3 \cdot 10^{-4}$	0.19	$1.6 \cdot 10^{-3}$
z^2	12.0	0	15.0	$1.9 \cdot 10^{-13}$	12.0	$-7.9 \cdot 10^{-4}$	12.0	$-7.9 \cdot 10^{-4}$
z	0	0	$4.5 \cdot 10^{-14}$	$-5.3 \cdot 10^{-14}$	0.023	$3.0 \cdot 10^{-5}$	0.023	$1.0 \cdot 10^{-4}$
1.0	0.32	0	0.42	$1.8 \cdot 10^{-15}$	0.32	$-2.7 \cdot 10^{-5}$	0.32	$-2.7 \cdot 10^{-5}$

Table 80: Case 1a (cpv MACA64 MR2022b), $r = 8$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$		LF, $\sigma_8 = 1.6 \cdot 10^{-09}$		AAA-sign-L200, $\sigma_8 = 7.1 \cdot 10^{-10}$		AAA-sign-L1000, $\sigma_8 = 7.1 \cdot 10^{-10}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^8	0	0	0	0	$-5.8 \cdot 10^{-11}$	$-1.2 \cdot 10^{-8}$	$2.5 \cdot 10^{-12}$	$1.4 \cdot 10^{-9}$
z^7	6.9	0	7.2	$-1.6 \cdot 10^{-14}$	6.9	$-2.1 \cdot 10^{-10}$	6.9	$-5.3 \cdot 10^{-11}$
z^6	0	0	$-5.1 \cdot 10^{-14}$	$4.2 \cdot 10^{-14}$	$-1.2 \cdot 10^{-9}$	$-2.6 \cdot 10^{-7}$	$6.3 \cdot 10^{-11}$	$2.9 \cdot 10^{-8}$
z^5	36.0	0	41.0	$-2.4 \cdot 10^{-14}$	36.0	$-3.3 \cdot 10^{-9}$	36.0	$-8.3 \cdot 10^{-10}$
z^4	0	0	$-1.8 \cdot 10^{-13}$	$1.9 \cdot 10^{-13}$	$-2.1 \cdot 10^{-9}$	$-4.8 \cdot 10^{-7}$	$1.3 \cdot 10^{-10}$	$5.5 \cdot 10^{-8}$
z^3	27.0	0	33.0	$1.8 \cdot 10^{-13}$	27.0	$-4.1 \cdot 10^{-9}$	27.0	$-1.0 \cdot 10^{-9}$
z^2	0	0	$-8.7 \cdot 10^{-14}$	$1.0 \cdot 10^{-13}$	$-6.3 \cdot 10^{-10}$	$-1.5 \cdot 10^{-7}$	$4.3 \cdot 10^{-11}$	$1.6 \cdot 10^{-8}$
z	2.9	0	3.7	$4.3 \cdot 10^{-14}$	2.9	$-6.2 \cdot 10^{-10}$	2.9	$-1.6 \cdot 10^{-10}$
1.0	0	0	$-2.1 \cdot 10^{-15}$	$4.2 \cdot 10^{-15}$	$-1.7 \cdot 10^{-11}$	$-3.9 \cdot 10^{-9}$	$1.3 \cdot 10^{-12}$	$4.3 \cdot 10^{-10}$
z^8	1.0	0	1.0	0	1.0	0	1.0	0
z^7	0	0	$-7.0 \cdot 10^{-15}$	$9.4 \cdot 10^{-16}$	$-3.9 \cdot 10^{-10}$	$-8.4 \cdot 10^{-8}$	$1.9 \cdot 10^{-11}$	$9.7 \cdot 10^{-9}$
z^6	21.0	0	23.0	$-4.4 \cdot 10^{-14}$	21.0	$-1.3 \cdot 10^{-9}$	21.0	$-3.2 \cdot 10^{-10}$
z^5	0	0	$-1.2 \cdot 10^{-13}$	$1.1 \cdot 10^{-13}$	$-2.0 \cdot 10^{-9}$	$-4.5 \cdot 10^{-7}$	$1.2 \cdot 10^{-10}$	$5.1 \cdot 10^{-8}$
z^4	39.0	0	46.0	$9.4 \cdot 10^{-14}$	39.0	$-4.8 \cdot 10^{-9}$	39.0	$-1.2 \cdot 10^{-9}$
z^3	0	0	$-1.6 \cdot 10^{-13}$	$1.7 \cdot 10^{-13}$	$-1.5 \cdot 10^{-9}$	$-3.4 \cdot 10^{-7}$	$9.7 \cdot 10^{-11}$	$3.8 \cdot 10^{-8}$
z^2	12.0	0	15.0	$1.3 \cdot 10^{-13}$	12.0	$-2.1 \cdot 10^{-9}$	12.0	$-5.4 \cdot 10^{-10}$
z	0	0	$-2.1 \cdot 10^{-14}$	$3.5 \cdot 10^{-14}$	$-1.5 \cdot 10^{-10}$	$-3.6 \cdot 10^{-8}$	$1.1 \cdot 10^{-11}$	$4.0 \cdot 10^{-9}$
1.0	0.32	0	0.42	$6.7 \cdot 10^{-15}$	0.32	$-7.6 \cdot 10^{-11}$	0.32	$-1.9 \cdot 10^{-11}$

Table 81: Case 1a (cpv PCWIN64 MR2025b), $r = 8$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$		LF, $\sigma_8 = 1.6 \cdot 10^{-09}$		AAA-sign-L200, $\sigma_8 = 7.1 \cdot 10^{-10}$		AAA-sign-L1000, $\sigma_8 = 7.1 \cdot 10^{-10}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^8	0	0	0	0	$-5.8 \cdot 10^{-11}$	$-1.2 \cdot 10^{-8}$	$2.5 \cdot 10^{-12}$	$1.4 \cdot 10^{-9}$
z^7	6.9	0	7.2	$-1.6 \cdot 10^{-14}$	6.9	$-2.1 \cdot 10^{-10}$	6.9	$-5.3 \cdot 10^{-11}$
z^6	0	0	$-5.1 \cdot 10^{-14}$	$4.2 \cdot 10^{-14}$	$-1.2 \cdot 10^{-9}$	$-2.6 \cdot 10^{-7}$	$6.3 \cdot 10^{-11}$	$2.9 \cdot 10^{-8}$
z^5	36.0	0	41.0	$-2.4 \cdot 10^{-14}$	36.0	$-3.3 \cdot 10^{-9}$	36.0	$-8.3 \cdot 10^{-10}$
z^4	0	0	$-1.8 \cdot 10^{-13}$	$1.9 \cdot 10^{-13}$	$-2.1 \cdot 10^{-9}$	$-4.8 \cdot 10^{-7}$	$1.3 \cdot 10^{-10}$	$5.5 \cdot 10^{-8}$
z^3	27.0	0	33.0	$1.8 \cdot 10^{-13}$	27.0	$-4.1 \cdot 10^{-9}$	27.0	$-1.0 \cdot 10^{-9}$
z^2	0	0	$-8.7 \cdot 10^{-14}$	$1.0 \cdot 10^{-13}$	$-6.3 \cdot 10^{-10}$	$-1.5 \cdot 10^{-7}$	$4.3 \cdot 10^{-11}$	$1.6 \cdot 10^{-8}$
z	2.9	0	3.7	$4.3 \cdot 10^{-14}$	2.9	$-6.2 \cdot 10^{-10}$	2.9	$-1.6 \cdot 10^{-10}$
1.0	0	0	$-2.1 \cdot 10^{-15}$	$4.2 \cdot 10^{-15}$	$-1.7 \cdot 10^{-11}$	$-3.9 \cdot 10^{-9}$	$1.3 \cdot 10^{-12}$	$4.3 \cdot 10^{-10}$
z^8	1.0	0	1.0	0	1.0	0	1.0	0
z^7	0	0	$-7.0 \cdot 10^{-15}$	$9.4 \cdot 10^{-16}$	$-3.9 \cdot 10^{-10}$	$-8.4 \cdot 10^{-8}$	$1.9 \cdot 10^{-11}$	$9.7 \cdot 10^{-9}$
z^6	21.0	0	23.0	$-4.4 \cdot 10^{-14}$	21.0	$-1.3 \cdot 10^{-9}$	21.0	$-3.2 \cdot 10^{-10}$
z^5	0	0	$-1.2 \cdot 10^{-13}$	$1.1 \cdot 10^{-13}$	$-2.0 \cdot 10^{-9}$	$-4.5 \cdot 10^{-7}$	$1.2 \cdot 10^{-10}$	$5.1 \cdot 10^{-8}$
z^4	39.0	0	46.0	$9.4 \cdot 10^{-14}$	39.0	$-4.8 \cdot 10^{-9}$	39.0	$-1.2 \cdot 10^{-9}$
z^3	0	0	$-1.6 \cdot 10^{-13}$	$1.7 \cdot 10^{-13}$	$-1.5 \cdot 10^{-9}$	$-3.4 \cdot 10^{-7}$	$9.7 \cdot 10^{-11}$	$3.8 \cdot 10^{-8}$
z^2	12.0	0	15.0	$1.3 \cdot 10^{-13}$	12.0	$-2.1 \cdot 10^{-9}$	12.0	$-5.4 \cdot 10^{-10}$
z	0	0	$-2.1 \cdot 10^{-14}$	$3.5 \cdot 10^{-14}$	$-1.5 \cdot 10^{-10}$	$-3.6 \cdot 10^{-8}$	$1.1 \cdot 10^{-11}$	$4.0 \cdot 10^{-9}$
1.0	0.32	0	0.42	$6.7 \cdot 10^{-15}$	0.32	$-7.6 \cdot 10^{-11}$	0.32	$-1.9 \cdot 10^{-11}$

Table 82: Case 1a (AlexReis PCWIN64 MR2025b), $r = 8$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$		LF, $\sigma_8 = 1.6 \cdot 10^{-09}$		AAA-sign-L200, $\sigma_8 = 7.1 \cdot 10^{-10}$		AAA-sign-L1000, $\sigma_8 = 7.1 \cdot 10^{-10}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^8	0	0	0	0	$-9.8 \cdot 10^{-11}$	$4.5 \cdot 10^{-8}$	$-5.8 \cdot 10^{-11}$	$-1.6 \cdot 10^{-7}$
z^7	6.9	0	7.2	$5.2 \cdot 10^{-17}$	6.9	$6.4 \cdot 10^{-10}$	6.9	$-1.4 \cdot 10^{-9}$
z^6	0	0	$-2.5 \cdot 10^{-14}$	$-1.7 \cdot 10^{-13}$	$-2.2 \cdot 10^{-9}$	$9.4 \cdot 10^{-7}$	$-9.7 \cdot 10^{-10}$	$-3.4 \cdot 10^{-6}$
z^5	36.0	0	41.0	$7.3 \cdot 10^{-14}$	36.0	$1.0 \cdot 10^{-8}$	36.0	$-2.2 \cdot 10^{-8}$
z^4	0	0	$-9.9 \cdot 10^{-15}$	$-5.9 \cdot 10^{-13}$	$-4.3 \cdot 10^{-9}$	$1.8 \cdot 10^{-6}$	$-1.4 \cdot 10^{-9}$	$-6.3 \cdot 10^{-6}$
z^3	27.0	0	33.0	$1.7 \cdot 10^{-13}$	27.0	$1.3 \cdot 10^{-8}$	27.0	$-2.7 \cdot 10^{-8}$
z^2	0	0	$8.3 \cdot 10^{-14}$	$-2.7 \cdot 10^{-13}$	$-1.3 \cdot 10^{-9}$	$5.3 \cdot 10^{-7}$	$-3.2 \cdot 10^{-10}$	$-1.9 \cdot 10^{-6}$
z	2.9	0	3.7	$2.7 \cdot 10^{-14}$	2.9	$1.9 \cdot 10^{-9}$	2.9	$-4.1 \cdot 10^{-9}$
1.0	0	0	$1.0 \cdot 10^{-14}$	$-8.0 \cdot 10^{-15}$	$-3.7 \cdot 10^{-11}$	$1.4 \cdot 10^{-8}$	$-6.2 \cdot 10^{-12}$	$-5.1 \cdot 10^{-8}$
z^8	1.0	0	1.0	0	1.0	0	1.0	0
z^7	0	0	$-7.1 \cdot 10^{-15}$	$-3.9 \cdot 10^{-14}$	$-7.0 \cdot 10^{-10}$	$3.1 \cdot 10^{-7}$	$-3.6 \cdot 10^{-10}$	$-1.1 \cdot 10^{-6}$
z^6	21.0	0	23.0	$1.1 \cdot 10^{-14}$	21.0	$3.9 \cdot 10^{-9}$	21.0	$-8.5 \cdot 10^{-9}$
z^5	0	0	$-2.9 \cdot 10^{-14}$	$-4.3 \cdot 10^{-13}$	$-3.8 \cdot 10^{-9}$	$1.6 \cdot 10^{-6}$	$-1.5 \cdot 10^{-9}$	$-5.8 \cdot 10^{-6}$
z^4	39.0	0	46.0	$1.4 \cdot 10^{-13}$	39.0	$1.5 \cdot 10^{-8}$	39.0	$-3.2 \cdot 10^{-8}$
z^3	0	0	$6.1 \cdot 10^{-14}$	$-5.1 \cdot 10^{-13}$	$-3.0 \cdot 10^{-9}$	$1.2 \cdot 10^{-6}$	$-8.6 \cdot 10^{-10}$	$-4.4 \cdot 10^{-6}$
z^2	12.0	0	15.0	$8.6 \cdot 10^{-14}$	12.0	$6.5 \cdot 10^{-9}$	12.0	$-1.4 \cdot 10^{-8}$
z	0	0	$5.0 \cdot 10^{-14}$	$-8.1 \cdot 10^{-14}$	$-3.3 \cdot 10^{-10}$	$1.3 \cdot 10^{-7}$	$-6.8 \cdot 10^{-11}$	$-4.7 \cdot 10^{-7}$
1.0	0.32	0	0.42	$2.6 \cdot 10^{-15}$	0.32	$2.3 \cdot 10^{-10}$	0.32	$-5.1 \cdot 10^{-10}$

Table 83: Case 1a (AlexReis PCWIN64perso MR2017a), $r = 8$, Z4: numerator (first lines) and denominator (last lines) coefficients.

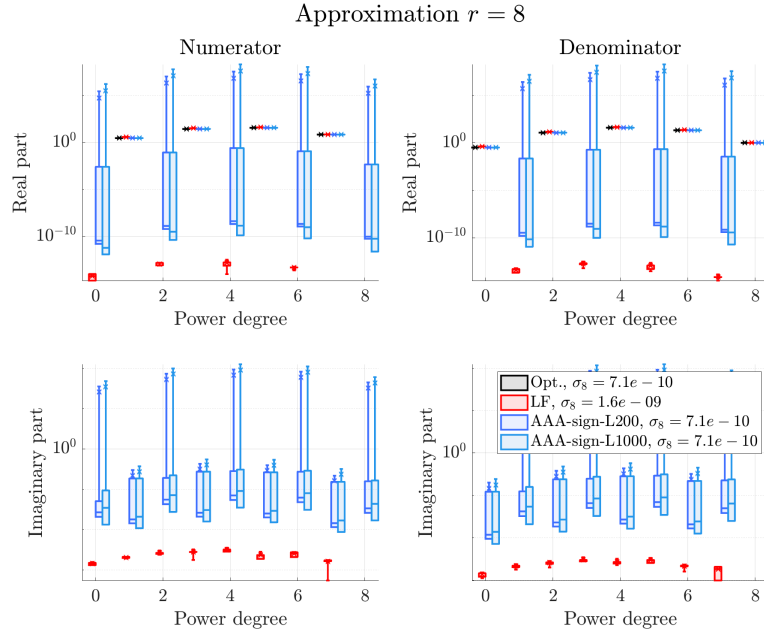


Figure 20: Case 1a, $r = 8$, Z4. Numerator and denominator dispersion over the different platforms.

A.6.2 Poles

The poles are given in the following tables. Each table gives the results obtained on different platform.

	Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$	LF, $\sigma_8 = 1.6 \cdot 10^{-09}$	AAA-sign-L200, $\sigma_8 = 7.1 \cdot 10^{-10}$	AAA-sign-L1000, $\sigma_8 = 7.1 \cdot 10^{-10}$
1.0	$6.66 \cdot 10^{-16} + 4.35_z$	$-4.06 \cdot 10^{-15} + 0.178_z$	$3.84 \cdot 10^{-12} + 6.42 \cdot 10^{-10}_z$	$5.64 \cdot 10^{-13} - 1.45 \cdot 10^{-10}_z$
2.0	$6.66 \cdot 10^{-16} - 4.35_z$	$-6.53 \cdot 10^{-16} - 0.178_z$	$-4.23 \cdot 10^{-12} - 0.359_z$	$-2.48 \cdot 10^{-12} + 0.359_z$
3.0	$2.22 \cdot 10^{-16} + 1.3_z$	$-7.33 \cdot 10^{-17} - 0.599_z$	$1.28 \cdot 10^{-11} + 0.359_z$	$3.91 \cdot 10^{-12} - 0.359_z$
4.0	$2.22 \cdot 10^{-16} - 1.3_z$	$-1.58 \cdot 10^{-15} + 0.599_z$	$-1.39 \cdot 10^{-11} - 0.866_z$	$-6.57 \cdot 10^{-12} + 0.866_z$
5.0	$1.39 \cdot 10^{-17} + 0.579_z$	$4.61 \cdot 10^{-15} - 1.34_z$	$2.73 \cdot 10^{-11} + 0.866_z$	$9.26 \cdot 10^{-12} - 0.866_z$
6.0	$1.39 \cdot 10^{-17} - 0.579_z$	$3.38 \cdot 10^{-15} + 1.34_z$	$-2.86 \cdot 10^{-11} - 2.09_z$	$-1.37 \cdot 10^{-11} + 2.09_z$
7.0	$1.47 \cdot 10^{-17} + 0.172_z$	$-4.63 \cdot 10^{-16} - 4.52_z$	$7.16 \cdot 10^{-11} + 2.09_z$	$2.47 \cdot 10^{-11} - 2.09_z$
8.0	$1.47 \cdot 10^{-17} - 0.172_z$	$1.25 \cdot 10^{-14} + 4.52_z$	$6.04 \cdot 10^6 - 1.21 \cdot 10^9_z$	$3.54 \cdot 10^7 + 5.44 \cdot 10^9_z$

Table 84: Case 1a (cpv MACA64 MR2023b), $r = 8$, Z4: poles.

	Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$	LF, $\sigma_8 = 1.6 \cdot 10^{-09}$	AAA-sign-L200, $\sigma_8 = 7.4 \cdot 10^{-10}$	AAA-sign-L1000, $\sigma_8 = 7.4 \cdot 10^{-10}$
1.0	$8.88 \cdot 10^{-16} + 4.35_z$	$-2.44 \cdot 10^{-15} - 0.178_z$	$-8.97 \cdot 10^{-4} + 0.172_z$	$-8.97 \cdot 10^{-4} + 0.172_z$
2.0	$8.88 \cdot 10^{-16} - 4.35_z$	$-5.6 \cdot 10^{-16} + 0.178_z$	$-9.0 \cdot 10^{-4} - 0.172_z$	$-9.0 \cdot 10^{-4} - 0.172_z$
3.0	$8.33 \cdot 10^{-17} + 1.3_z$	$-5.87 \cdot 10^{-17} - 0.599_z$	$-0.0011 + 0.579_z$	$-0.0011 + 0.579_z$
4.0	$8.33 \cdot 10^{-17} - 1.3_z$	$-3.93 \cdot 10^{-15} + 0.599_z$	$-0.00112 - 0.579_z$	$-0.00112 - 0.579_z$
5.0	$4.16 \cdot 10^{-17} + 0.579_z$	$1.3 \cdot 10^{-15} - 1.34_z$	$-0.002 + 1.3_z$	$-0.002 + 1.3_z$
6.0	$4.16 \cdot 10^{-17} - 0.579_z$	$-5.03 \cdot 10^{-15} + 1.34_z$	$-0.00203 - 1.3_z$	$-0.00203 - 1.3_z$
7.0	$-2.6 \cdot 10^{-18} + 0.172_z$	$1.06 \cdot 10^{-14} - 4.52_z$	$-0.0135 + 4.36_z$	$-0.0135 + 4.36_z$
8.0	$-2.6 \cdot 10^{-18} - 0.172_z$	$-2.57 \cdot 10^{-15} + 4.52_z$	$-0.0136 - 4.36_z$	$-0.0136 - 4.36_z$

Table 85: Case 1a (cpv MACA64 MR2022b), $r = 8$, Z4: poles.

	Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$	LF, $\sigma_8 = 1.6 \cdot 10^{-09}$	AAA-sign-L200, $\sigma_8 = 7.1 \cdot 10^{-10}$	AAA-sign-L1000, $\sigma_8 = 7.1 \cdot 10^{-10}$
1.0	$7.22 \cdot 10^{-16} + 4.35_z$	$1.37 \cdot 10^{-17} + 0.178_z$	$7.54 \cdot 10^{-13} - 0.172_z$	$8.64 \cdot 10^{-13} + 0.172_z$
2.0	$7.22 \cdot 10^{-16} - 4.35_z$	$1.31 \cdot 10^{-15} - 0.178_z$	$1.1 \cdot 10^{-11} + 0.172_z$	$-1.73 \cdot 10^{-12} - 0.172_z$
3.0	$1.94 \cdot 10^{-16} + 1.3_z$	$-1.07 \cdot 10^{-15} + 0.599_z$	$-9.06 \cdot 10^{-12} - 0.579_z$	$3.82 \cdot 10^{-12} + 0.579_z$
4.0	$1.94 \cdot 10^{-16} - 1.3_z$	$3.39 \cdot 10^{-15} - 0.599_z$	$2.57 \cdot 10^{-11} + 0.579_z$	$-4.98 \cdot 10^{-12} - 0.579_z$
5.0	$-4.16 \cdot 10^{-17} + 0.579_z$	$-1.82 \cdot 10^{-15} + 1.34_z$	$-2.0 \cdot 10^{-11} - 1.3_z$	$8.73 \cdot 10^{-12} + 1.3_z$
6.0	$-4.16 \cdot 10^{-17} - 0.579_z$	$3.47 \cdot 10^{-15} - 1.34_z$	$5.85 \cdot 10^{-11} + 1.3_z$	$-1.1 \cdot 10^{-11} - 1.3_z$
7.0	$1.73 \cdot 10^{-18} + 0.172_z$	$-5.1 \cdot 10^{-15} - 4.52_z$	$2.98 \cdot 10^{-11} - 4.35_z$	$2.57 \cdot 10^{-11} + 4.35_z$
8.0	$1.73 \cdot 10^{-18} - 0.172_z$	$6.85 \cdot 10^{-15} + 4.52_z$	$2.95 \cdot 10^{-10} + 4.35_z$	$-4.05 \cdot 10^{-11} - 4.35_z$

Table 86: Case 1a (cpv PCWIN64 MR2025b), $r = 8$, Z4: poles.

	Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$	LF, $\sigma_8 = 1.6 \cdot 10^{-09}$	AAA-sign-L200, $\sigma_8 = 7.1 \cdot 10^{-10}$	AAA-sign-L1000, $\sigma_8 = 7.1 \cdot 10^{-10}$
1.0	$7.22 \cdot 10^{-16} + 4.35_z$	$1.37 \cdot 10^{-17} + 0.178_z$	$7.54 \cdot 10^{-13} - 0.172_z$	$8.64 \cdot 10^{-13} + 0.172_z$
2.0	$7.22 \cdot 10^{-16} - 4.35_z$	$1.31 \cdot 10^{-15} - 0.178_z$	$1.1 \cdot 10^{-11} + 0.172_z$	$-1.73 \cdot 10^{-12} - 0.172_z$
3.0	$1.94 \cdot 10^{-16} + 1.3_z$	$-1.07 \cdot 10^{-15} + 0.599_z$	$-9.06 \cdot 10^{-12} - 0.579_z$	$3.82 \cdot 10^{-12} + 0.579_z$
4.0	$1.94 \cdot 10^{-16} - 1.3_z$	$3.39 \cdot 10^{-15} - 0.599_z$	$2.57 \cdot 10^{-11} + 0.579_z$	$-4.98 \cdot 10^{-12} - 0.579_z$
5.0	$-4.16 \cdot 10^{-17} + 0.579_z$	$-1.82 \cdot 10^{-15} + 1.34_z$	$-2.0 \cdot 10^{-11} - 1.3_z$	$8.73 \cdot 10^{-12} + 1.3_z$
6.0	$-4.16 \cdot 10^{-17} - 0.579_z$	$3.47 \cdot 10^{-15} - 1.34_z$	$5.85 \cdot 10^{-11} + 1.3_z$	$-1.1 \cdot 10^{-11} - 1.3_z$
7.0	$1.73 \cdot 10^{-18} + 0.172_z$	$-5.1 \cdot 10^{-15} - 4.52_z$	$2.98 \cdot 10^{-11} - 4.35_z$	$2.57 \cdot 10^{-11} + 4.35_z$
8.0	$1.73 \cdot 10^{-18} - 0.172_z$	$6.85 \cdot 10^{-15} + 4.52_z$	$2.95 \cdot 10^{-10} + 4.35_z$	$-4.05 \cdot 10^{-11} - 4.35_z$

Table 87: Case 1a (AlexReis PCWIN64 MR2025b), $r = 8$, Z4: poles.

	Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$	LF, $\sigma_8 = 1.6 \cdot 10^{-09}$	AAA-sign-L200, $\sigma_8 = 7.1 \cdot 10^{-10}$	AAA-sign-L1000, $\sigma_8 = 7.1 \cdot 10^{-10}$
1.0	$8.88 \cdot 10^{-16} + 4.35_z$	$-2.02 \cdot 10^{-15} - 0.178_z$	$-2.62 \cdot 10^{-12} + 0.172_z$	$-3.18 \cdot 10^{-11} - 0.172_z$
2.0	$8.88 \cdot 10^{-16} - 4.35_z$	$-2.06 \cdot 10^{-15} + 0.178_z$	$2.87 \cdot 10^{-11} - 0.172_z$	$3.64 \cdot 10^{-11} + 0.172_z$
3.0	$8.33 \cdot 10^{-17} + 1.3_z$	$2.21 \cdot 10^{-15} - 0.599_z$	$-3.53 \cdot 10^{-11} + 0.579_z$	$-1.11 \cdot 10^{-10} - 0.579_z$
4.0	$8.33 \cdot 10^{-17} - 1.3_z$	$4.26 \cdot 10^{-16} + 0.599_z$	$7.04 \cdot 10^{-11} - 0.579_z$	$1.2 \cdot 10^{-10} + 0.579_z$
5.0	$4.16 \cdot 10^{-17} + 0.579_z$	$3.13 \cdot 10^{-15} - 1.34_z$	$-8.23 \cdot 10^{-11} + 1.3_z$	$-2.46 \cdot 10^{-10} - 1.3_z$
6.0	$4.16 \cdot 10^{-17} - 0.579_z$	$-7.79 \cdot 10^{-16} + 1.34_z$	$1.56 \cdot 10^{-10} - 1.3_z$	$2.75 \cdot 10^{-10} + 1.3_z$
7.0	$-2.6 \cdot 10^{-18} + 0.172_z$	$3.57 \cdot 10^{-15} - 4.52_z$	$-1.22 \cdot 10^{-10} + 4.35_z$	$-7.23 \cdot 10^{-10} - 4.35_z$
8.0	$-2.6 \cdot 10^{-18} - 0.172_z$	$2.63 \cdot 10^{-15} + 4.52_z$	$6.85 \cdot 10^{-10} - 4.35_z$	$1.04 \cdot 10^{-9} + 4.35_z$

Table 88: Case 1a (AlexReis PCWIN64perso MR2017a), $r = 8$, Z4: poles.

A.6.3 Zeros

The zeros are given in the following tables. Each table gives the results obtained on different platform.

	Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$	LF, $\sigma_8 = 1.6 \cdot 10^{-09}$	AAA-sign-L200, $\sigma_8 = 7.1 \cdot 10^{-10}$	AAA-sign-L1000, $\sigma_8 = 7.1 \cdot 10^{-10}$
1.0	NaN	NaN	$-1.46 \cdot 10^{-13} - 0.172z$	$-9.08 \cdot 10^{-13} + 0.172z$
2.0	0	$-2.76 \cdot 10^{-15} - 2.73 \cdot 10^{-15}z$	$8.04 \cdot 10^{-12} + 0.172z$	$2.11 \cdot 10^{-12} - 0.172z$
3.0	$1.39 \cdot 10^{-16} + 2.09z$	$4.64 \cdot 10^{-16} - 0.371z$	$-8.67 \cdot 10^{-12} - 0.579z$	$-4.32 \cdot 10^{-12} + 0.579z$
4.0	$1.39 \cdot 10^{-16} - 2.09z$	$-3.67 \cdot 10^{-15} + 0.371z$	$1.88 \cdot 10^{-11} + 0.579z$	$6.15 \cdot 10^{-12} - 0.579z$
5.0	$-5.55 \cdot 10^{-17} + 0.866z$	$8.79 \cdot 10^{-16} - 0.897z$	$-2.05 \cdot 10^{-11} - 1.3z$	$-9.53 \cdot 10^{-12} + 1.3z$
6.0	$-5.55 \cdot 10^{-17} - 0.866z$	$1.64 \cdot 10^{-15} + 0.897z$	$4.14 \cdot 10^{-11} + 1.3z$	$1.43 \cdot 10^{-11} - 1.3z$
7.0	$-6.94 \cdot 10^{-18} + 0.359z$	$6.16 \cdot 10^{-15} - 2.17z$	$-2.26 \cdot 10^{-11} - 4.35z$	$-1.71 \cdot 10^{-11} + 4.35z$
8.0	$-6.94 \cdot 10^{-18} - 0.359z$	$4.41 \cdot 10^{-15} + 2.17z$	$1.86 \cdot 10^{-10} + 4.35z$	$6.27 \cdot 10^{-11} - 4.35z$

Table 89: Case 1a (cpv MACA64 MR2023b), $r = 8$, Z4: zeros.

	Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$	LF, $\sigma_8 = 1.6 \cdot 10^{-09}$	AAA-sign-L200, $\sigma_8 = 7.4 \cdot 10^{-10}$	AAA-sign-L1000, $\sigma_8 = 7.4 \cdot 10^{-10}$
1.0	NaN	NaN	$-8.78 \cdot 10^{-4} + 5.09 \cdot 10^{-8}z$	$-8.78 \cdot 10^{-4} - 2.54 \cdot 10^{-6}z$
2.0	0	$-1.63 \cdot 10^{-15} + 1.75 \cdot 10^{-15}z$	$-9.64 \cdot 10^{-4} + 0.359z$	$-9.64 \cdot 10^{-4} + 0.359z$
3.0	$-3.33 \cdot 10^{-16} + 2.09z$	$-2.19 \cdot 10^{-15} - 0.371z$	$-9.71 \cdot 10^{-4} - 0.359z$	$-9.71 \cdot 10^{-4} - 0.359z$
4.0	$-3.33 \cdot 10^{-16} - 2.09z$	$-1.01 \cdot 10^{-15} + 0.371z$	$-0.00138 + 0.866z$	$-0.00138 + 0.866z$
5.0	$-5.55 \cdot 10^{-17} + 0.866z$	$1.73 \cdot 10^{-15} - 0.897z$	$-0.0014 - 0.866z$	$-0.0014 - 0.866z$
6.0	$-5.55 \cdot 10^{-17} - 0.866z$	$-5.71 \cdot 10^{-15} + 0.897z$	$-0.0038 + 2.09z$	$-0.0038 + 2.09z$
7.0	$-6.25 \cdot 10^{-17} + 0.359z$	$2.2 \cdot 10^{-15} - 2.17z$	$-0.00385 - 2.09z$	$-0.00385 - 2.09z$
8.0	$-6.25 \cdot 10^{-17} - 0.359z$	$-2.01 \cdot 10^{-15} + 2.17z$	$-1500.0 + 32.5z$	$-1500.0 + 40.5z$

Table 90: Case 1a (cpv MACA64 MR2022b), $r = 8$, Z4: zeros.

	Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$	LF, $\sigma_8 = 1.6 \cdot 10^{-09}$	AAA-sign-L200, $\sigma_8 = 7.1 \cdot 10^{-10}$	AAA-sign-L1000, $\sigma_8 = 7.1 \cdot 10^{-10}$
1.0	NaN	NaN	$5.66 \cdot 10^{-12} + 1.34 \cdot 10^{-9}z$	$-4.28 \cdot 10^{-13} - 1.47 \cdot 10^{-10}z$
2.0	0	$5.49 \cdot 10^{-16} - 1.11 \cdot 10^{-15}z$	$-4.07 \cdot 10^{-12} - 0.359z$	$2.24 \cdot 10^{-12} + 0.359z$
3.0	$1.39 \cdot 10^{-16} + 2.09z$	$-4.33 \cdot 10^{-17} + 0.371z$	$1.74 \cdot 10^{-11} + 0.359z$	$-3.19 \cdot 10^{-12} - 0.359z$
4.0	$1.39 \cdot 10^{-16} - 2.09z$	$2.25 \cdot 10^{-15} - 0.371z$	$-1.44 \cdot 10^{-11} - 0.866z$	$5.83 \cdot 10^{-12} + 0.866z$
5.0	$-5.55 \cdot 10^{-17} + 0.866z$	$-2.49 \cdot 10^{-15} + 0.897z$	$3.78 \cdot 10^{-11} + 0.866z$	$-7.39 \cdot 10^{-12} - 0.866z$
6.0	$-5.55 \cdot 10^{-17} - 0.866z$	$3.87 \cdot 10^{-15} - 0.897z$	$-2.21 \cdot 10^{-11} - 2.09z$	$1.38 \cdot 10^{-11} + 2.09z$
7.0	$-6.94 \cdot 10^{-18} + 0.359z$	$8.84 \cdot 10^{-16} + 2.17z$	$1.05 \cdot 10^{-10} + 2.09z$	$-1.81 \cdot 10^{-11} - 2.09z$
8.0	$-6.94 \cdot 10^{-18} - 0.359z$	$2.13 \cdot 10^{-15} - 2.17z$	$2.7 \cdot 10^6 - 5.7 \cdot 10^8z$	$-8.64 \cdot 10^6 + 4.91 \cdot 10^9z$

Table 91: Case 1a (cpv PCWIN64 MR2025b), $r = 8$, Z4: zeros.

	Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$	LF, $\sigma_8 = 1.6 \cdot 10^{-09}$	AAA-sign-L200, $\sigma_8 = 7.1 \cdot 10^{-10}$	AAA-sign-L1000, $\sigma_8 = 7.1 \cdot 10^{-10}$
1.0	NaN	NaN	$5.66 \cdot 10^{-12} + 1.34 \cdot 10^{-9}z$	$-4.28 \cdot 10^{-13} - 1.47 \cdot 10^{-10}z$
2.0	0	$5.49 \cdot 10^{-16} - 1.11 \cdot 10^{-15}z$	$-4.07 \cdot 10^{-12} - 0.359z$	$2.24 \cdot 10^{-12} + 0.359z$
3.0	$1.39 \cdot 10^{-16} + 2.09z$	$-4.33 \cdot 10^{-17} + 0.371z$	$1.74 \cdot 10^{-11} + 0.359z$	$-3.19 \cdot 10^{-12} - 0.359z$
4.0	$1.39 \cdot 10^{-16} - 2.09z$	$2.25 \cdot 10^{-15} - 0.371z$	$-1.44 \cdot 10^{-11} - 0.866z$	$5.83 \cdot 10^{-12} + 0.866z$
5.0	$-5.55 \cdot 10^{-17} + 0.866z$	$-2.49 \cdot 10^{-15} + 0.897z$	$3.78 \cdot 10^{-11} + 0.866z$	$-7.39 \cdot 10^{-12} - 0.866z$
6.0	$-5.55 \cdot 10^{-17} - 0.866z$	$3.87 \cdot 10^{-15} - 0.897z$	$-2.21 \cdot 10^{-11} - 2.09z$	$1.38 \cdot 10^{-11} + 2.09z$
7.0	$-6.94 \cdot 10^{-18} + 0.359z$	$8.84 \cdot 10^{-16} + 2.17z$	$1.05 \cdot 10^{-10} + 2.09z$	$-1.81 \cdot 10^{-11} - 2.09z$
8.0	$-6.94 \cdot 10^{-18} - 0.359z$	$2.13 \cdot 10^{-15} - 2.17z$	$2.7 \cdot 10^6 - 5.7 \cdot 10^8z$	$-8.64 \cdot 10^6 + 4.91 \cdot 10^9z$

Table 92: Case 1a (AlexReis PCWIN64 MR2025b), $r = 8$, Z4: zeros.

	Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$	LF, $\sigma_8 = 1.6 \cdot 10^{-09}$	AAA-sign-L200, $\sigma_8 = 7.1 \cdot 10^{-10}$	AAA-sign-L1000, $\sigma_8 = 7.1 \cdot 10^{-10}$
1.0	NaN	NaN	$1.26 \cdot 10^{-11} - 4.9 \cdot 10^{-9}i$	$2.13 \cdot 10^{-12} + 1.75 \cdot 10^{-8}i$
2.0	0	$-2.74 \cdot 10^{-15} + 2.12 \cdot 10^{-15}i$	$-1.81 \cdot 10^{-11} + 0.359i$	$-6.83 \cdot 10^{-11} - 0.359i$
3.0	$-3.33 \cdot 10^{-16} + 2.09i$	$4.72 \cdot 10^{-17} - 0.371i$	$4.72 \cdot 10^{-11} - 0.359i$	$7.41 \cdot 10^{-11} + 0.359i$
4.0	$-3.33 \cdot 10^{-16} - 2.09i$	$-4.68 \cdot 10^{-16} + 0.371i$	$-5.58 \cdot 10^{-11} + 0.866i$	$-1.66 \cdot 10^{-10} - 0.866i$
5.0	$-5.55 \cdot 10^{-17} + 0.866i$	$3.63 \cdot 10^{-15} - 0.897i$	$1.03 \cdot 10^{-10} - 0.866i$	$1.81 \cdot 10^{-10} + 0.866i$
6.0	$-5.55 \cdot 10^{-17} - 0.866i$	$-3.34 \cdot 10^{-16} + 0.897i$	$-1.18 \cdot 10^{-10} + 2.09i$	$-3.85 \cdot 10^{-10} - 2.09i$
7.0	$-6.25 \cdot 10^{-17} + 0.359i$	$3.14 \cdot 10^{-15} - 2.17i$	$2.68 \cdot 10^{-10} - 2.09i$	$4.59 \cdot 10^{-10} + 2.09i$
8.0	$-6.25 \cdot 10^{-17} - 0.359i$	$2.22 \cdot 10^{-16} + 2.17i$	$3.42 \cdot 10^5 + 1.56 \cdot 10^8i$	$1.58 \cdot 10^4 - 4.34 \cdot 10^7i$

Table 93: Case 1a (AlexReis PCWIN64perso MR2017a), $r = 8$, Z4: zeros.

A.6.4 Poles and zeros visualisation (case cpv MACA64 MR2023b)

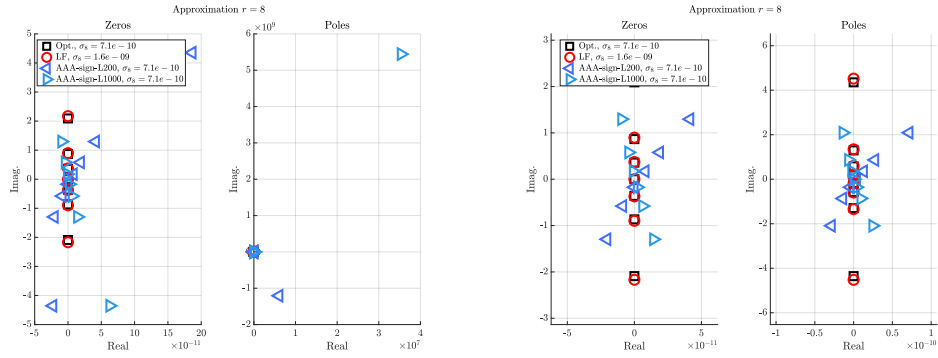


Figure 21: Case 1a (cpv MACA64 MR2023b), $r = 8$, Z4. Left frame: zeros (left) and poles (right). Right frame: same but discarding the largest real entry.

A.6.5 Approximation visualization

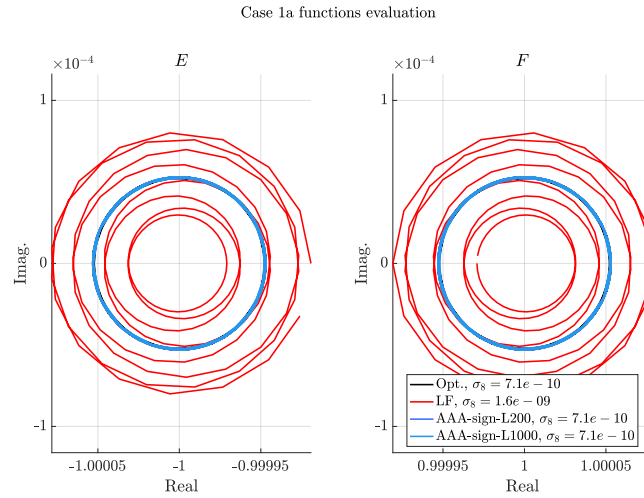


Figure 22: Case 1a (cpv MACA64 MR2023b), $r = 8$, Z4. E and F evaluations.

A.7 Approximation objective $r = 9$

A.7.1 Numerator and denominator coefficients

The numerator and denominator coefficients (both real and complex parts) are given in the following tables. Each table gives the results obtained on different platform.

Basis	Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$		LF, $\sigma_9 = 2.7 \cdot 10^{-10}$		AAA-sign-L200, $\sigma_9 = 5.1 \cdot 10^{-11}$		AAA-sign-L1000, $\sigma_9 = 5.1 \cdot 10^{-11}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^9	0	0	0	0	$-2.1 \cdot 10^3$	$1.4 \cdot 10^6$	-940.0	$-5.1 \cdot 10^5$
z^8	7.8	0	8.9	$5.0 \cdot 10^{-14}$	7.8	$-8.0 \cdot 10^{-4}$	7.8	$3.7 \cdot 10^{-4}$
z^7	0	0	$9.1 \cdot 10^{-13}$	$1.1 \cdot 10^{-12}$	$-5.7 \cdot 10^4$	$3.7 \cdot 10^7$	$-2.5 \cdot 10^4$	$-1.4 \cdot 10^7$
z^6	55.0	0	78.0	$1.6 \cdot 10^{-13}$	54.0	-0.016	55.0	$7.1 \cdot 10^{-3}$
z^5	0	0	$4.0 \cdot 10^{-12}$	$6.6 \cdot 10^{-12}$	$-1.5 \cdot 10^5$	$9.8 \cdot 10^7$	$-6.6 \cdot 10^4$	$-3.6 \cdot 10^7$
z^4	61.0	0	110.0	$3.2 \cdot 10^{-13}$	60.0	-0.027	62.0	0.012
z^3	0	0	$2.2 \cdot 10^{-12}$	$4.7 \cdot 10^{-12}$	$-7.4 \cdot 10^4$	$4.9 \cdot 10^7$	$-3.3 \cdot 10^4$	$-1.8 \cdot 10^7$
z^2	13.0	0	27.0	$3.1 \cdot 10^{-13}$	13.0	$-7.6 \cdot 10^{-3}$	13.0	$3.1 \cdot 10^{-3}$
z	0	0	$6.9 \cdot 10^{-14}$	$5.2 \cdot 10^{-13}$	$-6.0 \cdot 10^3$	$3.9 \cdot 10^6$	$-2.7 \cdot 10^3$	$-1.5 \cdot 10^6$
1.0	0.27	0	0.64	$9.3 \cdot 10^{-17}$	0.27	$-1.9 \cdot 10^{-4}$	0.28	$7.3 \cdot 10^{-5}$
z^9	1.0	0	1.0	0	1.0	0	1.0	0
z^8	0	0	$1.5 \cdot 10^{-13}$	$1.3 \cdot 10^{-13}$	$-1.6 \cdot 10^4$	$1.1 \cdot 10^7$	$-7.3 \cdot 10^3$	$-4.0 \cdot 10^6$
z^7	27.0	0	35.0	$1.7 \cdot 10^{-13}$	27.0	$-5.3 \cdot 10^{-3}$	27.0	$2.4 \cdot 10^{-3}$
z^6	0	0	$2.6 \cdot 10^{-12}$	$3.7 \cdot 10^{-12}$	$-1.1 \cdot 10^5$	$7.5 \cdot 10^7$	$-5.1 \cdot 10^4$	$-2.8 \cdot 10^7$
z^5	71.0	0	110.0	$1.0 \cdot 10^{-14}$	70.0	-0.026	72.0	0.012
z^4	0	0	$3.9 \cdot 10^{-12}$	$7.1 \cdot 10^{-12}$	$-1.3 \cdot 10^5$	$8.5 \cdot 10^7$	$-5.8 \cdot 10^4$	$-3.2 \cdot 10^7$
z^3	35.0	0	67.0	$4.4 \cdot 10^{-13}$	35.0	-0.018	36.0	$7.7 \cdot 10^{-3}$
z^2	0	0	$6.8 \cdot 10^{-13}$	$2.1 \cdot 10^{-12}$	$-2.8 \cdot 10^4$	$1.8 \cdot 10^7$	$-1.2 \cdot 10^4$	$-6.8 \cdot 10^6$
z	2.8	0	6.2	$5.9 \cdot 10^{-14}$	2.8	$-1.8 \cdot 10^{-3}$	2.9	$7.3 \cdot 10^{-4}$
1.0	0	0	$-6.9 \cdot 10^{-15}$	$6.8 \cdot 10^{-14}$	-570.0	$3.8 \cdot 10^5$	-260.0	$-1.4 \cdot 10^5$

Table 94: Case 1a (cpv MACA64 MR2023b), $r = 9$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$		LF, $\sigma_9 = 2.7 \cdot 10^{-10}$		AAA-sign-L200, $\sigma_9 = 5.2 \cdot 10^{-11}$		AAA-sign-L1000, $\sigma_9 = 5.2 \cdot 10^{-11}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^9	0	0	0	0	190.0	32.0	190.0	32.0
z^8	7.8	0	8.9	$-1.1 \cdot 10^{-13}$	8.3	0.29	8.3	0.29
z^7	0	0	$-1.7 \cdot 10^{-13}$	$3.8 \cdot 10^{-13}$	$5.2 \cdot 10^3$	850.0	$5.2 \cdot 10^3$	850.0
z^6	55.0	0	78.0	$-3.4 \cdot 10^{-12}$	65.0	6.0	65.0	6.0
z^5	0	0	$-3.2 \cdot 10^{-13}$	$3.3 \cdot 10^{-12}$	$1.4 \cdot 10^4$	$2.2 \cdot 10^3$	$1.4 \cdot 10^4$	$2.2 \cdot 10^3$
z^4	61.0	0	110.0	$-6.6 \cdot 10^{-12}$	81.0	11.0	81.0	11.0
z^3	0	0	$-7.6 \cdot 10^{-14}$	$2.8 \cdot 10^{-12}$	$6.8 \cdot 10^3$	$1.1 \cdot 10^3$	$6.8 \cdot 10^3$	$1.1 \cdot 10^3$
z^2	13.0	0	27.0	$-2.2 \cdot 10^{-12}$	19.0	3.2	19.0	3.2
z	0	0	$3.2 \cdot 10^{-14}$	$3.4 \cdot 10^{-13}$	550.0	89.0	550.0	89.0
1.0	0.27	0	0.64	$-4.4 \cdot 10^{-14}$	0.43	0.085	0.43	0.085
z^9	1.0	0	1.0	0	1.0	0	1.0	0
z^8	0	0	$-4.9 \cdot 10^{-14}$	$2.4 \cdot 10^{-14}$	$1.5 \cdot 10^3$	250.0	$1.5 \cdot 10^3$	250.0
z^7	27.0	0	35.0	$-1.0 \cdot 10^{-12}$	30.0	2.0	30.0	2.0
z^6	0	0	$-2.7 \cdot 10^{-13}$	$1.6 \cdot 10^{-12}$	$1.0 \cdot 10^4$	$1.7 \cdot 10^3$	$1.0 \cdot 10^4$	$1.7 \cdot 10^3$
z^5	71.0	0	110.0	$-6.0 \cdot 10^{-12}$	89.0	10.0	89.0	10.0
z^4	0	0	$-2.2 \cdot 10^{-13}$	$3.9 \cdot 10^{-12}$	$1.2 \cdot 10^4$	$1.9 \cdot 10^3$	$1.2 \cdot 10^4$	$1.9 \cdot 10^3$
z^3	35.0	0	67.0	$-4.8 \cdot 10^{-12}$	49.0	7.6	49.0	7.6
z^2	0	0	$2.4 \cdot 10^{-14}$	$1.3 \cdot 10^{-12}$	$2.5 \cdot 10^3$	410.0	$2.5 \cdot 10^3$	410.0
z	2.8	0	6.2	$-5.1 \cdot 10^{-13}$	4.3	0.79	4.3	0.79
1.0	0	0	$6.7 \cdot 10^{-15}$	$3.6 \cdot 10^{-14}$	52.0	8.6	52.0	8.6

Table 95: Case 1a (cpv MACA64 MR2022b), $r = 9$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$		LF, $\sigma_9 = 2.7 \cdot 10^{-10}$		AAA-sign-L200, $\sigma_9 = 5.1 \cdot 10^{-11}$		AAA-sign-L1000, $\sigma_9 = 5.1 \cdot 10^{-11}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^9	0	0	0	0	$2.5 \cdot 10^{-9}$	$-2.9 \cdot 10^{-6}$	$-3.5 \cdot 10^{-10}$	$-7.9 \cdot 10^{-7}$
z^8	7.8	0	8.9	$-2.6 \cdot 10^{-13}$	7.8	$3.9 \cdot 10^{-8}$	7.8	$1.9 \cdot 10^{-9}$
z^7	0	0	$3.5 \cdot 10^{-13}$	$-1.3 \cdot 10^{-13}$	$7.7 \cdot 10^{-8}$	$-7.9 \cdot 10^{-5}$	$-8.9 \cdot 10^{-9}$	$-2.1 \cdot 10^{-5}$
z^6	55.0	0	78.0	$-4.5 \cdot 10^{-12}$	55.0	$8.3 \cdot 10^{-7}$	55.0	$4.1 \cdot 10^{-8}$
z^5	0	0	$2.1 \cdot 10^{-12}$	$3.3 \cdot 10^{-13}$	$2.2 \cdot 10^{-7}$	$-2.1 \cdot 10^{-4}$	$-2.2 \cdot 10^{-8}$	$-5.6 \cdot 10^{-5}$
z^4	61.0	0	110.0	$-8.0 \cdot 10^{-12}$	61.0	$1.6 \cdot 10^{-6}$	61.0	$7.7 \cdot 10^{-8}$
z^3	0	0	$1.7 \cdot 10^{-12}$	$6.6 \cdot 10^{-13}$	$1.2 \cdot 10^{-7}$	$-1.0 \cdot 10^{-4}$	$-1.1 \cdot 10^{-8}$	$-2.8 \cdot 10^{-5}$
z^2	13.0	0	27.0	$-2.4 \cdot 10^{-12}$	13.0	$4.7 \cdot 10^{-7}$	13.0	$2.3 \cdot 10^{-8}$
z	0	0	$2.0 \cdot 10^{-13}$	$1.2 \cdot 10^{-13}$	$9.6 \cdot 10^{-9}$	$-8.2 \cdot 10^{-6}$	$-8.4 \cdot 10^{-10}$	$-2.2 \cdot 10^{-6}$
1.0	0.27	0	0.64	$-6.1 \cdot 10^{-14}$	0.27	$1.3 \cdot 10^{-8}$	0.27	$6.3 \cdot 10^{-10}$
z^9	1.0	0	1.0	0	1.0	0	1.0	0
z^8	0	0	$5.1 \cdot 10^{-14}$	$-5.3 \cdot 10^{-14}$	$2.1 \cdot 10^{-8}$	$-2.3 \cdot 10^{-5}$	$-2.6 \cdot 10^{-9}$	$-6.1 \cdot 10^{-6}$
z^7	27.0	0	35.0	$-1.5 \cdot 10^{-12}$	27.0	$2.7 \cdot 10^{-7}$	27.0	$1.3 \cdot 10^{-8}$
z^6	0	0	$1.1 \cdot 10^{-12}$	$-6.4 \cdot 10^{-14}$	$1.6 \cdot 10^{-7}$	$-1.6 \cdot 10^{-4}$	$-1.8 \cdot 10^{-8}$	$-4.3 \cdot 10^{-5}$
z^5	71.0	0	110.0	$-7.3 \cdot 10^{-12}$	71.0	$1.4 \cdot 10^{-6}$	71.0	$7.1 \cdot 10^{-8}$
z^4	0	0	$2.4 \cdot 10^{-12}$	$7.6 \cdot 10^{-13}$	$2.0 \cdot 10^{-7}$	$-1.8 \cdot 10^{-4}$	$-1.9 \cdot 10^{-8}$	$-4.8 \cdot 10^{-5}$
z^3	35.0	0	67.0	$-5.5 \cdot 10^{-12}$	35.0	$1.1 \cdot 10^{-6}$	35.0	$5.4 \cdot 10^{-8}$
z^2	0	0	$8.1 \cdot 10^{-13}$	$3.8 \cdot 10^{-13}$	$4.4 \cdot 10^{-8}$	$-3.8 \cdot 10^{-5}$	$-3.9 \cdot 10^{-9}$	$-1.0 \cdot 10^{-5}$
z	2.8	0	6.2	$-5.8 \cdot 10^{-13}$	2.8	$1.2 \cdot 10^{-7}$	2.8	$5.8 \cdot 10^{-9}$
1.0	0	0	$2.3 \cdot 10^{-14}$	$1.9 \cdot 10^{-14}$	$9.3 \cdot 10^{-10}$	$-7.9 \cdot 10^{-7}$	$-8.0 \cdot 10^{-11}$	$-2.1 \cdot 10^{-7}$

Table 96: Case 1a (cpv PCWIN64 MR2025b), $r = 9$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$		LF, $\sigma_9 = 2.7 \cdot 10^{-10}$		AAA-sign-L200, $\sigma_9 = 5.1 \cdot 10^{-11}$		AAA-sign-L1000, $\sigma_9 = 5.1 \cdot 10^{-11}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^9	0	0	0	0	$2.5 \cdot 10^{-9}$	$-2.9 \cdot 10^{-6}$	$-3.5 \cdot 10^{-10}$	$-7.9 \cdot 10^{-7}$
z^8	7.8	0	8.9	$-2.6 \cdot 10^{-13}$	7.8	$3.9 \cdot 10^{-8}$	7.8	$1.9 \cdot 10^{-9}$
z^7	0	0	$3.5 \cdot 10^{-13}$	$-1.3 \cdot 10^{-13}$	$7.7 \cdot 10^{-8}$	$-7.9 \cdot 10^{-5}$	$-8.9 \cdot 10^{-9}$	$-2.1 \cdot 10^{-5}$
z^6	55.0	0	78.0	$-4.5 \cdot 10^{-12}$	55.0	$8.3 \cdot 10^{-7}$	55.0	$4.1 \cdot 10^{-8}$
z^5	0	0	$2.1 \cdot 10^{-12}$	$3.3 \cdot 10^{-13}$	$2.2 \cdot 10^{-7}$	$-2.1 \cdot 10^{-4}$	$-2.2 \cdot 10^{-8}$	$-5.6 \cdot 10^{-5}$
z^4	61.0	0	110.0	$-8.0 \cdot 10^{-12}$	61.0	$1.6 \cdot 10^{-6}$	61.0	$7.7 \cdot 10^{-8}$
z^3	0	0	$1.7 \cdot 10^{-12}$	$6.6 \cdot 10^{-13}$	$1.2 \cdot 10^{-7}$	$-1.0 \cdot 10^{-4}$	$-1.1 \cdot 10^{-8}$	$-2.8 \cdot 10^{-5}$
z^2	13.0	0	27.0	$-2.4 \cdot 10^{-12}$	13.0	$4.7 \cdot 10^{-7}$	13.0	$2.3 \cdot 10^{-8}$
z	0	0	$2.0 \cdot 10^{-13}$	$1.2 \cdot 10^{-13}$	$9.6 \cdot 10^{-9}$	$-8.2 \cdot 10^{-6}$	$-8.4 \cdot 10^{-10}$	$-2.2 \cdot 10^{-6}$
1.0	0.27	0	0.64	$-6.1 \cdot 10^{-14}$	0.27	$1.3 \cdot 10^{-8}$	0.27	$6.3 \cdot 10^{-10}$
z^9	1.0	0	1.0	0	1.0	0	1.0	0
z^8	0	0	$5.1 \cdot 10^{-14}$	$-5.3 \cdot 10^{-14}$	$2.1 \cdot 10^{-8}$	$-2.3 \cdot 10^{-5}$	$-2.6 \cdot 10^{-9}$	$-6.1 \cdot 10^{-6}$
z^7	27.0	0	35.0	$-1.5 \cdot 10^{-12}$	27.0	$2.7 \cdot 10^{-7}$	27.0	$1.3 \cdot 10^{-8}$
z^6	0	0	$1.1 \cdot 10^{-12}$	$-6.4 \cdot 10^{-14}$	$1.6 \cdot 10^{-7}$	$-1.6 \cdot 10^{-4}$	$-1.8 \cdot 10^{-8}$	$-4.3 \cdot 10^{-5}$
z^5	71.0	0	110.0	$-7.3 \cdot 10^{-12}$	71.0	$1.4 \cdot 10^{-6}$	71.0	$7.1 \cdot 10^{-8}$
z^4	0	0	$2.4 \cdot 10^{-12}$	$7.6 \cdot 10^{-13}$	$2.0 \cdot 10^{-7}$	$-1.8 \cdot 10^{-4}$	$-1.9 \cdot 10^{-8}$	$-4.8 \cdot 10^{-5}$
z^3	35.0	0	67.0	$-5.5 \cdot 10^{-12}$	35.0	$1.1 \cdot 10^{-6}$	35.0	$5.4 \cdot 10^{-8}$
z^2	0	0	$8.1 \cdot 10^{-13}$	$3.8 \cdot 10^{-13}$	$4.4 \cdot 10^{-8}$	$-3.8 \cdot 10^{-5}$	$-3.9 \cdot 10^{-9}$	$-1.0 \cdot 10^{-5}$
z	2.8	0	6.2	$-5.8 \cdot 10^{-13}$	2.8	$1.2 \cdot 10^{-7}$	2.8	$5.8 \cdot 10^{-9}$
1.0	0	0	$2.3 \cdot 10^{-14}$	$1.9 \cdot 10^{-14}$	$9.3 \cdot 10^{-10}$	$-7.9 \cdot 10^{-7}$	$-8.0 \cdot 10^{-11}$	$-2.1 \cdot 10^{-7}$

Table 97: Case 1a (AlexReis PCWIN64 MR2025b), $r = 9$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$		LF, $\sigma_9 = 2.7 \cdot 10^{-10}$		AAA-sign-L200, $\sigma_9 = 6.2 \cdot 10^{-11}$		AAA-sign-L1000, $\sigma_9 = 6.2 \cdot 10^{-11}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^9	0	0	0	0	$3.7 \cdot 10^{-3}$	$-5.0 \cdot 10^{-3}$	$3.7 \cdot 10^{-3}$	$-5.0 \cdot 10^{-3}$
z^8	7.8	0	8.9	$6.4 \cdot 10^{-14}$	7.8	$-7.7 \cdot 10^{-3}$	7.8	$-7.7 \cdot 10^{-3}$
z^7	0	0	$-8.9 \cdot 10^{-13}$	$2.6 \cdot 10^{-13}$	-0.024	-0.086	-0.024	-0.086
z^6	55.0	0	78.0	$7.7 \cdot 10^{-13}$	55.0	-0.16	55.0	-0.16
z^5	0	0	$-4.1 \cdot 10^{-12}$	$3.9 \cdot 10^{-13}$	-0.4	-0.091	-0.4	-0.091
z^4	61.0	0	110.0	$1.1 \cdot 10^{-12}$	62.0	-0.3	62.0	-0.3
z^3	0	0	$-2.2 \cdot 10^{-12}$	$-7.7 \cdot 10^{-13}$	-0.37	0.023	-0.37	0.023
z^2	13.0	0	27.0	$1.9 \cdot 10^{-13}$	13.0	-0.092	13.0	-0.092
z	0	0	$-1.4 \cdot 10^{-13}$	$-1.4 \cdot 10^{-13}$	-0.043	$7.4 \cdot 10^{-3}$	-0.043	$7.4 \cdot 10^{-3}$
1.0	0.27	0	0.64	$6.0 \cdot 10^{-15}$	0.28	$-2.5 \cdot 10^{-3}$	0.28	$-2.5 \cdot 10^{-3}$
z^9	1.0	0	1.0	0	1.0	0	1.0	0
z^8	0	0	$-1.3 \cdot 10^{-13}$	$3.8 \cdot 10^{-14}$	0.011	-0.032	0.011	-0.032
z^7	27.0	0	35.0	$3.3 \cdot 10^{-13}$	27.0	-0.053	27.0	-0.053
z^6	0	0	$-2.5 \cdot 10^{-12}$	$6.1 \cdot 10^{-13}$	-0.18	-0.12	-0.18	-0.12
z^5	71.0	0	110.0	$1.3 \cdot 10^{-12}$	72.0	-0.28	72.0	-0.28
z^4	0	0	$-3.8 \cdot 10^{-12}$	$-1.6 \cdot 10^{-13}$	-0.49	-0.02	-0.49	-0.02
z^3	35.0	0	67.0	$6.3 \cdot 10^{-13}$	36.0	-0.21	36.0	-0.21
z^2	0	0	$-7.6 \cdot 10^{-13}$	$-4.7 \cdot 10^{-13}$	-0.17	0.021	-0.17	0.021
z	2.8	0	6.2	$4.1 \cdot 10^{-14}$	2.9	-0.023	2.9	-0.023
1.0	0	0	$-1.1 \cdot 10^{-14}$	$-1.7 \cdot 10^{-14}$	$-4.8 \cdot 10^{-3}$	$9.9 \cdot 10^{-4}$	$-4.8 \cdot 10^{-3}$	$9.9 \cdot 10^{-4}$

Table 98: Case 1a (AlexReis PCWIN64perso MR2017a), $r = 9$, Z4: numerator (first lines) and denominator (last lines) coefficients.

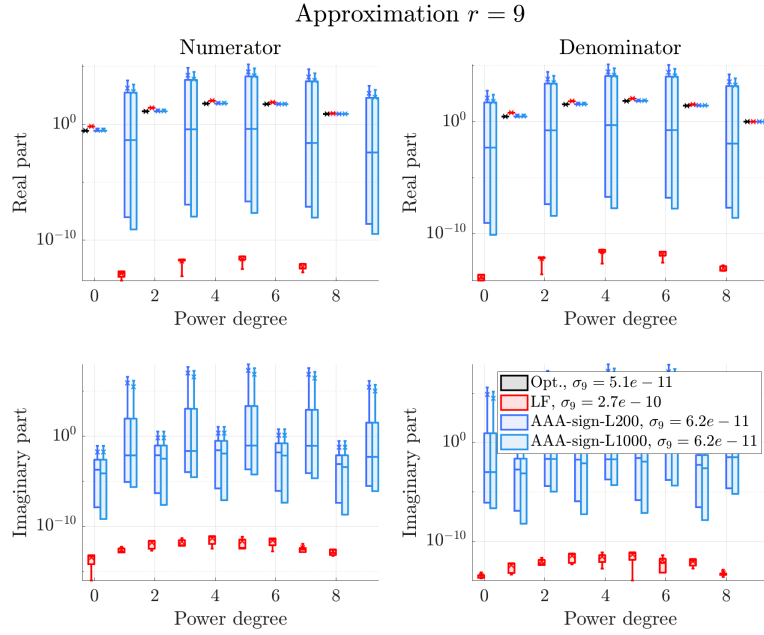


Figure 23: Case 1a, $r = 9$, Z4. Numerator and denominator dispersion over the different platforms.

A.7.2 Poles

The poles are given in the following tables. Each table gives the results obtained on different platform.

	Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$	LF, $\sigma_9 = 2.7 \cdot 10^{-10}$	AAA-sign-L200, $\sigma_9 = 5.1 \cdot 10^{-11}$	AAA-sign-L1000, $\sigma_9 = 5.1 \cdot 10^{-11}$
1.0	0	$1.11 \cdot 10^{-15} - 1.09 \cdot 10^{-14}_z$	$5.97 \cdot 10^{-10} - 0.153_z$	$-1.86 \cdot 10^{-10} + 0.153_z$
2.0	$-1.11 \cdot 10^{-15} + 4.91_z$	$-4.13 \cdot 10^{-15} + 0.338_z$	$-2.87 \cdot 10^{-10} + 0.153_z$	$1.01 \cdot 10^{-9} - 0.153_z$
3.0	$-1.11 \cdot 10^{-15} - 4.91_z$	$-3.5 \cdot 10^{-15} - 0.338_z$	$1.65 \cdot 10^{-9} - 0.5_z$	$-1.45 \cdot 10^{-9} + 0.5_z$
4.0	$-5.55 \cdot 10^{-17} + 1.5_z$	$-1.64 \cdot 10^{-14} + 0.789_z$	$-1.26 \cdot 10^{-9} + 0.5_z$	$2.51 \cdot 10^{-9} - 0.5_z$
5.0	$-5.55 \cdot 10^{-17} - 1.5_z$	$-6.96 \cdot 10^{-15} - 0.789_z$	$3.35 \cdot 10^{-9} - 1.03_z$	$-3.19 \cdot 10^{-9} + 1.03_z$
6.0	$-1.39 \cdot 10^{-17} + 0.727_z$	$-1.08 \cdot 10^{-14} + 1.67_z$	$-2.7 \cdot 10^{-9} + 1.03_z$	$5.04 \cdot 10^{-9} - 1.03_z$
7.0	$-1.39 \cdot 10^{-17} - 0.727_z$	$-2.77 \cdot 10^{-14} - 1.67_z$	$7.99 \cdot 10^{-9} - 2.38_z$	$-6.5 \cdot 10^{-9} + 2.38_z$
8.0	$1.73 \cdot 10^{-18} + 0.315_z$	$-5.92 \cdot 10^{-14} + 5.6_z$	$-6.02 \cdot 10^{-9} + 2.38_z$	$1.26 \cdot 10^{-8} - 2.38_z$
9.0	$1.73 \cdot 10^{-18} - 0.315_z$	$-2.58 \cdot 10^{-14} - 5.6_z$	$1.63 \cdot 10^4 - 1.07 \cdot 10^7_z$	$7310.0 + 4.0 \cdot 10^6_z$

Table 99: Case 1a (cpv MACA64 MR2023b), $r = 9$, Z4: poles.

	Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$	LF, $\sigma_9 = 2.7 \cdot 10^{-10}$	AAA-sign-L200, $\sigma_9 = 5.2 \cdot 10^{-11}$	AAA-sign-L1000, $\sigma_9 = 5.2 \cdot 10^{-11}$
1.0	0	$-1.08 \cdot 10^{-15} - 5.74 \cdot 10^{-15}_z$	$-7.88 \cdot 10^{-4} + 0.153_z$	$-7.88 \cdot 10^{-4} + 0.153_z$
2.0	$5.55 \cdot 10^{-16} + 4.91_z$	$1.48 \cdot 10^{-15} + 0.338_z$	$-8.16 \cdot 10^{-4} - 0.153_z$	$-8.16 \cdot 10^{-4} - 0.153_z$
3.0	$5.55 \cdot 10^{-16} - 4.91_z$	$-1.07 \cdot 10^{-15} - 0.338_z$	$-9.03 \cdot 10^{-4} + 0.5_z$	$-9.03 \cdot 10^{-4} + 0.5_z$
4.0	$1.11 \cdot 10^{-16} + 1.5_z$	$8.32 \cdot 10^{-15} + 0.789_z$	$-0.001 - 0.5_z$	$-0.001 - 0.5_z$
5.0	$1.11 \cdot 10^{-16} - 1.5_z$	$-6.52 \cdot 10^{-15} - 0.789_z$	$-0.0016 - 1.03_z$	$-0.0016 - 1.03_z$
6.0	$2.78 \cdot 10^{-17} + 0.727_z$	$2.26 \cdot 10^{-14} + 1.67_z$	$-0.00138 + 1.03_z$	$-0.00138 + 1.03_z$
7.0	$2.78 \cdot 10^{-17} - 0.727_z$	$-1.99 \cdot 10^{-14} - 1.67_z$	$-0.00476 - 2.38_z$	$-0.00476 - 2.38_z$
8.0	$1.73 \cdot 10^{-18} + 0.315_z$	$-6.04 \cdot 10^{-14} - 5.6_z$	$-0.0042 + 2.38_z$	$-0.0042 + 2.38_z$
9.0	$1.73 \cdot 10^{-18} - 0.315_z$	$1.06 \cdot 10^{-13} + 5.6_z$	$-1500.0 - 247.0_z$	$-1500.0 - 246.0_z$

Table 100: Case 1a (cpv MACA64 MR2022b), $r = 9$, Z4: poles.

	Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$	LF, $\sigma_9 = 2.7 \cdot 10^{-10}$	AAA-sign-L200, $\sigma_9 = 5.1 \cdot 10^{-11}$	AAA-sign-L1000, $\sigma_9 = 5.1 \cdot 10^{-11}$
1.0	0	$-3.67 \cdot 10^{-15} - 3.11 \cdot 10^{-15}_z$	$-3.27 \cdot 10^{-10} + 2.77 \cdot 10^{-7}_z$	$2.81 \cdot 10^{-11} + 7.54 \cdot 10^{-8}_z$
2.0	$-1.11 \cdot 10^{-15} + 4.91_z$	$-2.8 \cdot 10^{-15} + 0.338_z$	$1.28 \cdot 10^{-9} - 0.315_z$	$1.14 \cdot 10^{-10} - 0.315_z$
3.0	$-1.11 \cdot 10^{-15} - 4.91_z$	$-5.57 \cdot 10^{-15} - 0.338_z$	$-2.03 \cdot 10^{-9} + 0.315_z$	$-4.97 \cdot 10^{-11} + 0.315_z$
4.0	$-5.55 \cdot 10^{-17} + 1.5_z$	$3.63 \cdot 10^{-15} + 0.789_z$	$3.22 \cdot 10^{-9} - 0.727_z$	$2.35 \cdot 10^{-10} - 0.727_z$
5.0	$-5.55 \cdot 10^{-17} - 1.5_z$	$-1.15 \cdot 10^{-14} - 0.789_z$	$-4.32 \cdot 10^{-9} + 0.727_z$	$-1.36 \cdot 10^{-10} + 0.727_z$
6.0	$-1.39 \cdot 10^{-17} + 0.727_z$	$-2.13 \cdot 10^{-14} - 1.67_z$	$6.52 \cdot 10^{-9} - 1.5_z$	$5.02 \cdot 10^{-10} - 1.5_z$
7.0	$-1.39 \cdot 10^{-17} - 0.727_z$	$9.08 \cdot 10^{-15} + 1.67_z$	$-8.83 \cdot 10^{-9} + 1.5_z$	$-2.53 \cdot 10^{-10} + 1.5_z$
8.0	$1.73 \cdot 10^{-18} + 0.315_z$	$-1.42 \cdot 10^{-13} - 5.6_z$	$1.66 \cdot 10^{-8} - 4.91_z$	$2.31 \cdot 10^{-9} - 4.91_z$
9.0	$1.73 \cdot 10^{-18} - 0.315_z$	$1.23 \cdot 10^{-13} + 5.6_z$	$-3.32 \cdot 10^{-8} + 4.91_z$	$-1.3 \cdot 10^{-10} + 4.91_z$

Table 101: Case 1a (cpv PCWIN64 MR2025b), $r = 9$, Z4: poles.

	Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$	LF, $\sigma_9 = 2.7 \cdot 10^{-10}$	AAA-sign-L200, $\sigma_9 = 5.1 \cdot 10^{-11}$	AAA-sign-L1000, $\sigma_9 = 5.1 \cdot 10^{-11}$
1.0	0	$-3.67 \cdot 10^{-15} - 3.11 \cdot 10^{-15}_z$	$-3.27 \cdot 10^{-10} + 2.77 \cdot 10^{-7}_z$	$2.81 \cdot 10^{-11} + 7.54 \cdot 10^{-8}_z$
2.0	$-1.11 \cdot 10^{-15} + 4.91_z$	$-2.8 \cdot 10^{-15} + 0.338_z$	$1.28 \cdot 10^{-9} - 0.315_z$	$1.14 \cdot 10^{-10} - 0.315_z$
3.0	$-1.11 \cdot 10^{-15} - 4.91_z$	$-5.57 \cdot 10^{-15} - 0.338_z$	$-2.03 \cdot 10^{-9} + 0.315_z$	$-4.97 \cdot 10^{-11} + 0.315_z$
4.0	$-5.55 \cdot 10^{-17} + 1.5_z$	$3.63 \cdot 10^{-15} + 0.789_z$	$3.22 \cdot 10^{-9} - 0.727_z$	$2.35 \cdot 10^{-10} - 0.727_z$
5.0	$-5.55 \cdot 10^{-17} - 1.5_z$	$-1.15 \cdot 10^{-14} - 0.789_z$	$-4.32 \cdot 10^{-9} + 0.727_z$	$-1.36 \cdot 10^{-10} + 0.727_z$
6.0	$-1.39 \cdot 10^{-17} + 0.727_z$	$-2.13 \cdot 10^{-14} - 1.67_z$	$6.52 \cdot 10^{-9} - 1.5_z$	$5.02 \cdot 10^{-10} - 1.5_z$
7.0	$-1.39 \cdot 10^{-17} - 0.727_z$	$9.08 \cdot 10^{-15} + 1.67_z$	$-8.83 \cdot 10^{-9} + 1.5_z$	$-2.53 \cdot 10^{-10} + 1.5_z$
8.0	$1.73 \cdot 10^{-18} + 0.315_z$	$-1.42 \cdot 10^{-13} - 5.6_z$	$1.66 \cdot 10^{-8} - 4.91_z$	$2.31 \cdot 10^{-9} - 4.91_z$
9.0	$1.73 \cdot 10^{-18} - 0.315_z$	$1.23 \cdot 10^{-13} + 5.6_z$	$-3.32 \cdot 10^{-8} + 4.91_z$	$-1.3 \cdot 10^{-10} + 4.91_z$

Table 102: Case 1a (AlexReis PCWIN64 MR2025b), $r = 9$, Z4: poles.

	Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$	LF, $\sigma_9 = 2.7 \cdot 10^{-10}$	AAA-sign-L200, $\sigma_9 = 6.2 \cdot 10^{-11}$	AAA-sign-L1000, $\sigma_9 = 6.2 \cdot 10^{-11}$
1.0	0	$1.75 \cdot 10^{-15} + 2.77 \cdot 10^{-15}z$	$0.00167 - 3.27 \cdot 10^{-4}z$	$0.00167 - 3.27 \cdot 10^{-4}z$
2.0	$5.55 \cdot 10^{-16} + 4.91z$	$2.52 \cdot 10^{-15} - 0.338z$	$0.00193 + 0.316z$	$0.00193 + 0.316z$
3.0	$5.55 \cdot 10^{-16} - 4.91z$	$3.57 \cdot 10^{-15} + 0.338z$	$0.00131 - 0.316z$	$0.00131 - 0.316z$
4.0	$1.11 \cdot 10^{-16} + 1.5z$	$1.12 \cdot 10^{-14} + 0.789z$	$6.74 \cdot 10^{-4} - 0.728z$	$6.74 \cdot 10^{-4} - 0.728z$
5.0	$1.11 \cdot 10^{-16} - 1.5z$	$8.9 \cdot 10^{-15} - 0.789z$	$0.0021 + 0.728z$	$0.0021 + 0.728z$
6.0	$2.78 \cdot 10^{-17} + 0.727z$	$1.96 \cdot 10^{-14} + 1.67z$	$-9.43 \cdot 10^{-4} - 1.5z$	$-9.42 \cdot 10^{-4} - 1.5z$
7.0	$2.78 \cdot 10^{-17} - 0.727z$	$2.34 \cdot 10^{-14} - 1.67z$	$0.00202 + 1.5z$	$0.00202 + 1.5z$
8.0	$1.73 \cdot 10^{-18} + 0.315z$	$-1.72 \cdot 10^{-15} + 5.6z$	$-0.0147 - 4.91z$	$-0.0147 - 4.91z$
9.0	$1.73 \cdot 10^{-18} - 0.315z$	$5.66 \cdot 10^{-14} - 5.6z$	$-0.00509 + 4.94z$	$-0.00508 + 4.94z$

Table 103: Case 1a (AlexReis PCWIN64perso MR2017a), $r = 9$, Z4: poles.

A.7.3 Zeros

The zeros are given in the following tables. Each table gives the results obtained on different platform.

	Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$	LF, $\sigma_9 = 2.7 \cdot 10^{-10}$	AAA-sign-L200, $\sigma_9 = 5.1 \cdot 10^{-11}$	AAA-sign-L1000, $\sigma_9 = 5.1 \cdot 10^{-11}$
1.0	NaN	NaN	$1.51 \cdot 10^{-10} + 6.79 \cdot 10^{-8}z$	$4.01 \cdot 10^{-10} - 1.92 \cdot 10^{-7}z$
2.0	$-8.33 \cdot 10^{-17} + 2.38z$	$8.36 \cdot 10^{-16} + 0.163z$	$1.08 \cdot 10^{-9} - 0.315z$	$-7.87 \cdot 10^{-10} + 0.315z$
3.0	$-8.33 \cdot 10^{-17} - 2.38z$	$-1.5 \cdot 10^{-15} - 0.163z$	$-7.46 \cdot 10^{-10} + 0.315z$	$1.69 \cdot 10^{-9} - 0.315z$
4.0	$2.78 \cdot 10^{-17} + 1.03z$	$-1.15 \cdot 10^{-14} + 0.539z$	$2.36 \cdot 10^{-9} - 0.727z$	$-2.22 \cdot 10^{-9} + 0.727z$
5.0	$2.78 \cdot 10^{-17} - 1.03z$	$-3.68 \cdot 10^{-15} - 0.539z$	$-1.88 \cdot 10^{-9} + 0.727z$	$3.55 \cdot 10^{-9} - 0.727z$
6.0	$-6.07 \cdot 10^{-18} + 0.5z$	$-1.44 \cdot 10^{-14} + 1.13z$	$4.9 \cdot 10^{-9} - 1.5z$	$-4.53 \cdot 10^{-9} + 1.5z$
7.0	$-6.07 \cdot 10^{-18} - 0.5z$	$-1.54 \cdot 10^{-14} - 1.13z$	$-3.91 \cdot 10^{-9} + 1.5z$	$7.48 \cdot 10^{-9} - 1.5z$
8.0	$1.99 \cdot 10^{-17} + 0.153z$	$-2.2 \cdot 10^{-14} + 2.68z$	$1.81 \cdot 10^{-8} - 4.91z$	$-8.23 \cdot 10^{-9} + 4.91z$
9.0	$1.99 \cdot 10^{-17} - 0.153z$	$-3.49 \cdot 10^{-14} - 2.68z$	$-1.09 \cdot 10^{-8} + 4.91z$	$3.12 \cdot 10^{-8} - 4.91z$

Table 104: Case 1a (cpv MACA64 MR2023b), $r = 9$, Z4: zeros.

	Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$	LF, $\sigma_9 = 2.7 \cdot 10^{-10}$	AAA-sign-L200, $\sigma_9 = 5.2 \cdot 10^{-11}$	AAA-sign-L1000, $\sigma_9 = 5.2 \cdot 10^{-11}$
1.0	NaN	NaN	$-7.86 \cdot 10^{-4} - 2.65 \cdot 10^{-5}z$	$-7.86 \cdot 10^{-4} - 2.68 \cdot 10^{-5}z$
2.0	$-2.78 \cdot 10^{-16} + 2.38z$	$-2.12 \cdot 10^{-15} + 0.163z$	$-8.23 \cdot 10^{-4} + 0.315z$	$-8.23 \cdot 10^{-4} + 0.315z$
3.0	$-2.78 \cdot 10^{-16} - 2.38z$	$4.75 \cdot 10^{-16} - 0.163z$	$-8.82 \cdot 10^{-4} - 0.315z$	$-8.82 \cdot 10^{-4} - 0.315z$
4.0	$-4.16 \cdot 10^{-17} + 1.03z$	$6.99 \cdot 10^{-15} + 0.539z$	$-0.00121 - 0.726z$	$-0.00121 - 0.726z$
5.0	$-4.16 \cdot 10^{-17} - 1.03z$	$-5.34 \cdot 10^{-15} - 0.539z$	$-0.00106 + 0.726z$	$-0.00106 + 0.726z$
6.0	$1.73 \cdot 10^{-18} + 0.5z$	$8.08 \cdot 10^{-15} + 1.13z$	$-0.00243 - 1.5z$	$-0.00243 - 1.5z$
7.0	$1.73 \cdot 10^{-18} - 0.5z$	$-8.11 \cdot 10^{-15} - 1.13z$	$-0.00209 + 1.5z$	$-0.00209 + 1.5z$
8.0	$-8.67 \cdot 10^{-19} + 0.153z$	$-3.84 \cdot 10^{-14} - 2.68z$	$-0.017 - 4.91z$	$-0.017 - 4.91z$
9.0	$-8.67 \cdot 10^{-19} - 0.153z$	$5.6 \cdot 10^{-14} + 2.68z$	$-0.0159 + 4.91z$	$-0.0159 + 4.91z$

Table 105: Case 1a (cpv MACA64 MR2022b), $r = 9$, Z4: zeros.

	Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$	LF, $\sigma_9 = 2.7 \cdot 10^{-10}$	AAA-sign-L200, $\sigma_9 = 5.1 \cdot 10^{-11}$	AAA-sign-L1000, $\sigma_9 = 5.1 \cdot 10^{-11}$
1.0	NaN	NaN	$4.67 \cdot 10^{-10} - 0.153z$	$6.88 \cdot 10^{-11} - 0.153z$
2.0	$1.11 \cdot 10^{-16} + 2.38z$	$-3.53 \cdot 10^{-15} + 0.163z$	$-1.14 \cdot 10^{-9} + 0.153z$	$-1.08 \cdot 10^{-11} + 0.153z$
3.0	$1.11 \cdot 10^{-16} - 2.38z$	$-4.0 \cdot 10^{-15} - 0.163z$	$2.17 \cdot 10^{-9} - 0.5z$	$1.67 \cdot 10^{-10} - 0.5z$
4.0	$6.94 \cdot 10^{-17} + 1.03z$	$2.91 \cdot 10^{-16} + 0.539z$	$-3.05 \cdot 10^{-9} + 0.5z$	$-9.07 \cdot 10^{-11} + 0.5z$
5.0	$6.94 \cdot 10^{-17} - 1.03z$	$-8.48 \cdot 10^{-15} - 0.539z$	$4.57 \cdot 10^{-9} - 1.03z$	$3.34 \cdot 10^{-10} - 1.03z$
6.0	$2.08 \cdot 10^{-17} + 0.5z$	$4.72 \cdot 10^{-15} + 1.13z$	$-6.06 \cdot 10^{-9} + 1.03z$	$-1.89 \cdot 10^{-10} + 1.03z$
7.0	$2.08 \cdot 10^{-17} - 0.5z$	$-1.45 \cdot 10^{-14} - 1.13z$	$9.83 \cdot 10^{-9} - 2.38z$	$8.7 \cdot 10^{-10} - 2.38z$
8.0	$5.2 \cdot 10^{-18} + 0.153z$	$-4.7 \cdot 10^{-14} - 2.68z$	$-1.44 \cdot 10^{-8} + 2.38z$	$-3.2 \cdot 10^{-10} + 2.38z$
9.0	$5.2 \cdot 10^{-18} - 0.153z$	$3.31 \cdot 10^{-14} + 2.68z$	$-2310.0 - 2.66 \cdot 10^6z$	$4370.0 - 9.93 \cdot 10^6z$

Table 106: Case 1a (cpv PCWIN64 MR2025b), $r = 9$, Z4: zeros.

	Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$	LF, $\sigma_9 = 2.7 \cdot 10^{-10}$	AAA-sign-L200, $\sigma_9 = 5.1 \cdot 10^{-11}$	AAA-sign-L1000, $\sigma_9 = 5.1 \cdot 10^{-11}$
1.0	NaN	NaN	$4.67 \cdot 10^{-10} - 0.153z$	$6.88 \cdot 10^{-11} - 0.153z$
2.0	$1.11 \cdot 10^{-16} + 2.38z$	$-3.53 \cdot 10^{-15} + 0.163z$	$-1.14 \cdot 10^{-9} + 0.153z$	$-1.08 \cdot 10^{-11} + 0.153z$
3.0	$1.11 \cdot 10^{-16} - 2.38z$	$-4.0 \cdot 10^{-15} - 0.163z$	$2.17 \cdot 10^{-9} - 0.5z$	$1.67 \cdot 10^{-10} - 0.5z$
4.0	$6.94 \cdot 10^{-17} + 1.03z$	$2.91 \cdot 10^{-16} + 0.539z$	$-3.05 \cdot 10^{-9} + 0.5z$	$-9.07 \cdot 10^{-11} + 0.5z$
5.0	$6.94 \cdot 10^{-17} - 1.03z$	$-8.48 \cdot 10^{-15} - 0.539z$	$4.57 \cdot 10^{-9} - 1.03z$	$3.34 \cdot 10^{-10} - 1.03z$
6.0	$2.08 \cdot 10^{-17} + 0.5z$	$4.72 \cdot 10^{-15} + 1.13z$	$-6.06 \cdot 10^{-9} + 1.03z$	$-1.89 \cdot 10^{-10} + 1.03z$
7.0	$2.08 \cdot 10^{-17} - 0.5z$	$-1.45 \cdot 10^{-14} - 1.13z$	$9.83 \cdot 10^{-9} - 2.38z$	$8.7 \cdot 10^{-10} - 2.38z$
8.0	$5.2 \cdot 10^{-18} + 0.153z$	$-4.7 \cdot 10^{-14} - 2.68z$	$-1.44 \cdot 10^{-8} + 2.38z$	$-3.2 \cdot 10^{-10} + 2.38z$
9.0	$5.2 \cdot 10^{-18} - 0.153z$	$3.31 \cdot 10^{-14} + 2.68z$	$-2310.0 - 2.66 \cdot 10^6z$	$4370.0 - 9.93 \cdot 10^6z$

Table 107: Case 1a (AlexReis PCWIN64 MR2025b), $r = 9$, Z4: zeros.

	Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$	LF, $\sigma_9 = 2.7 \cdot 10^{-10}$	AAA-sign-L200, $\sigma_9 = 6.2 \cdot 10^{-11}$	AAA-sign-L1000, $\sigma_9 = 6.2 \cdot 10^{-11}$
1.0	NaN	NaN	$0.00181 + 0.153z$	$0.00181 + 0.153z$
2.0	$-2.78 \cdot 10^{-16} + 2.38z$	$2.26 \cdot 10^{-15} - 0.163z$	$0.00151 - 0.153z$	$0.00151 - 0.153z$
3.0	$-2.78 \cdot 10^{-16} - 2.38z$	$1.79 \cdot 10^{-15} + 0.163z$	$0.00202 + 0.501z$	$0.00202 + 0.501z$
4.0	$-4.16 \cdot 10^{-17} + 1.03z$	$3.99 \cdot 10^{-15} - 0.539z$	$0.00104 - 0.501z$	$0.00104 - 0.501z$
5.0	$-4.16 \cdot 10^{-17} - 1.03z$	$6.55 \cdot 10^{-15} + 0.539z$	$1.01 \cdot 10^{-4} - 1.03z$	$1.01 \cdot 10^{-4} - 1.03z$
6.0	$1.73 \cdot 10^{-18} + 0.5z$	$1.57 \cdot 10^{-14} + 1.13z$	$0.00213 + 1.03z$	$0.00213 + 1.03z$
7.0	$1.73 \cdot 10^{-18} - 0.5z$	$1.69 \cdot 10^{-14} - 1.13z$	$-0.00345 - 2.38z$	$-0.00345 - 2.38z$
8.0	$-8.67 \cdot 10^{-19} + 0.153z$	$2.22 \cdot 10^{-14} + 2.68z$	$0.00126 + 2.39z$	$0.00126 + 2.39z$
9.0	$-8.67 \cdot 10^{-19} - 0.153z$	$3.06 \cdot 10^{-14} - 2.68z$	$-741.0 - 1000.0z$	$-741.0 - 1000.0z$

Table 108: Case 1a (AlexReis PCWIN64perso MR2017a), $r = 9$, Z4: zeros.

A.7.4 Poles and zeros visualisation (case cpv MACA64 MR2023b)

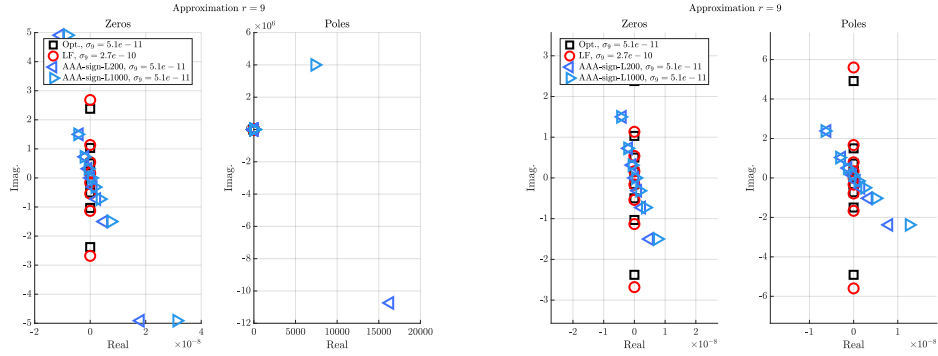


Figure 24: Case 1a (cpv MACA64 MR2023b), $r = 9$, Z4. Left frame: zeros (left) and poles (right). Right frame: same but discarding the largest real entry.

A.7.5 Approximation visualization

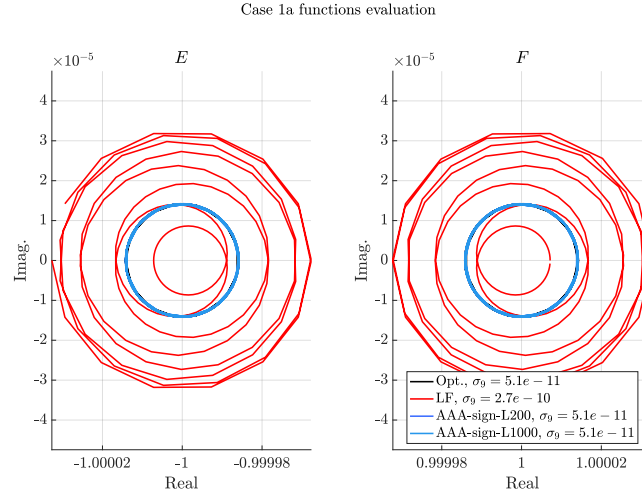


Figure 25: Case 1a (cpv MACA64 MR2023b), $r = 9$, Z4. E and F evaluations.

A.8 Approximation objective $r = 10$

A.8.1 Numerator and denominator coefficients

The numerator and denominator coefficients (both real and complex parts) are given in the following tables. Each table gives the results obtained on different platform.

	Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$		LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$		AAA-sign-L200, $\sigma_{10} = 4.3 \cdot 10^{-12}$		AAA-sign-L1000, $\sigma_{10} = 4.3 \cdot 10^{-12}$	
Basis	real	imag.	real	imag.	real	imag.	real	imag.
z^{10}	0	0	0	0	$7.7 \cdot 10^{-3}$	$3.1 \cdot 10^{-3}$	$7.7 \cdot 10^{-3}$	$3.1 \cdot 10^{-3}$
z^9	8.7	0	8.9	$4.8 \cdot 10^{-14}$	8.7	$4.6 \cdot 10^{-3}$	8.7	$4.6 \cdot 10^{-3}$
z^8	0	0	$-4.3 \cdot 10^{-13}$	$6.0 \cdot 10^{-13}$	0.14	0.07	0.14	0.069
z^7	78.0	0	85.0	$4.0 \cdot 10^{-13}$	78.0	0.13	78.0	0.13
z^6	0	0	$-7.8 \cdot 10^{-12}$	$4.4 \cdot 10^{-12}$	0.036	0.12	0.035	0.12
z^5	120.0	0	140.0	$2.5 \cdot 10^{-12}$	120.0	0.34	120.0	0.34
z^4	0	0	$-1.3 \cdot 10^{-11}$	$3.9 \cdot 10^{-12}$	-0.31	$-3.8 \cdot 10^{-3}$	-0.31	$-7.4 \cdot 10^{-3}$
z^3	44.0	0	54.0	$2.4 \cdot 10^{-12}$	44.0	0.17	44.0	0.17
z^2	0	0	$-2.9 \cdot 10^{-12}$	$8.8 \cdot 10^{-13}$	-0.1	-0.016	-0.11	-0.017
z	2.7	0	3.5	$2.6 \cdot 10^{-13}$	2.8	0.014	2.8	0.014
1.0	0	0	$-4.8 \cdot 10^{-14}$	$3.1 \cdot 10^{-14}$	$-2.7 \cdot 10^{-3}$	$-5.5 \cdot 10^{-4}$	$-2.7 \cdot 10^{-3}$	$-5.6 \cdot 10^{-4}$
z^{10}	1.0	0	1.0	0	1.0	$-3.4 \cdot 10^{-17}$	1.0	0
z^9	0	0	$3.0 \cdot 10^{-16}$	$4.9 \cdot 10^{-14}$	0.051	0.023	0.051	0.022
z^8	34.0	0	36.0	$1.9 \cdot 10^{-13}$	34.0	0.036	34.0	0.036
z^7	0	0	$-2.8 \cdot 10^{-12}$	$2.3 \cdot 10^{-12}$	0.17	0.12	0.17	0.12
z^6	120.0	0	130.0	$1.2 \cdot 10^{-12}$	120.0	0.26	120.0	0.26
z^5	0	0	$-1.3 \cdot 10^{-11}$	$5.3 \cdot 10^{-12}$	-0.2	0.062	-0.2	0.057
z^4	89.0	0	110.0	$3.3 \cdot 10^{-12}$	90.0	0.3	90.0	0.3
z^3	0	0	$-7.8 \cdot 10^{-12}$	$2.3 \cdot 10^{-12}$	-0.24	-0.026	-0.24	-0.028
z^2	14.0	0	18.0	$1.1 \cdot 10^{-12}$	14.0	0.064	14.0	0.064
z	0	0	$-5.9 \cdot 10^{-13}$	$2.5 \cdot 10^{-13}$	-0.026	$-4.7 \cdot 10^{-3}$	-0.026	$-4.9 \cdot 10^{-3}$
1.0	0.24	0	0.31	$3.2 \cdot 10^{-14}$	0.24	$1.4 \cdot 10^{-3}$	0.24	$1.4 \cdot 10^{-3}$

Table 109: Case 1a (cpv MACA64 MR2023b), $r = 10$, Z4: numerator (first lines) and denominator (last lines) coefficients.

	Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$		LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$		AAA-sign-L200, $\sigma_{10} = 3.6 \cdot 10^{-12}$		AAA-sign-L1000, $\sigma_{10} = 3.6 \cdot 10^{-12}$	
Basis	real	imag.	real	imag.	real	imag.	real	imag.
z^{10}	0	0	0	0	$4.2 \cdot 10^{-8}$	$-3.6 \cdot 10^{-5}$	$7.5 \cdot 10^{-8}$	$4.1 \cdot 10^{-6}$
z^9	8.7	0	8.9	$-1.1 \cdot 10^{-13}$	8.7	$-6.5 \cdot 10^{-7}$	8.7	$1.3 \cdot 10^{-7}$
z^8	0	0	$-1.6 \cdot 10^{-14}$	$-3.8 \cdot 10^{-12}$	$1.3 \cdot 10^{-6}$	$-1.2 \cdot 10^{-3}$	$2.6 \cdot 10^{-6}$	$1.4 \cdot 10^{-4}$
z^7	78.0	0	85.0	$-3.7 \cdot 10^{-12}$	78.0	$-1.8 \cdot 10^{-5}$	78.0	$3.5 \cdot 10^{-6}$
z^6	0	0	$1.9 \cdot 10^{-12}$	$-2.6 \cdot 10^{-11}$	$4.0 \cdot 10^{-6}$	$-4.3 \cdot 10^{-3}$	$9.1 \cdot 10^{-6}$	$4.9 \cdot 10^{-4}$
z^5	120.0	0	140.0	$-7.9 \cdot 10^{-12}$	120.0	$-4.8 \cdot 10^{-5}$	120.0	$9.3 \cdot 10^{-6}$
z^4	0	0	$4.0 \cdot 10^{-12}$	$-3.0 \cdot 10^{-11}$	$2.6 \cdot 10^{-6}$	$-3.2 \cdot 10^{-3}$	$6.9 \cdot 10^{-6}$	$3.7 \cdot 10^{-4}$
z^3	44.0	0	54.0	$-2.9 \cdot 10^{-12}$	44.0	$-2.4 \cdot 10^{-5}$	44.0	$4.7 \cdot 10^{-6}$
z^2	0	0	$1.2 \cdot 10^{-12}$	$-6.9 \cdot 10^{-12}$	$3.7 \cdot 10^{-7}$	$-5.2 \cdot 10^{-4}$	$1.1 \cdot 10^{-6}$	$6.0 \cdot 10^{-5}$
z	2.7	0	3.5	$-1.3 \cdot 10^{-13}$	2.7	$-2.0 \cdot 10^{-6}$	2.7	$3.9 \cdot 10^{-7}$
1.0	0	0	$3.2 \cdot 10^{-14}$	$-1.5 \cdot 10^{-13}$	$5.1 \cdot 10^{-9}$	$-8.7 \cdot 10^{-6}$	$1.9 \cdot 10^{-8}$	$1.0 \cdot 10^{-6}$
z^{10}	1.0	0	1.0	0	1.0	0	1.0	0
z^9	0	0	$-4.1 \cdot 10^{-14}$	$-5.2 \cdot 10^{-13}$	$3.5 \cdot 10^{-7}$	$-3.1 \cdot 10^{-4}$	$6.6 \cdot 10^{-7}$	$3.5 \cdot 10^{-5}$
z^8	34.0	0	36.0	$-1.1 \cdot 10^{-12}$	34.0	$-5.1 \cdot 10^{-6}$	34.0	$1.0 \cdot 10^{-6}$
z^7	0	0	$5.3 \cdot 10^{-13}$	$-1.3 \cdot 10^{-11}$	$2.8 \cdot 10^{-6}$	$-2.8 \cdot 10^{-3}$	$6.0 \cdot 10^{-6}$	$3.2 \cdot 10^{-4}$
z^6	120.0	0	130.0	$-6.8 \cdot 10^{-12}$	120.0	$-3.7 \cdot 10^{-5}$	120.0	$7.1 \cdot 10^{-6}$
z^5	0	0	$3.6 \cdot 10^{-12}$	$-3.4 \cdot 10^{-11}$	$3.9 \cdot 10^{-6}$	$-4.5 \cdot 10^{-3}$	$9.5 \cdot 10^{-6}$	$5.1 \cdot 10^{-4}$
z^4	89.0	0	110.0	$-6.0 \cdot 10^{-12}$	89.0	$-4.2 \cdot 10^{-5}$	89.0	$8.1 \cdot 10^{-6}$
z^3	0	0	$2.8 \cdot 10^{-12}$	$-1.8 \cdot 10^{-11}$	$1.2 \cdot 10^{-6}$	$-1.6 \cdot 10^{-3}$	$3.4 \cdot 10^{-6}$	$1.8 \cdot 10^{-4}$
z^2	14.0	0	18.0	$-8.5 \cdot 10^{-13}$	14.0	$-9.1 \cdot 10^{-6}$	14.0	$1.8 \cdot 10^{-6}$
z	0	0	$2.9 \cdot 10^{-13}$	$-1.5 \cdot 10^{-12}$	$6.5 \cdot 10^{-8}$	$-1.0 \cdot 10^{-4}$	$2.2 \cdot 10^{-7}$	$1.2 \cdot 10^{-5}$
1.0	0.24	0	0.31	$-8.1 \cdot 10^{-15}$	0.24	$-1.9 \cdot 10^{-7}$	0.24	$3.7 \cdot 10^{-8}$

Table 110: Case 1a (cpv MACA64 MR2022b), $r = 10$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$		LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$		AAA-sign-L200, $\sigma_{10} = 3.6 \cdot 10^{-12}$		AAA-sign-L1000, $\sigma_{10} = 3.6 \cdot 10^{-12}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^{10}	0	0	0	0	$2.4 \cdot 10^{-9}$	$4.3 \cdot 10^{-6}$	$2.2 \cdot 10^{-9}$	$-3.9 \cdot 10^{-8}$
z^9	8.7	0	8.9	$-2.5 \cdot 10^{-13}$	8.7	$-1.0 \cdot 10^{-7}$	8.7	$-1.3 \cdot 10^{-8}$
z^8	0	0	$1.6 \cdot 10^{-12}$	$1.8 \cdot 10^{-13}$	$5.9 \cdot 10^{-8}$	$1.5 \cdot 10^{-4}$	$7.2 \cdot 10^{-8}$	$-1.4 \cdot 10^{-6}$
z^7	78.0	0	85.0	$-6.0 \cdot 10^{-12}$	78.0	$-2.8 \cdot 10^{-6}$	78.0	$-3.7 \cdot 10^{-7}$
z^6	0	0	$1.1 \cdot 10^{-11}$	$2.3 \cdot 10^{-12}$	$1.4 \cdot 10^{-7}$	$5.0 \cdot 10^{-4}$	$2.4 \cdot 10^{-7}$	$-5.2 \cdot 10^{-6}$
z^5	120.0	0	140.0	$-1.8 \cdot 10^{-11}$	120.0	$-7.5 \cdot 10^{-6}$	120.0	$-9.7 \cdot 10^{-7}$
z^4	0	0	$1.2 \cdot 10^{-11}$	$2.9 \cdot 10^{-12}$	$7.2 \cdot 10^{-8}$	$3.8 \cdot 10^{-4}$	$1.8 \cdot 10^{-7}$	$-4.2 \cdot 10^{-6}$
z^3	44.0	0	54.0	$-1.1 \cdot 10^{-11}$	44.0	$-3.7 \cdot 10^{-6}$	44.0	$-4.9 \cdot 10^{-7}$
z^2	0	0	$3.0 \cdot 10^{-12}$	$8.7 \cdot 10^{-13}$	$7.8 \cdot 10^{-9}$	$6.0 \cdot 10^{-5}$	$2.8 \cdot 10^{-8}$	$-7.1 \cdot 10^{-7}$
z	2.7	0	3.5	$-1.1 \cdot 10^{-12}$	2.7	$-3.0 \cdot 10^{-7}$	2.7	$-3.9 \cdot 10^{-8}$
1.0	0	0	$7.9 \cdot 10^{-14}$	$2.1 \cdot 10^{-14}$	$9.9 \cdot 10^{-11}$	$1.0 \cdot 10^{-6}$	$4.6 \cdot 10^{-10}$	$-1.3 \cdot 10^{-8}$
z^{10}	1.0	0	1.0	0	1.0	0	1.0	0
z^9	0	0	$2.2 \cdot 10^{-13}$	$-7.1 \cdot 10^{-15}$	$1.8 \cdot 10^{-8}$	$3.7 \cdot 10^{-5}$	$1.9 \cdot 10^{-8}$	$-3.5 \cdot 10^{-7}$
z^8	34.0	0	36.0	$-1.8 \cdot 10^{-12}$	34.0	$-8.1 \cdot 10^{-7}$	34.0	$-1.1 \cdot 10^{-7}$
z^7	0	0	$5.5 \cdot 10^{-12}$	$1.0 \cdot 10^{-12}$	$1.1 \cdot 10^{-7}$	$3.3 \cdot 10^{-4}$	$1.6 \cdot 10^{-7}$	$-3.3 \cdot 10^{-6}$
z^6	120.0	0	130.0	$-1.3 \cdot 10^{-11}$	120.0	$-5.7 \cdot 10^{-6}$	120.0	$-7.4 \cdot 10^{-7}$
z^5	0	0	$1.4 \cdot 10^{-11}$	$3.1 \cdot 10^{-12}$	$1.2 \cdot 10^{-7}$	$5.2 \cdot 10^{-4}$	$2.5 \cdot 10^{-7}$	$-5.6 \cdot 10^{-6}$
z^4	89.0	0	110.0	$-1.7 \cdot 10^{-11}$	89.0	$-6.5 \cdot 10^{-6}$	89.0	$-8.4 \cdot 10^{-7}$
z^3	0	0	$7.6 \cdot 10^{-12}$	$1.9 \cdot 10^{-12}$	$2.9 \cdot 10^{-8}$	$1.9 \cdot 10^{-4}$	$8.7 \cdot 10^{-8}$	$-2.1 \cdot 10^{-6}$
z^2	14.0	0	18.0	$-4.6 \cdot 10^{-12}$	14.0	$-1.4 \cdot 10^{-6}$	14.0	$-1.8 \cdot 10^{-7}$
z	0	0	$7.3 \cdot 10^{-13}$	$2.2 \cdot 10^{-13}$	$1.3 \cdot 10^{-9}$	$1.2 \cdot 10^{-5}$	$5.4 \cdot 10^{-9}$	$-1.4 \cdot 10^{-7}$
1.0	0.24	0	0.31	$-1.2 \cdot 10^{-13}$	0.24	$-2.9 \cdot 10^{-8}$	0.24	$-3.8 \cdot 10^{-9}$

Table 111: Case 1a (cpv PCWIN64 MR2025b), $r = 10$, Z4: numerator (first lines) and denominator (last lines) coefficients.

Basis	Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$		LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$		AAA-sign-L200, $\sigma_{10} = 3.6 \cdot 10^{-12}$		AAA-sign-L1000, $\sigma_{10} = 3.6 \cdot 10^{-12}$	
	real	imag.	real	imag.	real	imag.	real	imag.
z^{10}	0	0	0	0	$2.4 \cdot 10^{-9}$	$4.3 \cdot 10^{-6}$	$2.2 \cdot 10^{-9}$	$-3.9 \cdot 10^{-8}$
z^9	8.7	0	8.9	$-2.5 \cdot 10^{-13}$	8.7	$-1.0 \cdot 10^{-7}$	8.7	$-1.3 \cdot 10^{-8}$
z^8	0	0	$1.6 \cdot 10^{-12}$	$1.8 \cdot 10^{-13}$	$5.9 \cdot 10^{-8}$	$1.5 \cdot 10^{-4}$	$7.2 \cdot 10^{-8}$	$-1.4 \cdot 10^{-6}$
z^7	78.0	0	85.0	$-6.0 \cdot 10^{-12}$	78.0	$-2.8 \cdot 10^{-6}$	78.0	$-3.7 \cdot 10^{-7}$
z^6	0	0	$1.1 \cdot 10^{-11}$	$2.3 \cdot 10^{-12}$	$1.4 \cdot 10^{-7}$	$5.0 \cdot 10^{-4}$	$2.4 \cdot 10^{-7}$	$-5.2 \cdot 10^{-6}$
z^5	120.0	0	140.0	$-1.8 \cdot 10^{-11}$	120.0	$-7.5 \cdot 10^{-6}$	120.0	$-9.7 \cdot 10^{-7}$
z^4	0	0	$1.2 \cdot 10^{-11}$	$2.9 \cdot 10^{-12}$	$7.2 \cdot 10^{-8}$	$3.8 \cdot 10^{-4}$	$1.8 \cdot 10^{-7}$	$-4.2 \cdot 10^{-6}$
z^3	44.0	0	54.0	$-1.1 \cdot 10^{-11}$	44.0	$-3.7 \cdot 10^{-6}$	44.0	$-4.9 \cdot 10^{-7}$
z^2	0	0	$3.0 \cdot 10^{-12}$	$8.7 \cdot 10^{-13}$	$7.8 \cdot 10^{-9}$	$6.0 \cdot 10^{-5}$	$2.8 \cdot 10^{-8}$	$-7.1 \cdot 10^{-7}$
z	2.7	0	3.5	$-1.1 \cdot 10^{-12}$	2.7	$-3.0 \cdot 10^{-7}$	2.7	$-3.9 \cdot 10^{-8}$
1.0	0	0	$7.9 \cdot 10^{-14}$	$2.1 \cdot 10^{-14}$	$9.9 \cdot 10^{-11}$	$1.0 \cdot 10^{-6}$	$4.6 \cdot 10^{-10}$	$-1.3 \cdot 10^{-8}$
z^{10}	1.0	0	1.0	0	1.0	0	1.0	0
z^9	0	0	$2.2 \cdot 10^{-13}$	$-7.1 \cdot 10^{-15}$	$1.8 \cdot 10^{-8}$	$3.7 \cdot 10^{-5}$	$1.9 \cdot 10^{-8}$	$-3.5 \cdot 10^{-7}$
z^8	34.0	0	36.0	$-1.8 \cdot 10^{-12}$	34.0	$-8.1 \cdot 10^{-7}$	34.0	$-1.1 \cdot 10^{-7}$
z^7	0	0	$5.5 \cdot 10^{-12}$	$1.0 \cdot 10^{-12}$	$1.1 \cdot 10^{-7}$	$3.3 \cdot 10^{-4}$	$1.6 \cdot 10^{-7}$	$-3.3 \cdot 10^{-6}$
z^6	120.0	0	130.0	$-1.3 \cdot 10^{-11}$	120.0	$-5.7 \cdot 10^{-6}$	120.0	$-7.4 \cdot 10^{-7}$
z^5	0	0	$1.4 \cdot 10^{-11}$	$3.1 \cdot 10^{-12}$	$1.2 \cdot 10^{-7}$	$5.2 \cdot 10^{-4}$	$2.5 \cdot 10^{-7}$	$-5.6 \cdot 10^{-6}$
z^4	89.0	0	110.0	$-1.7 \cdot 10^{-11}$	89.0	$-6.5 \cdot 10^{-6}$	89.0	$-8.4 \cdot 10^{-7}$
z^3	0	0	$7.6 \cdot 10^{-12}$	$1.9 \cdot 10^{-12}$	$2.9 \cdot 10^{-8}$	$1.9 \cdot 10^{-4}$	$8.7 \cdot 10^{-8}$	$-2.1 \cdot 10^{-6}$
z^2	14.0	0	18.0	$-4.6 \cdot 10^{-12}$	14.0	$-1.4 \cdot 10^{-6}$	14.0	$-1.8 \cdot 10^{-7}$
z	0	0	$7.3 \cdot 10^{-13}$	$2.2 \cdot 10^{-13}$	$1.3 \cdot 10^{-9}$	$1.2 \cdot 10^{-5}$	$5.4 \cdot 10^{-9}$	$-1.4 \cdot 10^{-7}$
1.0	0.24	0	0.31	$-1.2 \cdot 10^{-13}$	0.24	$-2.9 \cdot 10^{-8}$	0.24	$-3.8 \cdot 10^{-9}$

Table 112: Case 1a (AlexReis PCWIN64 MR2025b), $r = 10$, Z4: numerator (first lines) and denominator (last lines) coefficients.

	Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$		LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$		AAA-sign-L200, $\sigma_{10} = 4.3 \cdot 10^{-12}$		AAA-sign-L1000, $\sigma_{10} = 4.3 \cdot 10^{-12}$	
Basis	real	imag.	real	imag.	real	imag.	real	imag.
z^{10}	0	0	0	0	$-4.1 \cdot 10^{-3}$	$-1.7 \cdot 10^{-3}$	$-4.1 \cdot 10^{-3}$	$-1.7 \cdot 10^{-3}$
z^9	8.7	0	8.9	$2.6 \cdot 10^{-14}$	8.6	$-2.1 \cdot 10^{-3}$	8.6	$-2.1 \cdot 10^{-3}$
z^8	0	0	$9.9 \cdot 10^{-13}$	$-2.6 \cdot 10^{-12}$	-0.07	-0.035	-0.07	-0.035
z^7	78.0	0	85.0	$9.1 \cdot 10^{-13}$	77.0	-0.058	77.0	-0.058
z^6	0	0	$6.9 \cdot 10^{-12}$	$-2.3 \cdot 10^{-11}$	-0.022	-0.047	-0.022	-0.048
z^5	120.0	0	140.0	$1.5 \cdot 10^{-12}$	120.0	-0.15	120.0	-0.15
z^4	0	0	$6.3 \cdot 10^{-12}$	$-3.0 \cdot 10^{-11}$	0.14	0.02	0.14	0.019
z^3	44.0	0	54.0	$1.1 \cdot 10^{-14}$	43.0	-0.079	43.0	-0.079
z^2	0	0	$7.1 \cdot 10^{-13}$	$-7.0 \cdot 10^{-12}$	0.047	0.012	0.047	0.012
z	2.7	0	3.5	$-7.7 \cdot 10^{-14}$	2.7	$-6.4 \cdot 10^{-3}$	2.7	$-6.4 \cdot 10^{-3}$
1.0	0	0	$6.1 \cdot 10^{-15}$	$-1.5 \cdot 10^{-13}$	$1.2 \cdot 10^{-3}$	$3.3 \cdot 10^{-4}$	$1.2 \cdot 10^{-3}$	$3.3 \cdot 10^{-4}$
z^{10}	1.0	0	1.0	0	1.0	0	1.0	$3.3 \cdot 10^{-17}$
z^9	0	0	$1.1 \cdot 10^{-13}$	$-3.0 \cdot 10^{-13}$	-0.026	-0.012	-0.026	-0.012
z^8	34.0	0	36.0	$2.7 \cdot 10^{-13}$	34.0	-0.017	34.0	-0.017
z^7	0	0	$3.4 \cdot 10^{-12}$	$-1.0 \cdot 10^{-11}$	-0.087	-0.056	-0.087	-0.057
z^6	120.0	0	130.0	$1.6 \cdot 10^{-12}$	120.0	-0.12	120.0	-0.12
z^5	0	0	$8.0 \cdot 10^{-12}$	$-3.2 \cdot 10^{-11}$	0.088	-0.01	0.088	-0.012
z^4	89.0	0	110.0	$7.9 \cdot 10^{-13}$	87.0	-0.14	87.0	-0.14
z^3	0	0	$2.8 \cdot 10^{-12}$	$-1.8 \cdot 10^{-11}$	0.11	0.023	0.11	0.023
z^2	14.0	0	18.0	$-1.5 \cdot 10^{-13}$	14.0	-0.029	14.0	-0.029
z	0	0	$8.4 \cdot 10^{-14}$	$-1.5 \cdot 10^{-12}$	0.011	$3.1 \cdot 10^{-3}$	0.011	$3.0 \cdot 10^{-3}$
1.0	0.24	0	0.31	$-9.6 \cdot 10^{-15}$	0.23	$-6.2 \cdot 10^{-4}$	0.23	$-6.2 \cdot 10^{-4}$

Table 113: Case 1a (AlexReis PCWIN64perso MR2017a), $r = 10$, Z4: numerator (first lines) and denominator (last lines) coefficients.

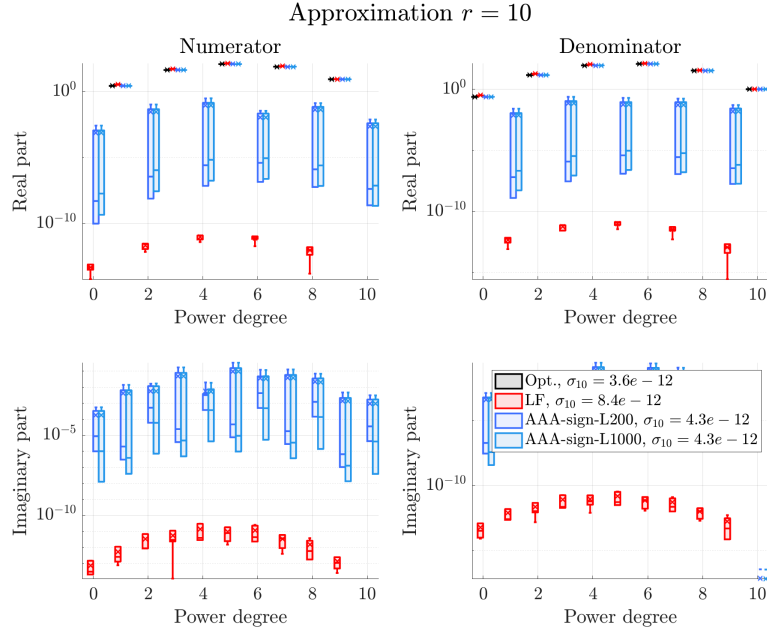


Figure 26: Case 1a, $r = 10$, Z4. Numerator and denominator dispersion over the different platforms.

A.8.2 Poles

The poles are given in the following tables. Each table gives the results obtained on different platform.

	Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$	LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$	AAA-sign-L200, $\sigma_{10} = 4.3 \cdot 10^{-12}$	AAA-sign-L1000, $\sigma_{10} = 4.3 \cdot 10^{-12}$
1.0	$2.22 \cdot 10^{-16} + 5.47i$	$1.27 \cdot 10^{-14} + 0.141i$	$0.00103 - 0.137i$	$0.00103 - 0.137i$
2.0	$2.22 \cdot 10^{-16} - 5.47i$	$1.93 \cdot 10^{-14} - 0.141i$	$8.67 \cdot 10^{-4} + 0.138i$	$8.68 \cdot 10^{-4} + 0.138i$
3.0	$2.78 \cdot 10^{-16} + 1.7i$	$1.97 \cdot 10^{-14} + 0.453i$	$0.00103 - 0.442i$	$0.00103 - 0.442i$
4.0	$2.78 \cdot 10^{-16} - 1.7i$	$3.14 \cdot 10^{-14} - 0.453i$	$5.19 \cdot 10^{-4} + 0.442i$	$5.2 \cdot 10^{-4} + 0.442i$
5.0	$-1.25 \cdot 10^{-16} + 0.866i$	$9.09 \cdot 10^{-16} + 0.891i$	$-2.36 \cdot 10^{-4} + 0.867i$	$-2.35 \cdot 10^{-4} + 0.867i$
6.0	$-1.25 \cdot 10^{-16} - 0.866i$	$2.9 \cdot 10^{-14} - 0.891i$	$7.39 \cdot 10^{-4} - 0.868i$	$7.4 \cdot 10^{-4} - 0.868i$
7.0	$-6.94 \cdot 10^{-18} + 0.441i$	$-7.64 \cdot 10^{-15} + 1.75i$	$-0.00261 + 1.7i$	$-0.00261 + 1.7i$
8.0	$-6.94 \cdot 10^{-18} - 0.441i$	$-1.98 \cdot 10^{-14} - 1.75i$	$-7.39 \cdot 10^{-4} - 1.7i$	$-7.38 \cdot 10^{-4} - 1.7i$
9.0	$-1.34 \cdot 10^{-17} + 0.137i$	$-2.66 \cdot 10^{-14} - 5.64i$	$-0.0286 + 5.47i$	$-0.0285 + 5.47i$
10.0	$-1.34 \cdot 10^{-17} - 0.137i$	$-5.92 \cdot 10^{-14} + 5.64i$	$-0.0228 - 5.49i$	$-0.0228 - 5.49i$

Table 114: Case 1a (cpv MACA64 MR2023b), $r = 10$, Z4: poles.

	Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$	LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$	AAA-sign-L200, $\sigma_{10} = 3.6 \cdot 10^{-12}$	AAA-sign-L1000, $\sigma_{10} = 3.6 \cdot 10^{-12}$
1.0	$6.66 \cdot 10^{-16} + 5.47i$	$-6.77 \cdot 10^{-15} - 0.141i$	$-1.38 \cdot 10^{-8} - 0.137i$	$-9.33 \cdot 10^{-9} + 0.137i$
2.0	$6.66 \cdot 10^{-16} - 5.47i$	$-1.0 \cdot 10^{-14} + 0.141i$	$9.86 \cdot 10^{-9} + 0.137i$	$-4.69 \cdot 10^{-9} - 0.137i$
3.0	$2.22 \cdot 10^{-16} + 1.7i$	$-3.47 \cdot 10^{-15} - 0.453i$	$-3.99 \cdot 10^{-8} - 0.441i$	$-1.58 \cdot 10^{-8} + 0.441i$
4.0	$2.22 \cdot 10^{-16} - 1.7i$	$-1.04 \cdot 10^{-14} + 0.453i$	$3.43 \cdot 10^{-8} + 0.441i$	$-1.28 \cdot 10^{-9} - 0.441i$
5.0	$-5.2 \cdot 10^{-17} + 0.866i$	$-3.18 \cdot 10^{-15} - 0.891i$	$-7.56 \cdot 10^{-8} - 0.866i$	$-2.7 \cdot 10^{-8} + 0.866i$
6.0	$-5.2 \cdot 10^{-17} - 0.866i$	$1.9 \cdot 10^{-15} + 0.891i$	$6.45 \cdot 10^{-8} + 0.866i$	$2.74 \cdot 10^{-10} - 0.866i$
7.0	$3.47 \cdot 10^{-18} + 0.441i$	$-1.86 \cdot 10^{-14} - 1.75i$	$-1.48 \cdot 10^{-7} - 1.7i$	$-5.77 \cdot 10^{-8} + 1.7i$
8.0	$3.47 \cdot 10^{-18} - 0.441i$	$3.05 \cdot 10^{-14} + 1.75i$	$1.16 \cdot 10^{-7} + 1.7i$	$-6.33 \cdot 10^{-9} - 1.7i$
9.0	$-1.52 \cdot 10^{-18} + 0.137i$	$-5.53 \cdot 10^{-14} - 5.64i$	$-5.63 \cdot 10^{-7} - 5.47i$	$-3.48 \cdot 10^{-7} + 5.47i$
10.0	$-1.52 \cdot 10^{-18} - 0.137i$	$1.18 \cdot 10^{-13} + 5.64i$	$2.67 \cdot 10^{-7} + 5.47i$	$-1.87 \cdot 10^{-7} - 5.47i$

Table 115: Case 1a (cpv MACA64 MR2022b), $r = 10$, Z4: poles.

	Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$	LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$	AAA-sign-L200, $\sigma_{10} = 3.6 \cdot 10^{-12}$	AAA-sign-L1000, $\sigma_{10} = 3.6 \cdot 10^{-12}$
1.0	$2.22 \cdot 10^{-16} + 5.47i$	$-1.13 \cdot 10^{-14} + 0.141i$	$1.68 \cdot 10^{-9} + 0.137i$	$-3.96 \cdot 10^{-10} - 0.137i$
2.0	$2.22 \cdot 10^{-16} - 5.47i$	$-3.06 \cdot 10^{-14} - 0.141i$	$-1.75 \cdot 10^{-9} - 0.137i$	$4.93 \cdot 10^{-11} + 0.137i$
3.0	$2.78 \cdot 10^{-16} + 1.7i$	$6.86 \cdot 10^{-15} + 0.453i$	$5.46 \cdot 10^{-9} + 0.441i$	$-9.24 \cdot 10^{-10} - 0.441i$
4.0	$2.78 \cdot 10^{-16} - 1.7i$	$-3.6 \cdot 10^{-14} - 0.453i$	$-5.53 \cdot 10^{-9} - 0.441i$	$5.0 \cdot 10^{-10} + 0.441i$
5.0	$-1.25 \cdot 10^{-16} + 0.866i$	$1.56 \cdot 10^{-14} + 0.891i$	$1.05 \cdot 10^{-8} + 0.866i$	$-1.72 \cdot 10^{-9} - 0.866i$
6.0	$-1.25 \cdot 10^{-16} - 0.866i$	$-4.39 \cdot 10^{-14} - 0.891i$	$-1.08 \cdot 10^{-8} - 0.866i$	$1.04 \cdot 10^{-9} + 0.866i$
7.0	$-6.94 \cdot 10^{-18} + 0.441i$	$8.07 \cdot 10^{-15} + 1.75i$	$2.0 \cdot 10^{-8} + 1.7i$	$-3.56 \cdot 10^{-9} - 1.7i$
8.0	$-6.94 \cdot 10^{-18} - 0.441i$	$-5.94 \cdot 10^{-14} - 1.75i$	$-2.13 \cdot 10^{-8} - 1.7i$	$1.79 \cdot 10^{-9} + 1.7i$
9.0	$-1.34 \cdot 10^{-17} + 0.137i$	$-1.74 \cdot 10^{-13} - 5.64i$	$5.76 \cdot 10^{-8} + 5.47i$	$-1.63 \cdot 10^{-8} - 5.47i$
10.0	$-1.34 \cdot 10^{-17} - 0.137i$	$1.08 \cdot 10^{-13} + 5.64i$	$-7.4 \cdot 10^{-8} - 5.47i$	$7.27 \cdot 10^{-10} + 5.47i$

Table 116: Case 1a (cpv PCWIN64 MR2025b), $r = 10$, Z4: poles.

	Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$	LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$	AAA-sign-L200, $\sigma_{10} = 3.6 \cdot 10^{-12}$	AAA-sign-L1000, $\sigma_{10} = 3.6 \cdot 10^{-12}$
1.0	$2.22 \cdot 10^{-16} + 5.47i$	$-1.13 \cdot 10^{-14} + 0.141i$	$1.68 \cdot 10^{-9} + 0.137i$	$-3.96 \cdot 10^{-10} - 0.137i$
2.0	$2.22 \cdot 10^{-16} - 5.47i$	$-3.06 \cdot 10^{-14} - 0.141i$	$-1.75 \cdot 10^{-9} - 0.137i$	$4.93 \cdot 10^{-11} + 0.137i$
3.0	$2.78 \cdot 10^{-16} + 1.7i$	$6.86 \cdot 10^{-15} + 0.453i$	$5.46 \cdot 10^{-9} + 0.441i$	$-9.24 \cdot 10^{-10} - 0.441i$
4.0	$2.78 \cdot 10^{-16} - 1.7i$	$-3.6 \cdot 10^{-14} - 0.453i$	$-5.53 \cdot 10^{-9} - 0.441i$	$5.0 \cdot 10^{-10} + 0.441i$
5.0	$-1.25 \cdot 10^{-16} + 0.866i$	$1.56 \cdot 10^{-14} + 0.891i$	$1.05 \cdot 10^{-8} + 0.866i$	$-1.72 \cdot 10^{-9} - 0.866i$
6.0	$-1.25 \cdot 10^{-16} - 0.866i$	$-4.39 \cdot 10^{-14} - 0.891i$	$-1.08 \cdot 10^{-8} - 0.866i$	$1.04 \cdot 10^{-9} + 0.866i$
7.0	$-6.94 \cdot 10^{-18} + 0.441i$	$8.07 \cdot 10^{-15} + 1.75i$	$2.0 \cdot 10^{-8} + 1.7i$	$-3.56 \cdot 10^{-9} - 1.7i$
8.0	$-6.94 \cdot 10^{-18} - 0.441i$	$-5.94 \cdot 10^{-14} - 1.75i$	$-2.13 \cdot 10^{-8} - 1.7i$	$1.79 \cdot 10^{-9} + 1.7i$
9.0	$-1.34 \cdot 10^{-17} + 0.137i$	$-1.74 \cdot 10^{-13} - 5.64i$	$5.76 \cdot 10^{-8} + 5.47i$	$-1.63 \cdot 10^{-8} - 5.47i$
10.0	$-1.34 \cdot 10^{-17} - 0.137i$	$1.08 \cdot 10^{-13} + 5.64i$	$-7.4 \cdot 10^{-8} - 5.47i$	$7.27 \cdot 10^{-10} + 5.47i$

Table 117: Case 1a (AlexReis PCWIN64 MR2025b), $r = 10$, Z4: poles.

	Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$	LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$	AAA-sign-L200, $\sigma_{10} = 4.3 \cdot 10^{-12}$	AAA-sign-L1000, $\sigma_{10} = 4.3 \cdot 10^{-12}$
1.0	$6.66 \cdot 10^{-16} + 5.47z$	$-2.79 \cdot 10^{-15} - 0.141z$	$-3.88 \cdot 10^{-4} + 0.137z$	$-3.88 \cdot 10^{-4} + 0.137z$
2.0	$6.66 \cdot 10^{-16} - 5.47z$	$4.68 \cdot 10^{-16} + 0.141z$	$-4.69 \cdot 10^{-4} - 0.137z$	$-4.69 \cdot 10^{-4} - 0.137z$
3.0	$2.22 \cdot 10^{-16} + 1.7z$	$-1.35 \cdot 10^{-14} - 0.453z$	$-2.48 \cdot 10^{-4} + 0.44z$	$-2.48 \cdot 10^{-4} + 0.44z$
4.0	$2.22 \cdot 10^{-16} - 1.7z$	$-4.37 \cdot 10^{-15} + 0.453z$	$-5.0 \cdot 10^{-4} - 0.44z$	$-5.0 \cdot 10^{-4} - 0.44z$
5.0	$-5.2 \cdot 10^{-17} + 0.866z$	$-2.21 \cdot 10^{-14} - 0.891z$	$-3.93 \cdot 10^{-4} - 0.863z$	$-3.94 \cdot 10^{-4} - 0.863z$
6.0	$-5.2 \cdot 10^{-17} - 0.866z$	$-1.84 \cdot 10^{-14} + 0.891z$	$7.24 \cdot 10^{-5} + 0.863z$	$7.23 \cdot 10^{-5} + 0.863z$
7.0	$3.47 \cdot 10^{-18} + 0.441z$	$-1.05 \cdot 10^{-14} - 1.75z$	$3.62 \cdot 10^{-4} - 1.69z$	$3.61 \cdot 10^{-4} - 1.69z$
8.0	$3.47 \cdot 10^{-18} - 0.441z$	$-2.61 \cdot 10^{-14} + 1.75z$	$0.00122 + 1.69z$	$0.00122 + 1.69z$
9.0	$-1.52 \cdot 10^{-18} + 0.137z$	$1.53 \cdot 10^{-14} - 5.64z$	$0.012 - 5.44z$	$0.012 - 5.44z$
10.0	$-1.52 \cdot 10^{-18} - 0.137z$	$-2.97 \cdot 10^{-14} + 5.64z$	$0.0147 + 5.46z$	$0.0147 + 5.46z$

Table 118: Case 1a (AlexReis PCWIN64perso MR2017a), $r = 10$, Z4: poles.

A.8.3 Zeros

The zeros are given in the following tables. Each table gives the results obtained on different platform.

	Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$	LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$	AAA-sign-L200, $\sigma_{10} = 4.3 \cdot 10^{-12}$	AAA-sign-L1000, $\sigma_{10} = 4.3 \cdot 10^{-12}$
1.0	NaN	NaN	$9.66 \cdot 10^{-4} + 1.92 \cdot 10^{-4}z$	$9.67 \cdot 10^{-4} + 1.96 \cdot 10^{-4}z$
2.0	0	$1.35 \cdot 10^{-14} - 8.67 \cdot 10^{-15}z$	$0.00105 - 0.282z$	$0.00105 - 0.282z$
3.0	$3.33 \cdot 10^{-16} + 2.67z$	$1.74 \cdot 10^{-14} + 0.289z$	$7.24 \cdot 10^{-4} + 0.282z$	$7.24 \cdot 10^{-4} + 0.282z$
4.0	$3.33 \cdot 10^{-16} - 2.67z$	$2.58 \cdot 10^{-14} - 0.289z$	$9.39 \cdot 10^{-4} - 0.63z$	$9.39 \cdot 10^{-4} - 0.63z$
5.0	$2.43 \cdot 10^{-16} + 1.19z$	$1.34 \cdot 10^{-14} + 0.647z$	$2.23 \cdot 10^{-4} + 0.63z$	$2.23 \cdot 10^{-4} + 0.63z$
6.0	$2.43 \cdot 10^{-16} - 1.19z$	$3.58 \cdot 10^{-14} - 0.647z$	$-0.00102 + 1.19z$	$-0.00102 + 1.19z$
7.0	$4.16 \cdot 10^{-17} + 0.629z$	$-5.1 \cdot 10^{-15} + 1.23z$	$3.05 \cdot 10^{-4} - 1.19z$	$3.06 \cdot 10^{-4} - 1.19z$
8.0	$4.16 \cdot 10^{-17} - 0.629z$	$6.08 \cdot 10^{-15} - 1.23z$	$-0.00688 + 2.67z$	$-0.00687 + 2.67z$
9.0	$-3.47 \cdot 10^{-18} + 0.281z$	$-2.24 \cdot 10^{-14} + 2.75z$	$-0.00398 - 2.67z$	$-0.00398 - 2.67z$
10.0	$-3.47 \cdot 10^{-18} - 0.281z$	$-3.62 \cdot 10^{-14} - 2.75z$	$-971.0 + 392.0z$	$-975.0 + 389.0z$

Table 119: Case 1a (cpv MACA64 MR2023b), $r = 10$, Z4: zeros.

	Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$	LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$	AAA-sign-L200, $\sigma_{10} = 3.6 \cdot 10^{-12}$	AAA-sign-L1000, $\sigma_{10} = 3.6 \cdot 10^{-12}$
1.0	NaN	NaN	$-1.87 \cdot 10^{-9} + 3.18 \cdot 10^{-6}z$	$-6.85 \cdot 10^{-9} - 3.69 \cdot 10^{-7}z$
2.0	0	$-9.07 \cdot 10^{-15} + 4.27 \cdot 10^{-14}z$	$-2.63 \cdot 10^{-8} - 0.281z$	$-1.22 \cdot 10^{-8} + 0.281z$
3.0	$3.33 \cdot 10^{-16} + 2.67z$	$-4.41 \cdot 10^{-15} - 0.289z$	$2.18 \cdot 10^{-8} + 0.281z$	$-2.83 \cdot 10^{-9} - 0.281z$
4.0	$3.33 \cdot 10^{-16} - 2.67z$	$-1.09 \cdot 10^{-14} + 0.289z$	$-5.58 \cdot 10^{-8} - 0.629z$	$-2.05 \cdot 10^{-8} + 0.629z$
5.0	$-2.22 \cdot 10^{-16} + 1.19z$	$-4.14 \cdot 10^{-15} - 0.647z$	$4.82 \cdot 10^{-8} + 0.629z$	$-1.51 \cdot 10^{-10} - 0.629z$
6.0	$-2.22 \cdot 10^{-16} - 1.19z$	$-5.14 \cdot 10^{-15} + 0.647z$	$-1.03 \cdot 10^{-7} - 1.19z$	$-3.76 \cdot 10^{-8} + 1.19z$
7.0	$-4.16 \cdot 10^{-17} + 0.629z$	$-4.28 \cdot 10^{-15} - 1.23z$	$8.56 \cdot 10^{-8} + 1.19z$	$-8.47 \cdot 10^{-10} - 1.19z$
8.0	$-4.16 \cdot 10^{-17} - 0.629z$	$9.49 \cdot 10^{-15} + 1.23z$	$-2.41 \cdot 10^{-7} - 2.67z$	$-1.08 \cdot 10^{-7} + 2.67z$
9.0	$1.73 \cdot 10^{-18} + 0.281z$	$-3.66 \cdot 10^{-14} - 2.75z$	$1.68 \cdot 10^{-7} + 2.67z$	$-2.91 \cdot 10^{-8} - 2.67z$
10.0	$1.73 \cdot 10^{-18} - 0.281z$	$6.7 \cdot 10^{-14} + 2.75z$	$-284.0 - 2.41 \cdot 10^5z$	$-3.94 \cdot 10^4 + 2.13 \cdot 10^6z$

Table 120: Case 1a (cpv MACA64 MR2022b), $r = 10$, Z4: zeros.

	Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$	LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$	AAA-sign-L200, $\sigma_{10} = 3.6 \cdot 10^{-12}$	AAA-sign-L1000, $\sigma_{10} = 3.6 \cdot 10^{-12}$
1.0	NaN	NaN	$-3.62 \cdot 10^{-11} - 3.64 \cdot 10^{-7}z$	$-1.7 \cdot 10^{-10} + 4.59 \cdot 10^{-9}z$
2.0	0	$-2.24 \cdot 10^{-14} - 5.81 \cdot 10^{-15}z$	$3.48 \cdot 10^{-9} + 0.281z$	$-6.42 \cdot 10^{-10} - 0.281z$
3.0	$3.33 \cdot 10^{-16} + 2.67z$	$-4.04 \cdot 10^{-16} + 0.289z$	$-3.55 \cdot 10^{-9} - 0.281z$	$2.69 \cdot 10^{-10} + 0.281z$
4.0	$3.33 \cdot 10^{-16} - 2.67z$	$-3.45 \cdot 10^{-14} - 0.289z$	$7.74 \cdot 10^{-9} + 0.629z$	$-1.27 \cdot 10^{-9} - 0.629z$
5.0	$2.43 \cdot 10^{-16} + 1.19z$	$-3.9 \cdot 10^{-14} - 0.647z$	$-7.85 \cdot 10^{-9} - 0.629z$	$7.51 \cdot 10^{-10} + 0.629z$
6.0	$2.43 \cdot 10^{-16} - 1.19z$	$1.22 \cdot 10^{-14} + 0.647z$	$1.43 \cdot 10^{-8} + 1.19z$	$-2.4 \cdot 10^{-9} - 1.19z$
7.0	$4.16 \cdot 10^{-17} + 0.629z$	$1.25 \cdot 10^{-14} + 1.23z$	$-1.48 \cdot 10^{-8} - 1.19z$	$1.38 \cdot 10^{-9} + 1.19z$
8.0	$4.16 \cdot 10^{-17} - 0.629z$	$-5.01 \cdot 10^{-14} - 1.23z$	$3.04 \cdot 10^{-8} + 2.67z$	$-6.13 \cdot 10^{-9} - 2.67z$
9.0	$-3.47 \cdot 10^{-18} + 0.281z$	$-8.32 \cdot 10^{-14} - 2.75z$	$-3.4 \cdot 10^{-8} - 2.67z$	$2.21 \cdot 10^{-9} + 2.67z$
10.0	$-3.47 \cdot 10^{-18} - 0.281z$	$2.31 \cdot 10^{-14} + 2.75z$	$-1130.0 + 2.0 \cdot 10^6z$	$-1.28 \cdot 10^7 - 2.24 \cdot 10^8z$

Table 121: Case 1a (cpv PCWIN64 MR2025b), $r = 10$, Z4: zeros.

	Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$	LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$	AAA-sign-L200, $\sigma_{10} = 3.6 \cdot 10^{-12}$	AAA-sign-L1000, $\sigma_{10} = 3.6 \cdot 10^{-12}$
1.0	NaN	NaN	$-3.62 \cdot 10^{-11} - 3.64 \cdot 10^{-7}i$	$-1.7 \cdot 10^{-10} + 4.59 \cdot 10^{-9}i$
2.0	0	$-2.24 \cdot 10^{-14} - 5.81 \cdot 10^{-15}i$	$3.48 \cdot 10^{-9} + 0.281i$	$-6.42 \cdot 10^{-10} - 0.281i$
3.0	$3.33 \cdot 10^{-16} + 2.67i$	$-4.04 \cdot 10^{-16} + 0.289i$	$-3.55 \cdot 10^{-9} - 0.281i$	$2.69 \cdot 10^{-10} + 0.281i$
4.0	$3.33 \cdot 10^{-16} - 2.67i$	$-3.45 \cdot 10^{-14} - 0.289i$	$7.74 \cdot 10^{-9} + 0.629i$	$-1.27 \cdot 10^{-9} - 0.629i$
5.0	$2.43 \cdot 10^{-16} + 1.19i$	$-3.9 \cdot 10^{-14} - 0.647i$	$-7.85 \cdot 10^{-9} - 0.629i$	$7.51 \cdot 10^{-10} + 0.629i$
6.0	$2.43 \cdot 10^{-16} - 1.19i$	$1.22 \cdot 10^{-14} + 0.647i$	$1.43 \cdot 10^{-8} + 1.19i$	$-2.4 \cdot 10^{-9} - 1.19i$
7.0	$4.16 \cdot 10^{-17} + 0.629i$	$1.25 \cdot 10^{-14} + 1.23i$	$-1.48 \cdot 10^{-8} - 1.19i$	$1.38 \cdot 10^{-9} + 1.19i$
8.0	$4.16 \cdot 10^{-17} - 0.629i$	$-5.01 \cdot 10^{-14} - 1.23i$	$3.04 \cdot 10^{-8} + 2.67i$	$-6.13 \cdot 10^{-9} - 2.67i$
9.0	$-3.47 \cdot 10^{-18} + 0.281i$	$-8.32 \cdot 10^{-14} - 2.75i$	$-3.4 \cdot 10^{-8} - 2.67i$	$2.21 \cdot 10^{-9} + 2.67i$
10.0	$-3.47 \cdot 10^{-18} - 0.281i$	$2.31 \cdot 10^{-14} + 2.75i$	$-1130.0 + 2.0 \cdot 10^6i$	$-1.28 \cdot 10^7 - 2.24 \cdot 10^8i$

Table 122: Case 1a (AlexReis PCWIN64 MR2025b), $r = 10$, Z4: zeros.

	Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$	LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$	AAA-sign-L200, $\sigma_{10} = 4.3 \cdot 10^{-12}$	AAA-sign-L1000, $\sigma_{10} = 4.3 \cdot 10^{-12}$
1.0	NaN	NaN	$-4.33 \cdot 10^{-4} - 1.26 \cdot 10^{-4}i$	$-4.33 \cdot 10^{-4} - 1.25 \cdot 10^{-4}i$
2.0	0	$-1.74 \cdot 10^{-15} + 4.16 \cdot 10^{-14}i$	$-3.29 \cdot 10^{-4} + 0.28i$	$-3.29 \cdot 10^{-4} + 0.28i$
3.0	$3.89 \cdot 10^{-16} + 2.67i$	$-6.24 \cdot 10^{-15} - 0.289i$	$-4.93 \cdot 10^{-4} - 0.281i$	$-4.93 \cdot 10^{-4} - 0.281i$
4.0	$3.89 \cdot 10^{-16} - 2.67i$	$1.17 \cdot 10^{-15} + 0.289i$	$-1.27 \cdot 10^{-4} + 0.627i$	$-1.27 \cdot 10^{-4} + 0.627i$
5.0	$2.43 \cdot 10^{-17} + 1.19i$	$-2.09 \cdot 10^{-14} - 0.647i$	$-4.76 \cdot 10^{-4} - 0.627i$	$-4.76 \cdot 10^{-4} - 0.627i$
6.0	$2.43 \cdot 10^{-17} - 1.19i$	$-1.33 \cdot 10^{-14} + 0.647i$	$-1.82 \cdot 10^{-4} - 1.19i$	$-1.82 \cdot 10^{-4} - 1.19i$
7.0	$4.16 \cdot 10^{-17} + 0.629i$	$-1.98 \cdot 10^{-14} - 1.23i$	$4.39 \cdot 10^{-4} + 1.19i$	$4.39 \cdot 10^{-4} + 1.19i$
8.0	$4.16 \cdot 10^{-17} - 0.629i$	$-2.29 \cdot 10^{-14} + 1.23i$	$0.00208 - 2.66i$	$0.00208 - 2.66i$
9.0	$2.6 \cdot 10^{-17} + 0.281i$	$4.44 \cdot 10^{-16} - 2.75i$	$0.00341 + 2.66i$	$0.0034 + 2.66i$
10.0	$2.6 \cdot 10^{-17} - 0.281i$	$-2.8 \cdot 10^{-14} + 2.75i$	$1810.0 - 759.0i$	$1810.0 - 763.0i$

Table 123: Case 1a (AlexReis PCWIN64perso MR2017a), $r = 10$, Z4: zeros.

A.8.4 Poles and zeros visualisation (case cpv MACA64 MR2023b)

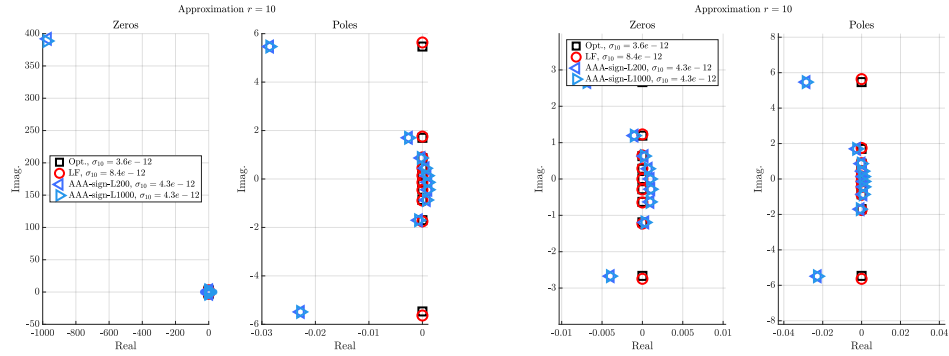


Figure 27: Case 1a (cpv MACA64 MR2023b), $r = 10$, Z4. Left frame: zeros (left) and poles (right). Right frame: same but discarding the largest real entry.

A.8.5 Approximation visualization

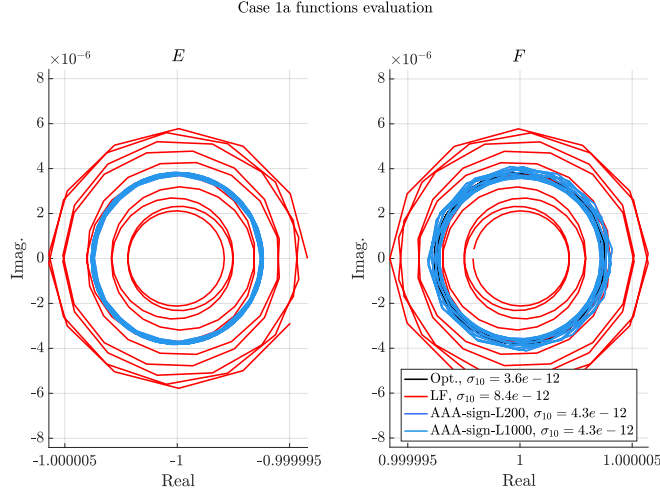


Figure 28: Case 1a (cpv MACA64 MR2023b), $r = 10$, Z4. E and F evaluations.

References

- [1] N. I. Akhiezer. *Elements of the theory of elliptic functions*, volume 79. American Mathematical Society, 1990.
- [2] A. C. Antoulas and B. D. O. Anderson. On the scalar rational interpolation problem. *IMA Journal of Mathematical Control and Information*, 3(2-3):61–88, 1986.
- [3] A. C. Antoulas, C. A. Beattie, and S. Gugercin. *Interpolatory Methods for Model Reduction*. Series in Computational Science and Engineering, SIAM, Philadelphia, 2020.
- [4] A. C. Antoulas, I. V. Gosea, and A. C. Ionita. Model reduction of bilinear systems in the Loewner framework. *SIAM Journal on Scientific Computing*, 38(5):B889–B916, 2016.
- [5] A. C. Antoulas, I. V. Gosea, and C. Poussot-Vassal. The Loewner framework for parametric systems: Taming the curse of dimensionality. e-print 2405.00495, arXiv, 2024. To appear in SIAM Review.
- [6] A. C. Antoulas, S. Lefteriu, and A. C. Ionita. *A Tutorial Introduction to the Loewner Framework for Model Reduction*, pages 335–376. Model Reduction and Approximation for Complex Systems, Computational Science & Engineering. SIAM, 2017.
- [7] P. D. Davis. *Interpolation and approximation*. Dover Publications, 1975.
- [8] T. A. Driscoll, N. Hale, and L. N. Trefethen. Chebfun Guide. <https://www.chebfun.org/docs/guide/>, 2014.
- [9] S. I. Filip, Y. Nakatsukasa, L. Trefethen, and B. Beckermann. Rational minimax approximation via adaptive barycentric interpolation. *SIAM Journal on Scientific Computing*, 40(4):A2427–A2455, 2018.

- [10] I. V. Gosea, C. Poussot-Vassal, and A. C. Antoulas. Data-driven modeling and control of large-scale dynamical systems in the Loewner framework: methodology and applications. In E. Zuazua and E. Trelat, editors, *Numerical Control: Part A*, volume 23 of *Handbook on Numerical Analysis*, chapter 15, pages 499–530. Elsevier, 2022.
- [11] A. C. Ionita and A. C. Antoulas. Data-driven parametrized model reduction in the Loewner framework. *SIAM Journal on Scientific Computing*, 36(3):A984–A1007, 2014.
- [12] A. Iske. *Approximation Theory and Algorithms for Data Analysis*. Texts in Applied Mathematics. Springer, 2018.
- [13] M-P Istace and J-P Thiran. On the third and fourth Zolotarev problems in the complex plane. *SIAM Journal on Numerical Analysis*, 32(1):249–259, 1995.
- [14] D. S. Karachalios, I. V. Gosea, and A. C. Antoulas. The Loewner framework for system identification and reduction. In *Model Reduction Handbook, Volume I: System- and Data-Driven Methods and Algorithms*, pages 181–228. De Gruyter, 2021.
- [15] D. S. Karachalios, I. V. Gosea, L. Gkimisis, and A. C. Antoulas. Data-driven quadratic modeling in the Loewner framework from input-output time-domain measurements. *SIAM Journal on Applied Dynamical Systems*, 24(1):457–500, 2025.
- [16] A J. Mayo and A C. Antoulas. A framework for the generalized realization problem. *Linear Algebra and Its Applications*, 426:634–662, 2007.
- [17] Y. Nakatsukasa and R. W. Freund. Computing fundamental matrix decompositions accurately via the matrix sign function in two iterations: The power of Zolotarev’s functions. *SIAM Review*, 58(3):461–493, 2016.
- [18] Y. Nakatsukasa, O. Sète, and L N. Trefethen. The AAA algorithm for rational approximation. *SIAM Journal on Scientific Computing*, 40(3):A1494–A1522, 2018.
- [19] Y. Nakatsukasa and L. N. Trefethen. An algorithm for real and complex rational minimax approximation. *SIAM Journal on Scientific Computing*, 42(5):A3157–A3179, 2020.
- [20] Y. Nakatsukasa and L. N. Trefethen. Applications of AAA rational approximation. Technical report, arXiv preprint, arXiv:2510.16237, 2020.
- [21] M. J. D. Powell. *Approximation theory and methods*. Cambridge University Press, 1981.
- [22] Gerhard Starke. Near-circularity for the rational Zolotarev problem in the complex plane. *Journal of Approximation Theory*, 70(1):115–130, 1992.
- [23] L N. Trefethen. *Approximation Theory and Approximation Practice*. SIAM, 2013.
- [24] L. N. Trefethen and H. Wilber. Computation of Zolotarev rational functions. *SIAM Journal on Scientific Computing*, 47(4):A2205–A2220, 2025.